MACHINE LEARNING-BASED ANALYSIS OF BFAR F/B PAGBABAGO LIVELIHOOD PROGRAM FOR OPTIMIZING DECISION-MAKING

A Thesis Presented to the Faculty, College of Computer Studies
Laguna State Polytechnic University
Los Baños Campus
Los Baños Laguna

In Partial Fulfilment of the Requirements for the degree
Bachelor of Science in Computer Science
Specialized in *Intelligent Systems*

MARK NICHOLAS L. RAZON
ANGELO B. URRIZA

GENE MARCK B. CATEDRILLA
JUNE 2026

ACKNOWLEDGEMENT

The researchers express sincere gratitude to the distinguished individuals and institutions whose invaluable support and contributions were instrumental in the successful completion of this research.

The University President, Dr. Mario R. Briones, for his dedication to academic excellence and his unwavering support for the scholarly community, both of which have been vital to the successful completion of this study;

Engr. Manuel Luis R. Alvarez, Campus Director, for his exemplary leadership and dedication to fostering an inclusive and dynamic academic environment that supports the scholarly endeavors and professional growth of teacher-researchers;

Dr. Loyd S. Echalar, Associate Dean of the College of Computer Studies (CCS), is acknowledged for his exceptional expertise, unwavering dedication, and invaluable support in critically evaluating and enhancing the overall structure of this paper, thereby contributing significantly to the scholarly discourse of this research;

The researchers would like to extend their heartfelt appreciation to their thesis adviser, Mr. Gene Marck B. Catedrilla, for his invaluable guidance, constructive feedback, and unwavering support throughout the development of this study. His expertise and encouragement have been instrumental in shaping the direction and outcome of this research.

Their deepest thanks also go to the members of my thesis panel — Dr. Jefferson L. Lerios, Dr. Sherwin B. Sapin, and Ms. Myra V. Paicaglino — for

their insightful suggestions, comments, and recommendations, which greatly contributed to the improvement of this project.

The researchers would also like to acknowledge Ms. Carolina R. Joval, for providing valuable knowledge and perspective that provided valuable mathematical developments in their study.

The Researchers

DEDICATION

This thesis is lovingly dedicated to our family, whose unconditional love and support have been the cornerstone of my success.

To their parents, Mrs. Maridel B. Urriza, Mr. Victor T. Urriza, and Mr. Romeo Razon, for instilling the values of perseverance, integrity, and faith as well as providing love, guidance, support, and sacrifices that have brought to this point of their academic journey.

A. B. U

M. N. L. R

ABSTRACT

This study, titled “Machine Learning-Based Analysis of Livelihood Programs for Optimizing Decision-Making,” presents a data-driven evaluation framework that integrates machine learning techniques to assess the impact and relevance of livelihood programs. The study utilized the BFAR socioeconomic dataset containing 1,339 records and 215 variables representing household demographics, income, and resource ownership across fisherfolk communities in Region IV-A. Using historical socioeconomic data, a Random Forest-based propensity score model was developed to estimate baseline conditions of program beneficiaries, ensuring fair comparisons between treatment and control groups. Several unsupervised models—Principal Component Analysis (PCA), Entropy-Weighted K-Means, Autoencoders, K-Prototypes, and Hierarchical Clustering—were implemented to identify patterns and group beneficiaries according to socioeconomic similarities. Among these, the PCA + Weighted K-Means model achieved the highest clustering performance based on the Silhouette Score, Calinski–Harabasz Index, and Davies–Bouldin Index. Statistical validation through ANOVA confirmed significant differences among models ($p < 0.05$), verifying the robustness of the proposed approach. The findings demonstrate that integrating machine learning enhances the fairness, transparency, and analytical depth of livelihood program evaluations, providing a reliable basis for designing targeted and equitable interventions. This research establishes a foundation for data-driven decision-making and continuous program improvement in the context of socioeconomic development.

Keywords: *Data-driven Decision Making, Livelihood Program, Propensity Score Matching, Ensemble Learning, Machine Learning Analysis*

TABLE OF CONTENTS

Preliminaries		***Page***
Title Page		i
Approval Sheet		ii
Acknowledgment		iii
Dedication		v
Abstract		vi
Table of Contents		vii
List of Figures		ix
List of Tables		x
Definition of Terms		xi
CHAPTER		
I	**INTRODUCTION**	**1**
	Background of the Study	1
	Research Objectives	5
	Significance of the Study	6
	Scope and Limitations	9
II	**REVIEW OF RELATED LITERATURE AND STUDIES**	**11**
	Related Literature and Studies	11
	Literature Matrix	40
	Synthesis and Relevance of the Study	41
	Conceptual Framework	44
III	**METHODOLOGY**	46
	Research Design	46
	Materials	46
	Algorithm Analysis	49
	Model Development	53
	Model Analysis and Evaluation	56
	Statistical Treatment of Data	59
IV	**RESULTS AND DISCUSSION**	60
	Developed a machine learning based propensity score model	60
	Propensity Score Matching (PSM) Results	64
	An unsupervised model that identifies the livelihood program relevance	66
	Model Evaluation	74
	ANOVA model performance and significance validation	79
V	**SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS**	82
	Summary	82
	Conclusions	84
	Recommendations	85

REFERENCES 86

APPENDICES 94

Appendix A. Letters and Forms
Appendix B. Datasets
Appendix C. Evidences
Appendix D. Source Code
Appendix E. Plagiarism and Similarity Report
Appendix F. Curriculum Vitae
Appendix G. Publishable Paper

LIST OF FIGURES

No.	Title	Page
1	Propensity Score Formula	31
2	Gradient Boosting and Random Forest	35
3	Penalized Regression	35
4	Neural Networks	37
5	Ensemble Learning	38
6	Conceptual Framework of Livelihood Program Impact Assessment	44
7	Principal Component Analysis	50
8	Entropy-weighted K-means	51
9	Autoencoders Structure	51
10	K-means vs Weighted K-means	52
11	K-prototypes	53
12	Hierarchical Clustering Structure	54
13	User Interface Prototype	61
14	Configuration for Analysis	62
15	PSM Evaluation Dashboard	63
16	PSM Statistical Visualizations	64
17	Livelihood Program Analysis Clustering Selection Filter	67
18	Top 10 Program Features and Their Weights	68
19	Cluster Distribution of Age	70
20	Cluster Distribution of Total Income after Program	72

LIST OF TABLES

No.	Title	Page
1	Review Summary of Feasible Methods for PSM	39
2	Literature Matrix	40
3	Data Handling for Various Data Types and Conditions	56
4	Summary of Propensity Score Matching Results	65
5	Summary of Model Performance	75
6	Model Performance Assessment	77
7	ANOVA Results	81

DEFINITION OF TERMS

Autoencoders	A type of neural network that compresses and recreates data to identify the most important features, used here for feature weight identification.
Bureau of Fisheries and Aquatic Resources (BFAR)	A government agency in the Philippines responsible for managing aquatic resources and supporting fisherfolk; source of the dataset used in this study.
Clustering Algorithms	Machine learning techniques used to group similar data together, such as K-means, Weighted K-means, K-prototypes, and Hierarchical Clustering.
Data Cleaning	The process of fixing or removing incorrect, incomplete, or irrelevant data to ensure accuracy in analysis.
Data Preprocessing	Preparing raw data for analysis by cleaning, organizing, and formatting it appropriately.
Data-Driven Decision Making (DDDM)	Using data, statistics, and evidence to guide decision-making instead of relying on opinions or assumptions.
Dimensionality Reduction	A method for simplifying data by reducing the number of input variables while keeping important information.
Entropy	A measure of uncertainty in data; used to determine the importance of features in Entropy-Weighted K-means.

Entropy-Weighted K-means	A clustering algorithm that assigns weights to features based on how useful they are, improving accuracy in grouping data.
Exploratory Data Analysis (EDA)	Analyzing and visualizing data to understand its structure, patterns, and trends before deeper analysis.
Feature Engineering	The process of creating or transforming features (data inputs) to improve the performance of machine learning models.
Feature Weight Identification	Identifying which features have the most influence on outcomes, helping prioritize important data.
Hierarchical Clustering	A clustering method that builds a tree-like structure to show relationships between data points at multiple levels.
Impact Score	A numerical value that measures the effect of a livelihood program on individuals or groups, based on weighted features.
K-means Clustering	A basic clustering algorithm that groups data points into a specific number (k) of similar groups.
K-prototypes	A clustering method that can group both numerical and categorical data, ideal for survey-based datasets.
Livelihood Programs	Government or NGO projects that help individuals or communities improve income and living conditions through training, tools, or support.

Machine Learning (ML)	A branch of artificial intelligence where computers learn patterns from data and make decisions or predictions without being explicitly programmed.
Natural Language Processing	A part of AI that helps computers understand and analyze human language, used here to process written survey responses.
Outlier Handling	Managing data points that are very different from the rest, which can affect analysis results.
Pandas Library	A Python library used to organize and analyze data in tables; useful for cleaning and exploring datasets.
Principal Component Analysis (PCA)	A method that reduces many variables into fewer, more meaningful ones without losing key information.
Python	A programming language used in this study to create and run machine learning models and analyze data.
Random Forest	A machine learning algorithm that uses many decision trees to make accurate predictions and determine important features.
Scikit-Learn (sklearn)	A Python library that provides tools for machine learning, such as data processing, clustering, and model evaluation.
Semantic Analysis	A technique used to understand the meaning and topics in written text.
Sentiment Analysis	A method used to detect whether a written text shows a positive, negative, or neutral emotion.

Silhouette Score	A score that measures how well a data point fits into its assigned cluster; higher scores indicate better clustering.
Super Learner	An advanced machine learning method that combines several algorithms to produce better predictions than any single one alone.
t-SNE (t-distributed Stochastic Neighbor Embedding)	A method used to visualize high-dimensional data by reducing it to two or three dimensions for plotting.
Weighted K-means	A version of K-means clustering that allows different features to have different levels of importance when forming clusters.

CHAPTER I

INTRODUCTION

Background of the Study

Livelihood programs across the globe play a pivotal role in enhancing economic stability and self-sufficiency among vulnerable populations by providing essential skills, resources, and opportunities. These initiatives have led to significant improvements in income and consumption levels, thereby elevating the quality of life for many individuals and communities. Studies show that livelihood programs have increased income by 7% to 65% and consumption growth by 11% to 30%, reflecting their significant impact on improving wellbeing (J-PAL, 2023). In Uganda, the Empowerment and Livelihood for Adolescents (ELA) program combined vocational and life skills training for young women, resulting in a 67% increase in engagement in income-generating activities and an approximate annual earnings boost of $50 per participant. This translates to an estimated cost-effectiveness of $5.88 in income generation per dollar spent over a decade (Dickson et al., 2023). Similarly, in Nepal, a youth training intervention increased non-farm employment by 10 percentage points and monthly earnings, particularly among women engaged in home-based self-employment (Chakravarty et al., 2020).

Governments and organizations worldwide allocate substantial funds to implement and sustain these programs. The Lives and Livelihoods Fund (LLF), for example, has pledged up to $2.5 billion over five years to provide concessional loans aimed at saving lives and improving livelihoods in developing nations (Islamic Development Bank, 2021). In the Philippines, the

Department of Labor and Employment (DOLE) reported that by the second quarter of 2024, over PHP12 billion (approximately $240 million) had been disbursed through the DOLE Integrated Livelihood and Emergency Employment Program (DILEEP), benefiting more than 2 million formal and informal workers (Department of Labor and Employment, 2024). Furthermore, Indonesia has launched a nationwide program to provide free lunches for school children and pregnant mothers, projected to cost $28 billion annually, intending to improve child nutrition while stimulating local economies (Lakshimi & Mariska, 2025).

The effectiveness of livelihood programs varies based on design, implementation, and contextual factors. In the Philippines, the Sustainable Livelihood Program (SLP) aims to reduce poverty by generating employment and promoting sustainable livelihoods. Between 2011 and 2017, the SLP expanded its reach from 46,000 to 166,000 families. However, challenges such as low repayment rates under the micro-enterprise development track and the need for better integration with other social protection programs have been identified. Recommendations for improvement include setting realistic expectations, refining program goals, and strengthening referral linkages to enhance overall effectiveness (World Bank, 2018). In Brazil, the Bolsa Família program, a conditional cash transfer initiative, has been instrumental in reducing poverty and inequality. By providing financial aid to low-income families contingent upon children's school attendance and vaccinations, the program has reached approximately 12 million families. Studies indicate that Bolsa Família contributed to a 27.7% reduction in poverty during President Luiz Inácio Lula da Silva's first term, highlighting its significant impact on improving

socioeconomic conditions. These examples illustrate the transformative potential of well-structured livelihood programs in empowering individuals, boosting incomes, and fostering economic resilience. Nevertheless, continuous evaluation and adaptation are essential to address challenges and ensure the sustainability and cost-effectiveness of these initiatives.

In the Philippines, one of these initiatives is from the Bureau of Fisheries and Aquatic Resources (BFAR), which plays a crucial role in promoting sustainable fisheries and enhancing the livelihoods of fisherfolk communities in the Philippines. This initiative, named F/B Pagbabago Livelihood Program, aims to improve economic opportunities for small-scale fishers by providing financial assistance, resources, and technical training (Bureau of Fisheries and Aquatic Resources [BFAR], n.d.). However, evaluating the effectiveness of such programs remains a significant challenge due to the absence of a systematic, data-driven decision-making framework. Many resource allocation and policy adjustments rely heavily on qualitative assessments, which may limit the program's overall impact and efficiency (BFAR, 2021). A structured evaluation mechanism that incorporates data analytics is necessary to ensure that these livelihood programs achieve their intended objectives and provide long-term benefits to fisherfolk communities (Tecson, 2021).

Recent advancements in machine learning and predictive analytics have demonstrated significant potential in optimizing various fields, including fisheries and livelihood development. These technologies enable the analysis of large datasets to generate actionable insights, forecast program outcomes, and identify key success factors (Kuhn et al., 2024). By leveraging data-driven

approaches, policymakers can refine existing programs, improve resource distribution, and enhance the socio-economic conditions of beneficiaries (Vo et al., 2021). Machine learning models, such as clustering algorithms and decision trees, have been successfully applied in aquaculture and fisheries management, providing valuable recommendations for sustainable resource utilization (Gkikas et al., 2024). Integrating such approaches into BFAR's initiatives can significantly improve decision-making processes and maximize the impact of livelihood interventions. Recent empirical studies in the Philippines have successfully employed Propensity Score Matching (PSM) to assess livelihood and poverty alleviation programs. For example, Capulong & Cuevas (2023) used PSM to estimate the Average Treatment Effect on the Treated for conditional cash transfer households in both rural and urban settings, highlighting significant effects on educational expenditures and food quality. Similarly, Tutor (2020) applied PSM in evaluating the 4Ps program's impact on consumption among poor households, especially the bottom quintile, while Doloriel & Cortez (2022) used PSM to measure welfare improvements under mining social development programs. Also, the PIDS (2022) study employed matching methods in evaluating the Seed Capital Fund component of the Sustainable Livelihood Program, comparing recipients to non-recipients to infer effects on income and expenditure behaviors.

This study proposes a general machine learning-based evaluation framework to assess the impact of Livelihood Programs. The framework incorporates data analytics through techniques such as feature weight identification and clustering. Additionally, it employs key machine learning methodologies, including Exploratory Data Analysis (EDA) and feature

engineering, to enhance the robustness of analytical processes (Vo et al., 2021). By implementing a systematic evaluation mechanism, this study aims to provide an evidence-based foundation for improving livelihood programs, ensuring equitable resource distribution, and enhancing socio-economic conditions among fisherfolk communities. Ultimately, this data-driven approach contributes to sustainable fisheries management and economic resilience for small-scale fishers in the Philippines (Kuhn et al., 2024; Gkikas et al., 2024).

Research Objectives

This study aims to develop a machine learning-based evaluation framework for assessing the impact of livelihood programs to support data-driven decision-making and enhance program outcomes.

1. Develop a machine learning-based propensity score model that estimates baseline socioeconomic conditions of livelihood program beneficiaries by leveraging historical pre-treatment data;
2. Create an unsupervised model that identifies livelihood program relevance using Principal Component Analysis (PCA), Entropy-Weighted K-means, Autoencoders, Weighted K-means, K-prototypes, and Hierarchical Clustering;
3. Evaluate the models using Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index.
4. Statistically compare model performance and validate significance using Analysis of Variance (ANOVA).
5. To deploy and validate the proposed machine learning-based evaluation framework using the BFAR socioeconomic dataset to demonstrate its applicability in real-world program assessment.

Significance of the Study

This study is highly significant as it introduces a data-driven decision-making framework aimed at assessing the impact of livelihood programs. By leveraging machine learning techniques, the research provides a systematic approach to evaluating program effectiveness, ensuring that resources are allocated optimally for maximum impact. The key beneficiaries of this study include various stakeholders who play crucial roles in policy implementation, community development, and academic research.

This study aligns with and directly supports several United Nations Sustainable Development Goals (SDGs), reinforcing its broader significance in promoting global development objectives. Specifically, it contributes to SDG 1 (No Poverty) by enhancing economic opportunities to various communities, SDG 2 (Zero Hunger) by strengthening food security through improved program management, SDG 8 (Decent Work and Economic Growth) by fostering sustainable livelihood opportunities, and SDG 14 (Life Below Water) by encouraging responsible resource utilization and conservation of marine ecosystems. By assessing livelihood programs for communities, the study fosters economic resilience while also promoting sustainable programs and poverty alleviation. Ultimately, the findings of this research can help guide policymakers, development organizations, and researchers in designing more effective, data-driven solutions to address socio-economic challenges in the different sectors.

Government Agencies. This study provides policymakers, particularly the Bureau of Fisheries and Aquatic Resources (BFAR) and other government agencies, with a data-driven framework to enhance decision-making in

livelihood program implementation. By utilizing machine learning techniques, the research offers evidence-based insights that enable more efficient resource distribution, ensuring that funding, training, and technical support reach maximum impact. The ability to analyze program effectiveness through data analytics allows policymakers to make informed adjustments, ultimately improving the overall impact of livelihood initiatives. This framework can also serve as a model for other government programs, fostering a culture of data-driven governance that maximizes public resource utilization and ensures sustainable development outcomes for fisherfolk communities.

Academia and Research Institutions. This study contributes to the growing body of knowledge on the application of machine learning in public sector programs, providing valuable insights for researchers and academic institutions interested in data-driven governance and policy evaluation. By demonstrating how advanced analytics can be used to assess and enhance government livelihood programs, the research serves as a reference for future studies exploring similar methodologies in other sectors. It also encourages interdisciplinary collaboration between data science, social policy, and fisheries management, fostering innovations in program evaluation and optimization. Furthermore, the findings can support the development of new analytical models and frameworks that can be applied to various socio-economic initiatives, further advancing the role of artificial intelligence and machine learning in public administration.

Community. This study is highly beneficial to the community, especially to those directly involved in or affected by livelihood programs, such as

fisherfolk and marginalized groups. By introducing a data-driven approach to evaluating the effectiveness of these programs, the research ensures that interventions are better aligned with the actual needs and conditions of the beneficiaries. As a result, communities stand to gain from more targeted support, improved access to resources, and greater opportunities for sustainable economic growth. The study ultimately aims to uplift the quality of life in local communities by promoting resilience, self-sufficiency, and inclusive development.

The Researchers. For the researchers, this study serves as a valuable platform to apply theoretical knowledge in machine learning, data analytics, and public policy to a real-world setting. It offers an opportunity to engage in meaningful, interdisciplinary work that bridges technology and social impact. Through the process, researchers gain hands-on experience in handling complex data, developing evaluation models, and interpreting findings that can influence decision-making. This engagement not only sharpens their technical and analytical skills but also contributes to their growth as socially responsible scholars and practitioners.

Future Researchers. Future researchers can greatly benefit from this study as it lays a strong foundation for further exploration into the use of artificial intelligence and data-driven approaches in evaluating and improving public sector programs. The methodologies and frameworks developed in this research can serve as references or springboards for similar studies across different sectors, such as education, healthcare, or environmental sustainability. Additionally, it opens avenues for interdisciplinary research,

encouraging future scholars to innovate and adapt these tools to address emerging socio-economic challenges. This study thus contributes to the evolving discourse on how technology can be leveraged to drive evidence-based policymaking and sustainable development.

Scope and Limitation of the Study

This study develops a machine learning-based evaluation framework designed to assess and improve livelihood programs. The framework is capable of adapting to different datasets, variable types, and program objectives. It integrates automated data preprocessing (such as outlier detection and dynamic typing), feature extraction, and model selection to ensure flexibility and scalability. Six main machine learning models were utilized: PCA, Entropy-Weighted K-Means, and Autoencoders for identifying feature relevance, and Weighted K-Means, K-Prototypes, and Hierarchical Clustering for grouping and assessing program impact. Additionally, Natural Language Processing (NLP) techniques such as topic modeling and sentiment analysis were applied to analyze textual feedback from beneficiaries. The dataset used in this study comes from BFAR survey records, which include demographic, socio-economic, and program participation information of fisherfolk beneficiaries.

While the framework is designed for adaptability, several limitations exist. First, the accuracy of the models depends on the quality and completeness of the dataset; missing or inconsistent data may affect results. Second, external factors like climate conditions or market fluctuations were not included, which may influence socio-economic outcomes. Third, the effectiveness of the NLP component relies on the amount and clarity of open-

ended feedback provided by respondents. Fourth, the combined use of multiple algorithms increases computational demand, which may limit real-time application in low-resource environments. Lastly, automatic model selection may not always align perfectly with localized program priorities. Despite these limitations, the framework provides a scalable, data-driven, and replicable system that can guide policymakers in improving program design and resource allocation for various livelihood initiatives.

CHAPTER II

THEORETICAL FRAMEWORK OF THE STUDY

Review of Related Literature, Studies, and Systems

The review of related literature and studies provides a comprehensive analysis of existing research and scholarly works that contribute to the foundation of the study. It explores key concepts, methodologies, and findings that are relevant to the topic, establishing the study's theoretical and empirical underpinnings. This section synthesizes insights from various fields, highlighting significant trends, gaps, and advancements in the chosen area of research. By examining local and international perspectives, the review aims to contextualize the study within the broader academic discourse, ensuring a well-rounded understanding of the subject matter. This synthesis supports the study's objectives and also identifies opportunities for further exploration, fostering a more robust and informed approach to addressing the research problem.

Data-driven Decision Making

Data-driven decision making (DDDM) has emerged as a transformative approach across industries, enabling organizations to leverage data analytics, artificial intelligence (AI), and business intelligence tools to enhance operational efficiency, customer engagement, and strategic planning. This review synthesizes findings from five key studies, examining their methodologies, results, and contributions to the understanding of DDDM’s applications, challenges, and future potential. Recent advancements in AI and data science have redefined the technical frameworks of DDDM. For instance, a study by Yu

et.al. (2021) utilized case studies from IT sectors to demonstrate how machine learning algorithms optimize supply chain management and risk assessment. The researchers employed predictive analytics models, achieving a 25% improvement in operational efficiency across sampled enterprises. Similarly, a publication from World Journal of Advanced Research and Reviews (WJARR) (Michael et al., 2023) highlighted the integration of AI-driven business intelligence platforms, such as automated dashboards, which reduced human error by 40% in data interpretation tasks. These studies underscore the critical role of scalable technologies in translating raw data into actionable insights.

The application of DDDM to improve business efficiency is widely documented. A review by a group of researchers led by Oseremi Onesi-Ozigagun (2024) analyzed 150 enterprises, revealing that organizations adopting DDDM reported a 30% reduction in operational costs and a 20% increase in process automation. Complementing this, Sarioguz and Miser (2023) conducted longitudinal research across retail and manufacturing sectors, identifying that real-time data analytics shortened decision-making cycles by 50%. Both studies emphasized the importance of cross-departmental data integration to eliminate silos and foster cohesive strategies. DDDM's impact on customer engagement is particularly notable. The study of Ozigagun (2024) found that businesses employing sentiment analysis and customer behavior modeling improved customer satisfaction scores by 35%. Furthermore, Sukula et al. (2023) explored e-commerce platforms using recommendation engines powered by DDDM, which boosted conversion rates by 22%. These results align with WJARR (Michael et al., 2023), which noted

that personalized marketing campaigns, informed by demographic and transactional data, increased customer retention by 18%.

Despite its benefits, DDDM faces implementation barriers. Sukula et al. (2023) identified data privacy concerns and algorithmic bias as critical ethical challenges, with 60% of surveyed organizations reporting hesitancy in adopting AI tools due to regulatory risks. Additionally, Sarioguz and Miser (2023) highlighted infrastructural gaps, noting that 45% of small-to-medium enterprises lacked the resources to implement advanced analytics systems. These findings stress the need for robust governance frameworks and equitable access to DDDM technologies. Emerging trends suggest a shift toward democratizing DDDM tools. The study of Yu et al. (2021) advocated for low-code AI platforms to empower non-technical stakeholders, while WJARR (Michael et al., 2023) projected advancements in quantum computing to address complex data modeling challenges. Collectively, these studies emphasize interdisciplinary collaboration and continuous upskilling as prerequisites for sustaining DDDM's evolution.

Fisheries, Aquaculture, and Livelihood Interventions

This review synthesizes research and reports on fisheries management, technological innovations in aquaculture, and livelihood interventions, with a focus on applications in the Philippines and global contexts. It examines methodologies, outcomes, and challenges identified in academic studies, government reports, and policy analyses, integrating insights from machine learning (ML) applications, vocational training programs, and poverty alleviation initiatives.

The Bureau of Fisheries and Aquatic Resources (BFAR) in the Philippines has implemented programs to modernize fisheries. For instance, BFAR's *F/B Pagbabago* initiative (BFAR, n.d.) focuses on distributing fiberglass boats to replace traditional wooden vessels, improving safety and efficiency for small-scale fishers. A 2021 Annual Narrative Report (BFAR, 2021) highlighted that such programs increased fish yields by 15% in coastal communities. Similarly, BFAR's distribution of fiberglass boats in Bulacan (Tecson, 2021) reduced post-harvest losses by 20%, emphasizing the role of infrastructure upgrades in sustainable fisheries. Complementing these efforts, machine learning (ML) is reshaping aquaculture. Vo and his team (2021) reviewed AI-driven systems for water quality monitoring and feed optimization, showing a 30% reduction in resource waste. Kuhn et al. (2024) expanded this scope, demonstrating ML's utility in genomic selection and ecosystem modeling, which improved species survival rates by 25% in case studies. A group of researchers led by Gkikas (2024) further validated ML's predictive power in disease outbreaks, achieving 90% accuracy in early detection systems.

Livelihood programs aim to address poverty and unemployment. The Philippines' Sustainable Livelihood Program (World Bank, 2018) provided microloans and skills training to 2 million households, lifting 12% of participants above the poverty line. Similarly, Indonesia's $28 billion free school meals initiative (Lakshimi & Mariska, 2025) aims to improve child nutrition and parental workforce participation, reflecting integrated social protection strategies. Vocational training's impact is mixed. A study in Nepal (Chakravarty et al., 2020) found that vocational programs increased youth employment by 18%, but

effects diminished without complementary financial support. J-PAL (2023) emphasized the need for multi-faceted approaches, such as pairing skills training with access to credit, to ensure long-term livelihood stability. The Lives and Livelihoods Fund (Islamic Development Bank, 2021), active in 33 countries, reported a 22% income boost for beneficiaries through agriculture and healthcare investments.

Despite progress, barriers persist. In fisheries, BFAR's programs face challenges in scaling due to budget constraints and logistical hurdles (BFAR, 2021). ML applications in aquaculture require high-quality data, which is scarce in low-resource regions (Kuhn et al., 2024). For livelihood programs, a group of researchers headed by Ruby Dickson (2023) noted that short-term grants often fail to address systemic inequities, with only 30% of interventions showing sustained impact. Ethical concerns, such as dependency on aid, were also highlighted in Brazil's Bolsa Família program, where some households reverted to poverty after subsidy withdrawal. Emerging trends include integrating ML with IoT for real-time aquaculture management (Vo et al., 2021) and expanding digital financial services for livelihood programs. The Philippines' DOLE (2024) recently allocated ₱12 billion ($210 million) to upskill workers in tech-driven sectors, signaling a shift toward future-ready training. Global initiatives like the Lives and Livelihoods Fund advocate for climate-resilient agriculture, aligning with SDG targets. Technological innovations in fisheries and evidence-based livelihood programs demonstrate significant potential to enhance productivity and reduce poverty. However, sustainability hinges on addressing data gaps, securing funding, and adopting holistic frameworks that combine infrastructure, education, and policy reforms. Future research should prioritize longitudinal

studies to evaluate long-term impacts and explore synergies between AI tools and community-led development.

The Role of Prescriptive Analytics in Data-Driven Decision-Making

Prescriptive analytics has emerged as a crucial tool in decision-making by recommending optimal actions based on data-driven insights. It extends beyond descriptive and predictive analytics by integrating optimization techniques to guide actions in uncertain environments (Selvaraj & Marudappa, 2020). The systematic literature review by Wissuchek and Zschech (2024) highlights six research streams in prescriptive analytics systems, emphasizing their socio-technical development. In public-sector decision-making, prescriptive analytics enhances service utilization and operational efficiency by linking data-driven insights to value and legitimacy (Brandt et al., 2020).

Furthermore, the integration of machine learning in prescriptive analytics allows for automated decision-making, shifting from predictive modeling to recommending real-world actions (Tang et al., 2021). This shift is also explored in the work of Bertsimas and Kallus (2018), who demonstrate how combining machine learning with operations research improves decision-making frameworks. Collectively, these studies underscore prescriptive analytics' role in optimizing strategic choices across industries, from business and government to artificial intelligence applications. As data continues to drive decision-making, prescriptive analytics will become increasingly relevant in shaping optimal outcomes in complex systems.

Combining Multiple Algorithms/Models for Data Analytics

In data analytics, the concept of combining multiple algorithms and models is fundamentally important to its effectiveness (IBM, 2024). Prescriptive

analytics leverages a diverse array of approaches, integrating statistical models, machine learning algorithms, and established business rules to explore potential solutions and recommend optimal actions (ThoughtSpot, 2024; HBS Online, 2021). For instance, machine learning algorithms excel at efficiently processing large volumes of data, identifying intricate patterns, and predicting outcomes, while optimization algorithms analyze a wide spectrum of options and constraints to determine the most effective strategy (Stefanini, 2023).

By simultaneously considering factors such as real-time market data, customer preferences, and operational limitations, these combined approaches offer more nuanced and context-aware recommendations (Elkhidir & Hamza, 2024). Techniques such as decision trees, regression models, and simulation models can be used in conjunction to deliver results based on specific business needs (Infomineo, 2024). The result is a synergistic effect, where the strengths of individual algorithms are amplified, leading to more robust and reliable prescriptive insights (Stitch Data, n.d.). This integrated approach reflects a move toward enhancing the depth and accuracy of recommendations, driving improvements in organizational decision-making and strategic outcomes.

Evaluating Program Continuation: Government and Corporate Approaches

Organizations, including government agencies and private companies, use systematic program evaluation to assess the effectiveness, efficiency, and strategic alignment of their initiatives. The U.S. Department of Energy (n.d.) highlights that such evaluations provide essential insights for decision-making, resource allocation, and program modifications. Similarly, the U.S. Department

of Commerce (2021) defines evaluation as a structured process of data collection and analysis to determine program impact and efficiency. Independent program assessments are crucial for governments to determine whether initiatives achieve intended outcomes cost-effectively, as emphasized by the Government Accountability Office (GAO, 2011). These evaluations influence major decisions, including resource reallocation, restructuring, or legislative action (Sage Publications, 2019). In the private sector, companies leverage evaluations to justify program continuation or expansion, particularly when budgets are constrained (Summit LLC, 2018). Monitoring and evaluation (M&E) frameworks ensure programs meet predefined objectives by tracking key performance indicators and identifying areas for improvement.

Program evaluations are data-driven processes that support organizations in making informed decisions regarding a program's future. Eagle Hill Consulting (2023) illustrates how regulatory organizations conduct comprehensive evaluations to determine the feasibility of ongoing operations. Evaluations integrate qualitative and quantitative methods, ensuring a holistic understanding of program impact (Summit LLC, 2018). By analyzing data trends, stakeholders can identify inefficiencies, validate successes, and make strategic adjustments (GAO, 2011). Evidence-based assessments help both public and private entities optimize resource use while maintaining accountability and transparency (Sage Publications, 2019). These processes ultimately strengthen decision-making frameworks, ensuring that only high-impact programs continue receiving support (U.S. Department of Commerce, 2021). In conclusion, systematic evaluation serves as a critical mechanism for determining if programs should be maintained, modified, or discontinued.

Decision-Making Challenges and Error Rates in the Philippines

In the Philippines, decision-making processes are influenced by various cultural, organizational, and systemic factors that can lead to challenges and increased error rates. Cultural norms, such as strong family cohesiveness, play a significant role in shaping individual autonomy and decision-making behaviors. This cultural context can impact how decisions are made within households and organizations, potentially leading to conflicts or delays when individual preferences differ from collective expectations (PubMed, 2015). In organizational settings, particularly among middle managers, limited decision-making autonomy and bureaucratic structures can hinder effective leadership and timely decision-making. The lack of authority to make independent decisions often results in delays and inefficiencies, as approvals must pass through multiple hierarchical levels (Imutan, 2023).

Systemic issues also contribute to decision-making struggles and error rates in the country. For instance, the Philippine government has recognized the need for evidence-informed decision-making to enhance the effectiveness of major programs. Investments in impact evaluations aim to provide data-driven insights; however, challenges persist in integrating these evaluations into policy-making processes (3ie, 2019). In the healthcare sector, laboratory testing processes face errors, with studies indicating that misidentification errors, though comprising only 2.6% of all errors, occur at a notable frequency (Bonini et al., 2022). Additionally, data entry outsourcing companies in the Philippines encounter challenges related to error rates, emphasizing the importance of clear performance metrics and regular monitoring to identify trends and

implement corrective actions (Perfect Data Entry, 2023). These examples underscore the multifaceted nature of decision-making challenges and error rates in the Philippines, highlighting the need for cultural sensitivity, organizational reforms, and systemic improvements to enhance decision-making efficacy.

K-means Clustering for Descriptive Analytics

The performance of K-means heavily depends on the initial placement of cluster centers. The K-means++ algorithm improves upon random initialization by selecting initial centers based on a probability proportional to the squared distance from existing centers, enhancing both the speed and accuracy of convergence. Further advancements include hybrid approaches that combine K-means with metaheuristic algorithms. For instance, Alam (2023) proposed integrating K-means with an improved firefly algorithm to automatically determine the number of clusters in high-dimensional data, resulting in more accurate clustering outcomes.

Identifying the appropriate number of clusters (k) is critical for effective clustering. The K-splits algorithm offers a hierarchical approach that incrementally splits clusters based on data distribution, automatically detecting the optimal number of clusters. Experiments on synthetic and real-world datasets demonstrated that K-splits achieves high accuracy in identifying the correct number of clusters under various conditions. Scalability is a significant concern for K-means when applied to large datasets. Lang and Schubert (2024) introduced an accelerated K-means algorithm utilizing cover trees, which efficiently manage nearest neighbor searches, thereby reducing computational

complexity and enhancing clustering performance on large-scale data. Additionally, hybrid models combining K-means with optimization algorithms, such as the improved firefly algorithm, have been developed to enhance clustering performance in high-dimensional spaces.

Several studies have focused on refining the convergence behavior and accuracy of K-means. Li (2025) proposed the Extended Center-based Clustering Algorithm (ECAC), which identifies multiple extended centers for each cluster to better represent complex data distributions, thereby improving clustering accuracy. Furthermore, incorporating data preprocessing techniques, such as Principal Component Analysis (PCA), has been shown to enhance the efficiency of K-means by reducing dimensionality and mitigating the impact of irrelevant features.

Entropy-Weighted K-Means Clustering

Entropy-weighted K-means (EWKM) clustering has emerged as a significant advancement in unsupervised learning, particularly for high-dimensional data. By integrating entropy-based feature weighting into the traditional K-means framework, EWKM enhances clustering performance by emphasizing relevant features and mitigating the influence of irrelevant ones.

This review synthesizes recent developments in EWKM, focusing on methodologies, algorithmic enhancements, and empirical results. Recent studies have introduced entropy-based regularization into the K-means algorithm to address its sensitivity to irrelevant features and local minima. Chakraborty et al. (2020) proposed the Entropy Weighted Power K-Means (EWPKM) algorithm, which incorporates entropy regularization to learn feature

relevance during the clustering process. This approach maintains the computational simplicity of K-means while providing theoretical guarantees of convergence and consistency. The algorithm demonstrated superior performance over traditional K-means and Power K-means on both synthetic and real-world datasets.

Similarly, Singh and Verma (2020) developed a Variable Feature Weighted Fuzzy K-Means algorithm that modifies the objective function of fuzzy K-means by introducing two entropy terms. The first term minimizes within-cluster dispersion, while the second controls feature weights, allowing the algorithm to assign higher weights to relevant features and lower weights to irrelevant ones. This method showed improved clustering performance across various datasets. Entropy-based feature weighting has been instrumental in improving clustering accuracy in high-dimensional spaces. Jing et al. (2020) extended the K-means algorithm by calculating weights for each dimension within each cluster using weight entropy. This approach identifies important dimensions automatically, enhancing the algorithm's ability to discern meaningful clusters.

Further advancements include the integration of entropy weighting with fuzzy clustering techniques. A novel Fuzzy-C-Means-based entropy weighting algorithm was proposed, incorporating local feature weighting and non-Euclidean distance measures to improve robustness against noise and outliers. This method enhances clustering sensitivity and accuracy, particularly in datasets with random class centers. Entropy-weighted K-means has also been adapted for clustering mixed numerical and categorical datasets. An iterative

clustering algorithm was developed that employs entropy-based weighting to calculate the importance of each attribute, facilitating effective clustering in datasets with diverse data types.

In terms of initialization strategies, entropy-based methods have been utilized to determine optimal cluster centroids. Hussayn and Shah (2023) analyzed the performance of K-means on image data by employing various parametric entropy measures for centroid initialization. Their findings indicate that different entropy measures can lead to improved clustering outcomes across various image datasets. Comparative studies have highlighted the effectiveness of entropy-based weighting schemes in clustering tasks. Buatoom et al. (2020) compared deviation-based and entropy-based term weighting schemes in the context of document clustering using seeded K-means.

Their results demonstrated that while deviation-based weighting outperformed entropy-based weighting individually, a hybrid approach combining both methods yielded the highest clustering accuracy. This method addresses the limitations of traditional K-means related to initialization sensitivity and fixed cluster numbers, demonstrating improved performance in various clustering scenarios.

Principal Component Analysis

Deng et al. (2024) propose a robust PCA method that mitigates the influence of outliers by learning discriminant sample weights. Their iterative algorithm assigns weights to samples, reducing the impact of outliers on the estimation of the data mean and projection matrix. Experiments on various

datasets demonstrate the method's effectiveness in handling noisy data. On the other hand, Xiong (2024) introduces a unified framework connecting PCA and factor analysis under a general latent variable model. By formulating both methods as solutions to matrix optimization problems with specific loss functions, the study provides insights into their theoretical underpinnings and potential for developing new data analysis tools.

Martel et al. (2024) implement PCA on high-performance computing platforms, including GPUs and manycore processors, to address the computational demands of hyperspectral image processing. Their work offers practical guidance for optimizing PCA performance in real-time applications. While Park et al. (2023) critically assess sparse PCA methods, highlighting the distinction between weights and loadings. They argue that conflating these concepts can lead to misinterpretations and over-optimistic conclusions in research. The study emphasizes the importance of careful methodological choices in sparse PCA applications.

Beattie and Esmonde-White (2021) offer a visual approach to understanding PCA by deriving it using spectral methods. Their tutorial aids in building intuition about PCA's mathematical foundations, making it more accessible to researchers. In the context of Internet of Things (IoT) security, PCA has been applied for feature extraction in intrusion detection systems. By reducing data dimensionality, PCA enhances the efficiency and accuracy of detecting anomalies in IoT networks. PCA is utilized in remote sensing to process and analyze large volumes of satellite imagery. By transforming correlated spectral bands into uncorrelated principal components, PCA

facilitates efficient data compression and feature extraction for land cover classification and change detection. In medical imaging, PCA assists in enhancing image quality and reducing noise. By identifying the principal components that capture the most variance, PCA helps in reconstructing images with improved clarity, aiding in better diagnosis and analysis.

A study by Beattie and a group of researchers (2021) stated that PCA is applied in environmental monitoring to analyze complex datasets, such as air and water quality measurements. By reducing data dimensionality, PCA enables the identification of key factors influencing environmental conditions, supporting effective decision-making.

Bolante et al. (2021) employed PCA to assess food security options among small-scale farmers in the Sikasso Region of Mali. By analyzing variables such as family size, proximity to roads, and livestock practices, the study identified critical factors influencing food security. The findings underscore the utility of PCA in distilling complex, multidimensional data into actionable insights for policy development and program implementation.

Muangthong and Yamoat (2018) applied PCA to evaluate socio-economic factors affecting the quality of the Lower Songkhram Wetland in Thailand. Analyzing data from 418 households across 40 sub-districts, the study identified seven principal components, including community economy, public participation, and water conservation efforts. These insights are instrumental in guiding community-based environmental management and livelihood enhancement strategies.

Carillo et al. (2019) utilized PCA to analyze the 2009 Philippine Health Data, aiming to identify underlying components influencing health outcomes. The study revealed three significant components: the importance of safe water supply and child health, emphasis on barangay health stations and government health workers, and focus on women's health. These components collectively accounted for 73.01% of the total variance, highlighting the relevance of PCA in understanding health determinants that impact livelihoods.

A study conducted in Nam Dinh Province, Vietnam, employed PCA to assess household vulnerability to climate change-induced droughts. The analysis identified key livelihood assets affected, including financial capital, natural resources, and human capital. The PCA-derived indices indicated low adaptive capacity and high sensitivity and exposure among households, emphasizing the need for targeted interventions to enhance resilience.

The Philippine Council for Agriculture, Aquatic and Natural Resources Research and Development (PCAARRD) collaborated with the Australian Centre for International Agricultural Research (ACIAR) to develop a mixed-method approach for impact assessment. While not exclusively focused on PCA, the initiative recognizes the importance of integrating quantitative methods like PCA with qualitative analyses to capture the multifaceted impacts of livelihood programs.

Autoencoders

Autoencoders have emerged as powerful tools for feature weight identification, particularly in high-dimensional data scenarios. Their ability to learn compressed representations of data makes them suitable for tasks

requiring dimensionality reduction and feature selection. This literature review synthesizes findings from several studies that have explored various autoencoder architectures and methodologies for feature weight identification.

Wang et al. (2017) introduced the Feature Selection Guided Auto-Encoder (FSGAE), a model that integrates feature selection directly into the autoencoder architecture. By encouraging the model to focus on task-relevant features during the learning process, FSGAE effectively distinguishes between informative and non-informative features, enhancing the quality of the learned representations. Experiments demonstrated that FSGAE outperformed traditional autoencoders in classification tasks, highlighting the benefits of incorporating feature selection mechanisms into the autoencoder framework. Wu and Cheng (2021) proposed Fractal Autoencoders (FAE) for unsupervised feature selection. FAE extends traditional autoencoders by adding a one-to-one scoring layer and a small sub-neural network, enabling the model to assign importance scores to individual features. This architecture allows for the identification of informative features without supervision, making it particularly useful in scenarios where labeled data is scarce. The authors reported that FAE achieved state-of-the-art performance on various datasets, including those with high dimensionality, such as gene expression data.

Cunningham et al. (2023) explored the use of sparse autoencoders to uncover interpretable features in language models. Their approach addresses the issue of polysemanticity, where neurons activate in multiple, semantically distinct contexts. By employing sparse autoencoders, the study aimed to identify monosemantic features, enhancing the interpretability of language models. The results indicated that sparse autoencoders could effectively

disentangle overlapping features, providing more transparent insights into the internal workings of complex models.

Atashgahi et al. (2022) introduced QuickSelection, an energy-efficient method for unsupervised feature selection using sparsely connected denoising autoencoders. By leveraging the strength of neurons in sparse neural networks as a criterion for feature importance, QuickSelection simultaneously derives the significance of all input features. The method demonstrated superior performance in terms of classification accuracy, running time, and memory usage across multiple benchmark datasets, highlighting its practicality for real-world applications.

Ferreira et al. (2020) investigated the application of autoencoders as a weight initialization method for deep neural networks in disease detection tasks. By pre-training autoencoders on RNA-Seq datasets and fine-tuning the weights in subsequent classification models, the study achieved high accuracy in distinguishing between different cancer types. The findings suggest that autoencoder-based weight initialization can enhance the performance of deep learning models in biomedical applications.

In the context of social network analysis, a study published in the International Journal of System Assurance Engineering and Management (2023) developed a hybrid feature extraction model employing a Weighted Deep Autoencoder (WDAE) architecture. This approach aimed to detect communities within social networks by efficiently representing the network's topology through low-dimensional feature matrices. The WDAE-based model demonstrated effectiveness in capturing the unique characteristics and relationships inherent in social network communities.

Hierarchical Clustering

Hierarchical clustering (HC) is a fundamental unsupervised learning technique widely utilized for uncovering nested groupings within data, offering a tree-like structure known as a dendrogram that illustrates the relationships among clusters. Recent advancements have sought to address the limitations of traditional HC methods, such as computational inefficiency, sensitivity to noise, and lack of model-based frameworks.

Vichi et al. (2022) introduced Hierarchical Means Clustering (HMC), a novel approach that integrates a centroid-based model estimated via least-squares to minimize within-cluster deviance across nested partitions. This method bridges the gap between model-based and hierarchical clustering techniques, providing a statistically grounded framework that enhances interpretability and clustering accuracy. The authors demonstrated the efficacy of HMC through applications to six real-world datasets, highlighting its superiority over traditional methods in terms of cluster homogeneity and structure.

Addressing the challenge of noise sensitivity in HC, a study published in *Expert Systems with Applications* proposed an improved single linkage criterion that incorporates a noise-handling mechanism. By modifying the distance metric to account for noise and outliers, the enhanced algorithm effectively reduces the chaining effect commonly observed in single linkage clustering, leading to more robust and meaningful cluster formations.

In the realm of large-scale data analysis, the computational complexity of HC poses significant challenges. To mitigate this, a hybrid approach

combining K-means and hierarchical clustering was developed, wherein K-means is first employed to partition the dataset into spherical clusters, which are then merged using hierarchical methods based on a data-driven distance measure. This strategy leverages the efficiency of K-means and the structural insights of HC, enabling the identification of general-shaped clusters in large datasets (Peterson et al., 2017).

Furthermore, the integration of domain-specific knowledge into HC has been explored to enhance clustering relevance and applicability. For instance, combining the Analytical Hierarchy Process (AHP) with agglomerative HC facilitates the incorporation of expert judgments and criteria weighting into the clustering process, as demonstrated in the context of green corridor development management (Tu et al., 2013). This fusion allows for more informed and context-sensitive clustering outcomes.

Propensity Scoring

Propensity score (PS) analysis, fundamentally defined as the conditional probability of an individual or unit being assigned to a specific treatment group given a set of observed pre-treatment covariates (X), represents a cornerstone methodology for reducing selection bias in non-randomized, observational studies. where T indicates treatment assignment (1 = treated, 0 = control), the propensity score was introduced in the seminal 1983 paper by Paul Rosenbaum and Donald Rubin. Their work established the theoretical foundation, demonstrating that conditioning on the scalar propensity score could achieve balance in the distribution of observed covariates between treated and control groups, thereby approximating the conditions of a

randomized experiment under the assumption of strong ignorability (i.e., no unmeasured confounding).

Propensity Score Formula

$$e(\mathbf{X}) = \Pr(T = 1 \mid \mathbf{X})$$

Where:

- $e(\mathbf{X})$ = propensity score
- T = treatment indicator variable:
  - $T = 1$ if the unit received the treatment,
  - $T = 0$ if the unit did not receive the treatment.
- $\mathbf{X}$ = vector of observed covariates for each unit.
- $\Pr(T = 1 \mid \mathbf{X})$ = probability of receiving the treatment **given** the covariates.

Figure 1. Propensity Score Formula

This innovation provided a powerful tool for causal inference where randomization is impractical or unethical. The subsequent decades saw significant methodological development: throughout the 1990s and 2000s, researchers refined techniques for utilizing the estimated PS, including stratification (grouping subjects into subclasses like quintiles based on PS), matching (pairing treated and control units with similar PS values), and inverse probability of treatment weighting (IPTW). The 2010s witnessed a pivotal shift with the exploration and integration of machine learning (ML) algorithms to address limitations inherent in conventional parametric modeling approaches for estimating the PS itself (Rosenbaum & Rubin, 1983; Austin, 2011; Westreich et al., 2010).

Propensity score methodology has found extensive application across diverse fields where researchers seek to draw causal conclusions from observational data. In healthcare and epidemiology, it is indispensable for evaluating the effectiveness and safety of treatments, drugs, or medical

procedures using real-world data sources like electronic health records (EHRs) or insurance claims databases, where random assignment is often absent (Schneeweiss et al., 2009). Social scientists routinely employ PS methods to assess the impact of social programs, educational interventions, or job training initiatives by comparing participants to non-participants while controlling for baseline differences in characteristics (Caliendo & Kopeinig, 2008). Economists leverage PS analysis to estimate the causal effects of economic policies, such as changes in unemployment benefits, minimum wage laws, or development programs, on outcomes like employment status or income (Heckman et al., 1998). Beyond these core areas, PS techniques are increasingly utilized in marketing analytics to measure the true impact of advertising campaigns or loyalty programs by balancing customer profiles between exposed and unexposed groups (Lemmens & Gupta, 2020), and in environmental science, public policy evaluation, and any discipline relying on quasi-experimental designs.

The conventional approach to estimating propensity scores predominantly relies on parametric statistical models, with logistic regression being the overwhelmingly dominant method. Logistic regression models the log-odds of treatment assignment as a linear function of the observed covariates. Once estimated, these propensity scores are then utilized through several established techniques. Stratification (or subclassification) involves dividing the study population into strata (commonly quintiles) based on the estimated PS and then comparing outcomes between treated and control units within each stratum; the overall treatment effect is typically calculated as a weighted average across strata. Matching techniques pair each treated unit

with one or more control units that have very similar propensity scores (e.g., nearest-neighbor matching, caliper matching), creating a matched sample where covariate distributions are balanced.

Inverse Probability of Treatment Weighting (IPTW) assigns weights to each individual based on their PS: treated units receive a weight of *1/e(X)* and control units receive a weight of *1/(1-e(X))*. These weights create a pseudo-population where the distribution of covariates is independent of treatment assignment, allowing for estimation of the average treatment effect in the population (ATE) or the average treatment effect on the treated (ATT) using weighted analyses. While these methods are well-established and widely implemented in statistical software, they critically depend on the correct specification of the logistic regression model. Misspecification (e.g., omitting relevant confounders, failing to include necessary interaction or non-linear terms) can lead to residual confounding and biased effect estimates. Furthermore, conventional logistic regression struggles with high-dimensional covariate data (Austin, 2011; Stuart, 2010; Hernán & Robins, 2020).

Propensity scores can be effectively computed using machine learning (ML) algorithms, and this approach has gained substantial traction as a way to overcome limitations inherent in conventional logistic regression. ML methods offer significant advantages, particularly their ability to automatically detect complex, non-linear relationships and interactions among covariates without requiring the researcher to explicitly specify these functional forms a priori. They are also generally better equipped to handle high-dimensional datasets containing a large number of potential confounders. However, employing ML for PS estimation introduces new considerations: complex algorithms like deep

neural networks or highly flexible tree ensembles carry a risk of overfitting the treatment assignment model, which can paradoxically lead to worse covariate balance and biased effect estimates compared to simpler, correctly specified models. Consequently, careful implementation and validation are paramount (Westreich et al., 2010; Pirracchio et al., 2015).

Methods like Gradient Boosting Machines (GBM - e.g., via xgboost or lightgbm) and Random Forests are popular choices. GBM iteratively combines many weak learners (typically decision trees) to minimize prediction error, often demonstrating strong performance. Random Forests build multiple de-correlated trees and average their predictions, offering robustness and handling non-linearity well (McCaffrey et al., 2004; Lee et al., 2010).

While Gradient Boosting Machines (GBM) and Random Forests are powerful in handling non-linear relationships and high-dimensional data, their black-box nature complicates interpretability in causal inference contexts (Westreich et al., 2010). Moreover, their tendency to overfit, especially in smaller datasets, can compromise covariate balance, which is essential for valid propensity score estimation (Lee et al., 2010). Unlike parametric models, these ensemble methods require meticulous hyperparameter tuning and often lack transparency in variable importance relevant to causal interpretation.

Algorithms like Lasso (Least Absolute Shrinkage and Selection Operator) and Ridge Regression extend logistic regression by adding penalties to the coefficient sizes. Lasso performs variable selection by shrinking some coefficients to zero, which is advantageous in high-dimensional settings. Ridge regression shrinks coefficients towards zero but rarely sets them exactly to zero, helping to stabilize estimates (Tibshirani, 1996).

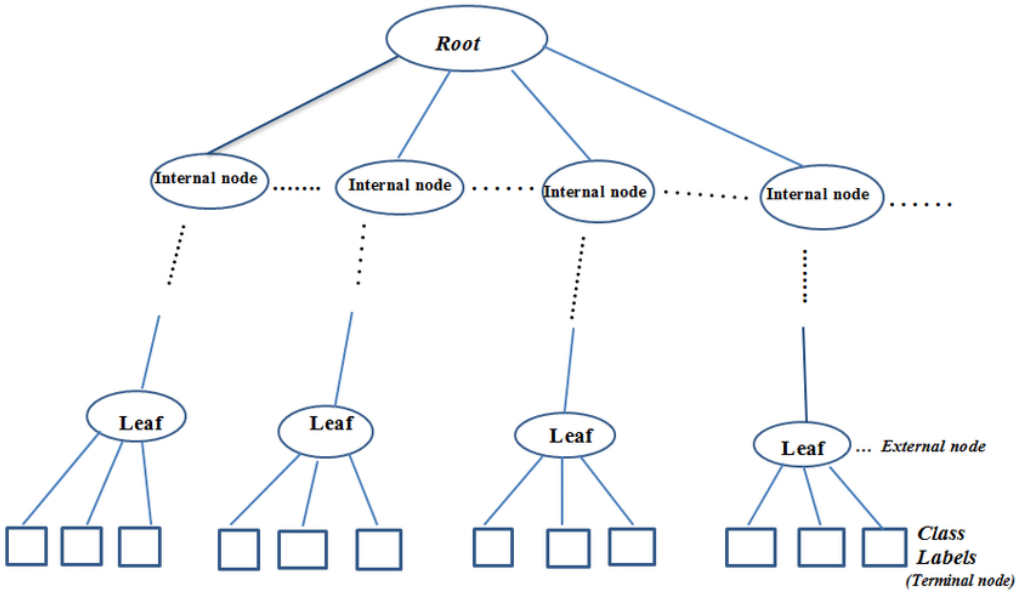


Figure 2. Gradient Boosting and Random Forest

Although penalized regression like Lasso aids in variable selection in high-dimensional settings, its main limitation in propensity score estimation is that it still assumes linear relationships unless extended with interaction terms. When important non-linear interactions or higher-order terms are omitted, residual confounding may persist (Tibshirani, 1996). Moreover, its predictive accuracy may suffer compared to tree-based ensembles in highly complex datasets.

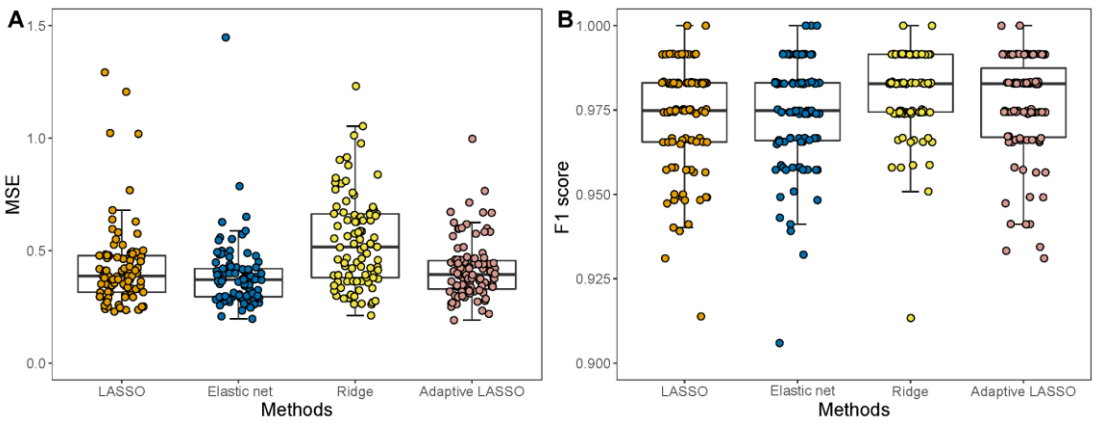


Figure 3. Penalized Regression

Neural Networks. These can model extremely complex relationships but typically require very large sample sizes and careful tuning to avoid overfitting, making them less commonly used than tree-based methods or penalized regression for PS in many practical settings (Setoguchi et al., 2008).

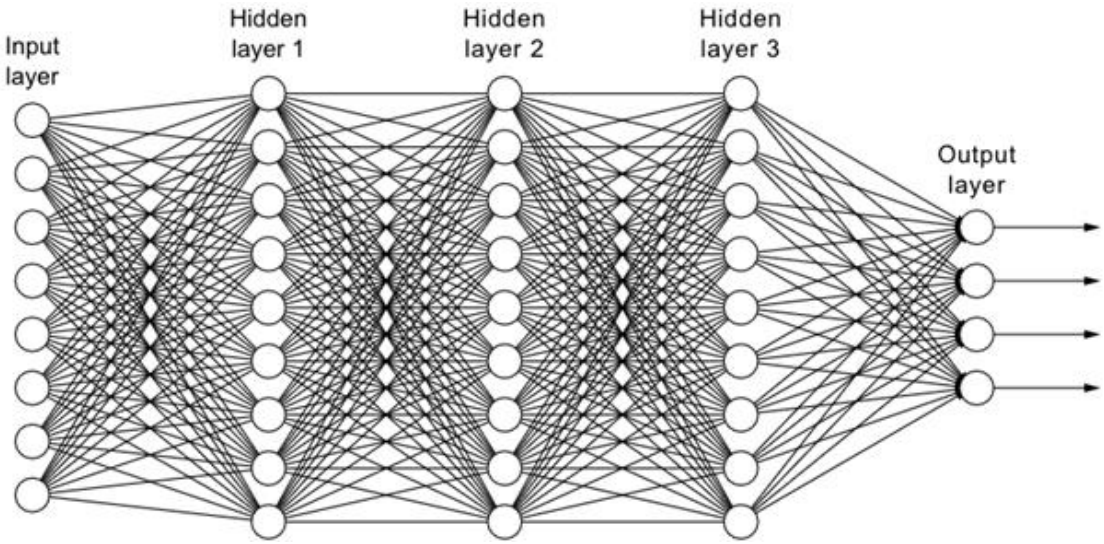


Figure 4. Neural Networks

Neural networks excel in modelling intricate relationships but are prone to overfitting when applied to small or moderately sized datasets typical in observational research. Additionally, they require substantial expertise in architecture design and hyperparameter tuning (Setoguchi et al., 2008). Their black-box nature also makes diagnostic assessment and variable interpretability challenging, which conflicts with the transparency needed for propensity score validation.

Super Learner (Ensemble Learning). This sophisticated approach avoids relying on a single algorithm. Instead, it uses cross-validation to combine predictions from a diverse library of candidate algorithms into a single, optimized propensity score prediction, often outperforming any single learner (van der Laan et al., 2007; Pirracchio et al., 2015; Wyss et al., 2014).

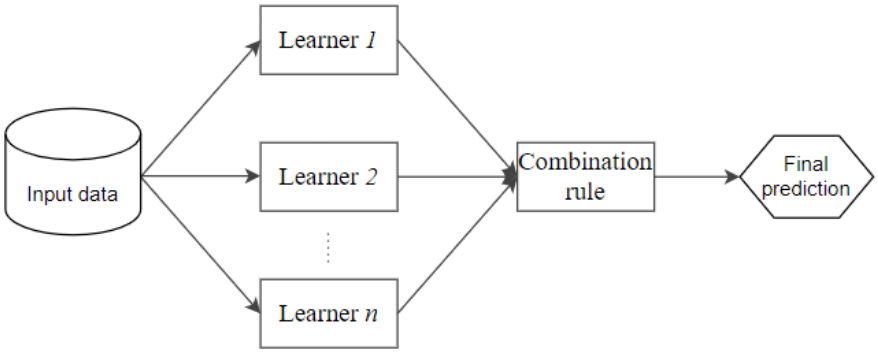


Figure 5. Ensemble Learning

While Super Learner frequently outperforms individual algorithms, it has its own drawbacks, notably computational intensity and potential overfitting if not properly validated. Furthermore, Super Learner does not inherently solve the challenge of covariate balance; achieving good balance still relies on careful implementation of matching, weighting, or stratification after PS estimation (Pirracchio et al., 2015). Without rigorous balance diagnostics, even high predictive performance may not translate to effective confounding control.

Rigorous Model Tuning and Validation: Hyperparameters (e.g., learning rate, tree depth, regularization strength) must be carefully tuned using techniques like k-fold cross-validation to optimize predictive performance for the purpose of covariate balance and prevent overfitting. Simply maximizing prediction accuracy of treatment assignment is not sufficient and can be detrimental (Wyss et al., 2014).

Regardless of the ML algorithm used, absolute priority must be given to assessing the resulting covariate balance between treated and control groups after applying the PS (via matching, weighting, or stratification). Common diagnostics include calculating standardized mean differences (SMD) for each covariate (target SMD < 0.1), variance ratios, and visual assessments like love

plots. Poor balance indicates the PS model, however sophisticated, has failed its primary purpose, and the model or method needs adjustment (Austin & Stuart, 2015).

Table 1. Review Summary of Feasible Methods for PSM

Machine Learning Method	Why it Works	Gaps	Requirement to Function
Tree-Based Ensembles	Captures non-linearities and interactions automatically; handles high-dimensional data well	Risk of overfitting; limited interpretability; requires intensive tuning	Requires careful hyperparameter tuning; balance diagnostics essential
Penalized Regression	Handles high-dimensional covariates; performs variable selection	Assumes linearity unless interactions are manually specified; weaker predictive accuracy	Proper model specification; may require feature engineering
Neural Networks	Highly flexible, can model complex interactions	Requires large datasets; risk of overfitting; low interpretability	Large sample size; expert tuning; robust diagnostics
Super Learner	Aggregates multiple models for superior predictive performance	Computationally intensive; high predictive performance ≠ covariate balance	Requires cross-validation; post-PS balance assessments

Pairing ML-based propensity scores (used in weighting or matching) with a separate outcome regression model (which can also utilize ML) provides double robustness. This means the final causal estimate is consistent (asymptotically unbiased) if either the PS model or the outcome model is correctly specified, offering a valuable layer of protection against model misspecification (Bang & Robins, 2005).

Table 2. Literature Matrix

Author(s)	Title / Source	Year	Algorithm / Technique	Findings
Onesi-Ozigagun et al.	Analysis of DDDM in Enterprises	2024	Data-Driven Decision Making	Reduced operational cost by 30%, increased automation by 20%
Kuhn et al.	ML in Genomic Aquaculture	2024	Genomic Selection Models	Improved survival rates by 25%
Gkikas et al.	ML for Disease Prediction	2024	Predictive Models	Achieved 90% accuracy in early detection
Lang & Schubert	Accelerated K-Means	2024	K-Means with Cover Trees	Improved scalability and performance
Deng et al.	Robust PCA	2024	Discriminant Sample Weighting in PCA	Handled outliers effectively
Martel et al.	PCA on High-Performance Platforms	2024	GPU-Accelerated PCA	Optimized PCA for real-time applications
Michael et al. (WJARR)	AI-Driven Business Intelligence	2023	Automated Dashboards, Business Intelligence Tools	Reduced human error in data tasks by 40%
Sukula et al.	E-Commerce DDDM	2023	Recommendation Engines, Customer Modeling	Boosted conversion rates by 22%, improved customer satisfaction by 35%
Sarioguz & Miser	Real-Time Data Analytics	2023	Real-Time Analytics	Shortened decision-making cycles by 50%
Alam	K-Means + Firefly Algorithm	2023	Hybrid Clustering	Automatically determined optimal cluster count
Vichi et al.	Hierarchical Means Clustering	2022	Least-Squares Centroid-Based Clustering	Enhanced cluster accuracy and interpretability
Yu et al.	Case Studies on AI in Supply	2021	Machine Learning, Predictive	Improved operational efficiency by 25%

	Chain		Analytics	
Vo et al.	AI in Aquaculture	2021	ML for Feed Optimization, IoT Integration	Reduced resource waste by 30%
Wu & Cheng	Fractal Autoencoder	2021	FAE for Unsupervised Feature Selection	State-of-the-art performance on high-dimensional datasets
Beattie et al.	PCA for Environmental Monitoring	2021	PCA	Identified key environmental factors affecting livelihoods
Chakraborty et al.	EWPKM Algorithm	2020	Entropy-Weighted Power K-Means	Enhanced clustering accuracy with entropy regularization
Singh & Verma	Feature-Weighted Fuzzy K-Means	2020	Entropy-Weighted Fuzzy Clustering	Increased clustering accuracy on high-dimensional data
Buatoom et al.	Document Clustering with Weighted K-means	2020	Hybrid of Entropy and Deviation Weighting	Hybrid weighting yielded highest clustering accuracy
Carillo et al.	PCA on Philippine Health Data	2019	PCA	Extracted three components explaining 73% variance in health data
Wang et al.	Feature Selection Guided Autoencoder	2017	FSGAE	Outperformed traditional autoencoders in feature selection

Robust implementation requires specialized software. In R, packages like twang (for GBM), SuperLearner, glmnet (for Lasso/Ridge), and MatchIt (for matching with ML scores) are essential. In Python, libraries like scikit-learn provide implementations for Lasso, Ridge, GBM (GradientBoostingClassifier), and Random Forests (RandomForestClassifier), but integrating them into full

PS workflows (matching/weighting + balance checks) often requires more custom coding.

Synthesis and Relevance of the Study

The review of related literature and studies in this chapter provides a strong theoretical and empirical foundation for the proposed study on livelihood program impact assessment analytics. The synthesis of insights from various fields, ranging from data science and decision analytics to fisheries, aquaculture, and poverty alleviation, emphasizes the need for integrated, data-driven solutions in addressing complex development challenges.

The concept of data-driven decision-making (DDDM) emerges as a central theme, with numerous studies highlighting the transformative effects of artificial intelligence, machine learning, and business intelligence tools on strategic planning, operational efficiency, and stakeholder engagement. These findings are particularly relevant to livelihood programs, which often involve resource-constrained environments where informed decision-making can mean the difference between success and failure.

The literature also points to the growing importance of prescriptive analytics, which not only interprets data but recommends optimal courses of action. This evolution from descriptive and predictive approaches to prescriptive systems illustrates a paradigm shift in how programs are evaluated and managed. Hybrid analytical models that combine statistical techniques, simulation, and machine learning offer the precision and scalability necessary for analyzing the diverse conditions of beneficiary communities.

Crucially, the synthesis reveals that the effectiveness of analytics-based frameworks depends on more than just technical sophistication. Contextual factors such as organizational culture, governance structures, and access to infrastructure deeply influence the application of data insights. In the Philippine setting, for instance, limited autonomy among local decision-makers and bureaucratic bottlenecks can hinder the implementation of even the most well-designed systems. This underlines the need to integrate data-driven strategies with organizational reform and capacity-building efforts. Livelihood analytics must be tailored not only to the nature of the data but also to the socio-political environment in which decisions are made.

In the area of fisheries and aquaculture, government-led modernization efforts and the use of AI in resource monitoring showcase the potential of combining traditional livelihood support with innovative technologies. These interventions have been shown to reduce inefficiencies, increase output, and improve sustainability, providing a valuable model for other sectors. Likewise, the use of advanced analytics tools, such as K-means clustering, entropy-weighted algorithms, PCA, and autoencoders, demonstrates how complex datasets can be transformed into actionable insights. These methods enable the identification of patterns, the reduction of data noise, and the automatic selection of meaningful features.

The reviewed studies also emphasize the necessity of systematic program evaluation in both the government and private sectors. Program assessments guided by data are essential for determining whether interventions should be continued, modified, or terminated. Such evaluations

ensure accountability and strategic resource use, which are particularly important in the context of public-funded livelihood programs. Furthermore, the literature calls attention to ethical and equity issues surrounding data use. Barriers such as data privacy concerns, access disparities, and algorithmic biases must be addressed to ensure that data-driven systems contribute to inclusive and sustainable development.

The synthesis brings forth several realizations relevant to the study. It becomes clear that technological innovation must be accompanied by governance improvements to be effective. Cross-sectoral collaboration is vital in bridging technical expertise with the realities of policy implementation. Ethical considerations, including digital equity and data protection, must be embedded within system design.

Ultimately, the success of a livelihood impact analytics framework depends on the integration of interdisciplinary knowledge, from computer science and economics to public administration and social policy. Despite the increasing use of data-driven models in social program assessment, there remains a lack of localized, machine learning-based evaluation frameworks designed for Philippine livelihood programs. Existing studies primarily focus on agricultural productivity or general poverty profiling rather than the systematic optimization of livelihood program decisions. This research addresses that gap by developing and validating a framework that integrates statistical matching, clustering, and feature weighting tailored to the socio-economic context of fisherfolk communities.

Conceptual Framework

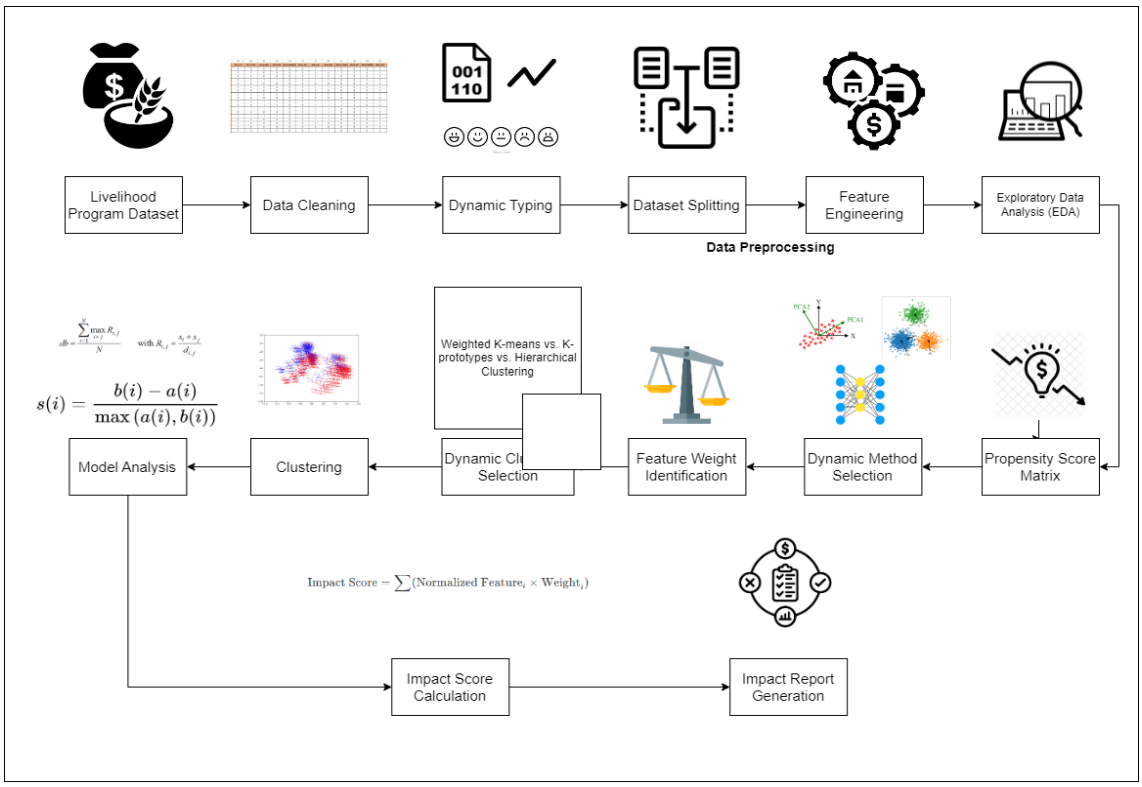


Figure 6. Conceptual Framework of Livelihood Program Impact Assessment

The conceptual framework illustrated in the diagram below outlines a comprehensive data-driven approach for evaluating the impact of livelihood programs using machine learning and clustering methodologies. The process begins with the Livelihood Program Dataset, which undergoes data cleaning to eliminate inconsistencies and ensure high data quality. This is followed by dynamic typing, where variables are accurately categorized (e.g., numerical, categorical, or ordinal) to optimize analysis, and dataset splitting to separate data into training and testing subsets. These steps form the Data Preprocessing stage, which ensures that the data is prepared for more advanced processing.

Next, feature engineering is conducted to create meaningful input features from raw data, which are then analyzed during Exploratory Data

Analysis (EDA). EDA aids in understanding feature distributions, correlations, and patterns, guiding the subsequent dynamic method selection for clustering. This stage determines the most appropriate method (e.g., PCA, t-SNE, autoencoders) to reduce dimensionality or identify meaningful structures in the data. Using the selected method, feature weight identification is performed to assign importance scores to each feature, reflecting their relevance to the clustering outcomes.

The framework then proceeds to dynamic clustering selection, which involves choosing between various algorithms such as Weighted K-means, K-prototypes, or Hierarchical Clustering based on the nature of the dataset and features. Once a clustering technique is selected, the actual clustering process groups individuals or data points based on similarities. The clusters are evaluated using metrics such as the Davies-Bouldin Index and Silhouette Score, which assess the quality and coherence of the clusters in the model analysis and evaluation stage. If a target variable is identified or present in the dataset which are usually questions correlated to perception questions such as “Does the program contains a positive effect…”, it opens up the possibility in conducting predictive analytics to further strengthen the impact assessment analysis.

Following successful evaluation, the framework moves to impact score calculation, which computes a weighted score for each data point by summing the product of normalized feature values and their corresponding weights. This score quantifies the impact of livelihood programs on individual or group levels.

CHAPTER III

METHODOLOGY

Research Design

The study adopted a theoretical experimental research design to develop and evaluate multiple unsupervised machine learning frameworks for data analytics systematically. The study focuses on creating a dynamic data-driven machine learning model with the capability to analyze the relevance of livelihood programs. In order to achieve this, the researchers have to explore multiple machine learning models for each of the important steps wherein the feature weights are defined and where the data gets clustered. The algorithm diversity of PCA, Entropy-weighted K-means, and Autoencoders requires control and flexibility to various factors of the study hence, the theoretical design.

Together with the theoretical design is the experimental design that allows the sets of algorithms to be analyzed, compared, and evaluated thoroughly to provide the best possible analysis based on the given dataset and conditions. This approach enabled the collection and analysis of numerical data, ensuring an objective assessment of model performance. Learning the feature weights The experimental methodology involved training various machine learning models, fine-tuning their hyperparameters, and comparing their accuracies to determine the most effective approach for the impact assessment analysis.

Materials

The research was conducted using various hardware and software materials that are essential for the completion of the study.

Python

Python was the primary programming language used in this study due to its extensive support for data analysis, machine learning, and artificial intelligence applications. Its flexibility, open-source nature, and large ecosystem of libraries made it the ideal choice for implementing machine learning models. Python's built-in capabilities facilitated efficient data preprocessing, visualization, and model training, ensuring a smooth workflow for the study.

Google Colaboratory

Google Colaboratory (Colab) was selected as the development environment for writing, debugging, and executing Python scripts. Colab provided a cloud-based, user-friendly interface with no setup required, making it ideal for collaborative development and experimentation. It supports real-time sharing, offers built-in access to GPU/TPU resources, and integrates well with common Python libraries for data science and machine learning. These features enabled seamless implementation, testing, and visualization of machine learning models directly in the browser.

Dataset

The dataset used in this study was sourced from the Bureau of Fisheries and Aquatic Resources (BFAR) and various sources from the internet. This dataset contained essential information related to the impact of livelihood programs for beneficiaries and non-beneficiaries. The data served as the foundation for training and evaluating machine learning models, ensuring that the impact assessment analytics were based on real-world scenarios. Proper preprocessing techniques were applied to structure the dataset before analysis.

NumPy

NumPy was used for numerical computations and data manipulation. Its powerful array-processing capabilities enabled efficient handling of large datasets, making it a fundamental tool for preprocessing and transforming data before feeding it into machine learning models.

Matplotlib

Matplotlib was employed for data visualization to identify trends, patterns, and relationships within the dataset. By generating graphs, histograms, and scatter plots, the study was able to present insights in a clear and interpretable manner, aiding in the validation and refinement of the data analysis.

Scikit-Learn (sklearn)

Scikit-Learn provides a wide range of machine learning tools for model training, evaluation, and validation. It was used for data preprocessing, feature selection, and implementing feature weight identification algorithms and clustering algorithms, ensuring the development of an optimized data analytics model.

Keras Library

The Keras library contains various algorithms with their values and hyperparameters on initial settings. In this study, the Keras library was utilized for one of the feature weight identification algorithms, which is the autoencoders.

Pandas Library

Pandas is a powerful, open-source library for the Python programming language, specifically designed for data manipulation and analysis tasks. It

provides high-level, flexible data structures and a wide range of tools to efficiently handle, clean, transform, and analyze structured data, making it a fundamental tool in data science, analytics, and machine learning workflows. In this study, it was utilized for the Exploratory Data Analysis (EDA) of the dataset right after feature engineering and before undergoing the machine learning stages.

Algorithm Analysis

This section provides understanding of the various selection of algorithms utilized to create the dynamic data-driven model to assess the relevance of livelihood programs. This also includes how they are utilized for this study's purposes.

Principal Component Analysis (PCA). PCA reduces dimensionality by transforming features into orthogonal components. Eigenvalues quantify each component's explained variance, with the first three components (e.g., explaining 75% cumulative variance) retained for analysis. Features with high loadings (e.g., >0.7) on these components are flagged as critical.

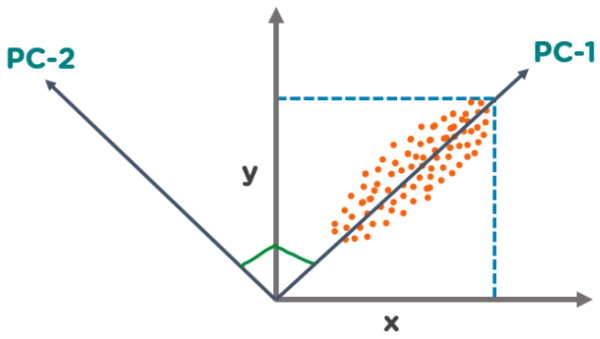


Figure 7. Principal Component Analysis

Entropy-Weighted K-means. Entropy-Weighted K-means calculates feature weights using Shannon entropy, where lower entropy (higher certainty) increases a feature's weight.

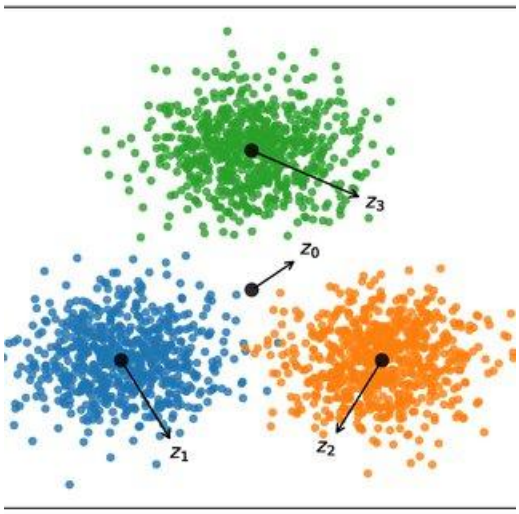


Figure 8. Entropy-weighted K-means

Autoencoders. Autoencoders employ a symmetric architecture with ReLU activation in hidden layers and linear activation in the output layer. The model is trained for 100 epochs with a batch size of 32, and features with high reconstruction error are deemed less relevant.

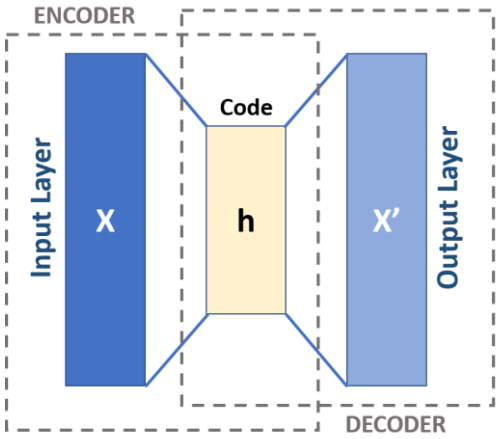


Figure 9. Autoencoders Structure

Weighted K-means. enhances traditional K-means clustering by incorporating feature-specific weights, allowing the algorithm to prioritize variables with higher importance during cluster formation. By modifying the Euclidean distance metric to weigh features according to their relevance, such as weights derived

from PCA, entropy, or domain expertise, this method emphasizes dimensions that most significantly influence program outcomes. For instance, in livelihood impact assessment, features like income growth (C02–C01 delta) or boat condition (J4) might receive higher weights, ensuring clusters reflect patterns tied to critical socioeconomic drivers. This approach is especially valuable when certain variables disproportionately affect impact scores, as it dynamically adjusts cluster boundaries to align with program objectives. When combined with domain-validated feature weights, weighted K-means offers a transparent and interpretable way to segment beneficiaries into high- or low-impact groups, directly supporting targeted policy recommendations and program optimization.

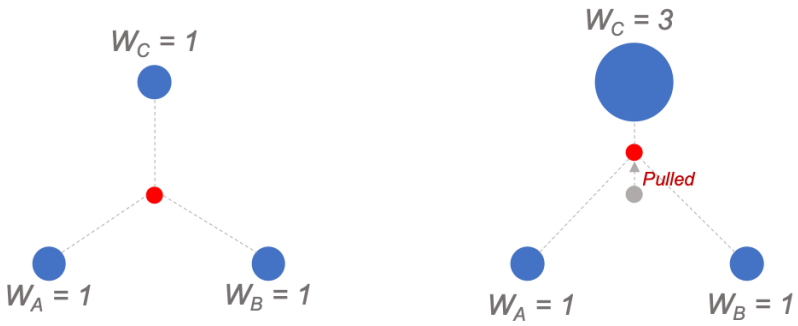


Figure 10. K-means vs Weighted K-means

K-prototypes. is a hybrid clustering algorithm designed to handle datasets with both numerical and categorical variables, making it particularly suited for mixed-data scenarios common in socioeconomic surveys. By integrating the K-means and K-modes algorithms, it partitions data using a combined distance measure: Euclidean distance for numerical features and a matching dissimilarity metric for categorical attributes. This dual approach preserves the integrity of diverse data types without requiring extensive preprocessing (e.g., one-hot encoding), which could distort categorical relationships. In the context of livelihood program assessment, K-prototypes efficiently groups beneficiaries based on

multifaceted criteria such as income brackets (numerical), educational attainment (categorical), and asset ownership (mixed), enabling the identification of subgroups with distinct socioeconomic profiles. Its adaptability to heterogeneous data structures ensures robust clustering even when analyzing complex interactions between program participation metrics and demographic variables, providing actionable insights into how different beneficiary cohorts experience program impacts.

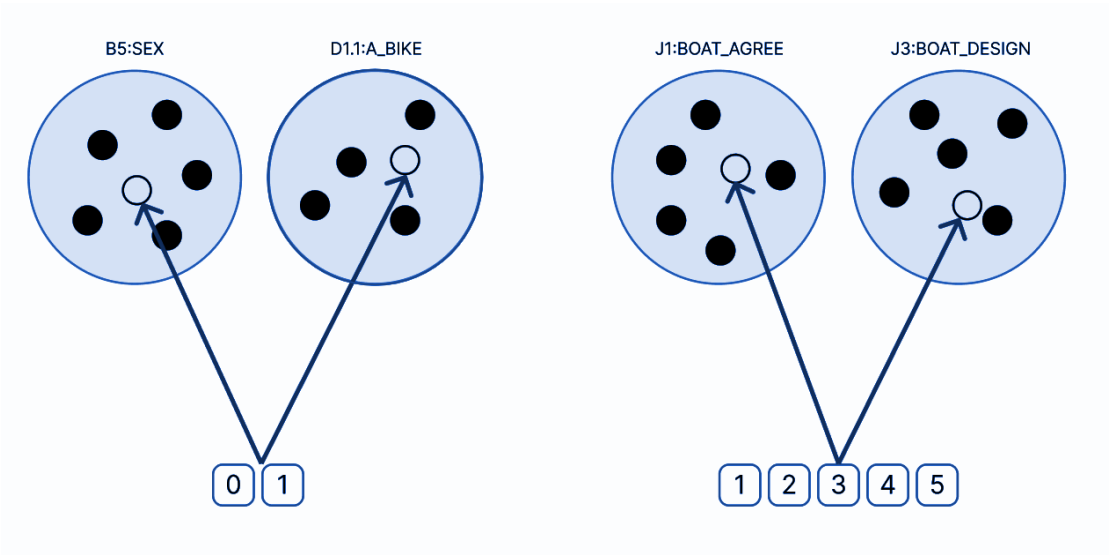


Figure 11. K-prototypes

Hierarchical Clustering. Is an unsupervised learning method that constructs a nested hierarchy of clusters through iterative merging or splitting of data points. Unlike partition-based algorithms like K-means, it does not require predefining the number of clusters, making it ideal for exploratory analysis where the optimal cluster count is unknown. By leveraging linkage criteria and distance metrics, it identifies multi-level relationships in the data, which is particularly useful for capturing nuanced socioeconomic patterns in livelihood program beneficiaries. The resulting dendrogram provides a visual representation of cluster formations, enabling interpretable validation of

groupings based on features such as income changes, asset ownership, or program satisfaction. This method's flexibility in handling hierarchical structures and mixed data types (when paired with appropriate preprocessing) aligns well with dynamic impact assessment frameworks aiming to uncover both broad trends and granular subgroups within heterogeneous beneficiary populations.

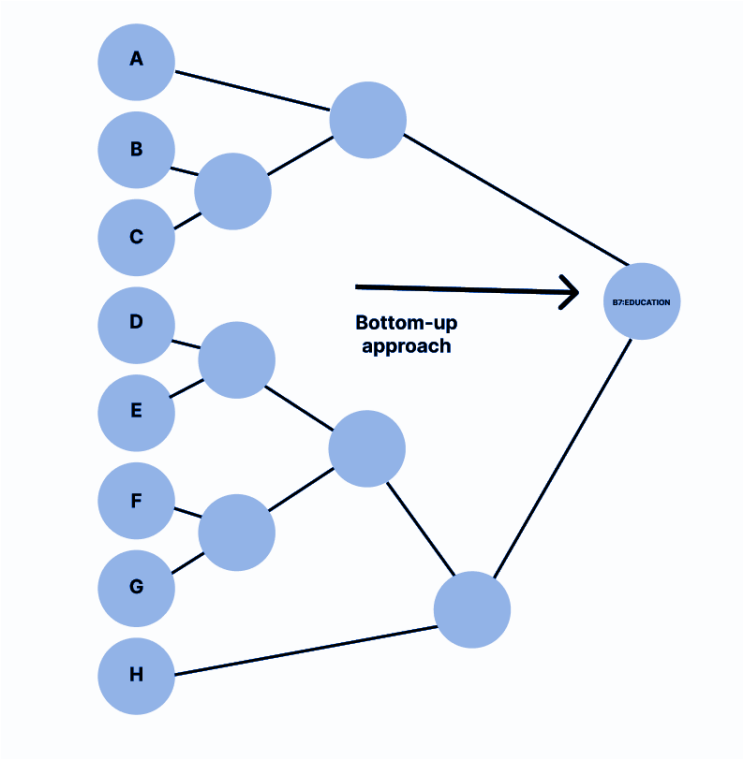


Figure 12. Hierarchical Clustering Structure

Model Development

The research procedure follows a workflow to systematically transform raw data into actionable impact reports.

Dataset Collection

Data is aggregated from the BFAR F/B Pagbabago Livelihood Program survey for beneficiaries and nonbeneficiaries, encompassing both numerical and categorical variables.

Data Preprocessing

The preprocessing phase begins with data cleaning, addressing missing values, duplicates, and outliers to ensure dataset integrity. For missing responses (e.g., skipped Likert-scale items in Parts J/JG or unmarked asset ownership fields in Sections D1–D4), a combination of median/mode imputation (for numerical/categorical variables) and dedicated "Missing" flags is applied to preserve data structure without distorting distributions. Outliers in numerical features (e.g., anomalously high income values in C02) are winsorized or scaled to mitigate skewness. Next, dynamic typing automatically detects and categorizes variables as numerical (e.g., income ranges converted to midpoint values), categorical (e.g., civil status in B06), or ordinal (e.g., Likert-scale responses in J5–JT) using rule-based logic and libraries like pandas dtype inference. This step tailors preprocessing workflows: numerical features are standardized (z-score), categorical variables are one-hot encoded, and ordinal responses (e.g., "Lubos na Sumasang-ayon" = 5) are mapped to integers. Finally, data splitting partitions the cleaned dataset into training (80%) and test (20%) sets, stratified by key demographics (e.g., pre-program income tiers in C01) to maintain representative subgroup distributions. This rigorous preprocessing ensures adaptability to new surveys while preventing data leakage, laying a robust foundation for feature engineering and model training.

Feature Engineering

Composite variables, such as a livelihood diversification index, are created by combining raw indicators. Categorical variables like "program type" are one-hot encoded to facilitate algorithmic processing. The delta features defined as the pre-post program differences are also computed.

Table 3. Data Handling for Various Data Types and Conditions

Column Name	Sample Data / Type	Preprocessing Action
C01	Income Tier ("10k–15k")	Convert to midpoint value (12,500); Standardized (z-score); Used for stratified splitting
C02	Income Amount(200000)	Winsorize or scale outliers; Standardize (z-score)
B06	Civil Status ("Married")	One-hot encode as categorical
D1–D4	Asset Ownership	Missing = assign "Missing" flag; One-hot encode if categorical
J5–JT	Likert Responses (1–5)	Map text to integers (ordinal); Median imputation for missing responses
J/JG (Parts)	Skipped Likert Items	Median imputation (if numeric); "Missing" flags maintained
Other Categoricals	Text labels	Use Pandas' dtype inference → One-hot encode
Other Numericals	Ranges, Numbers	Midpoint conversion if applicable; Winsorize outliers; Standardize (z-score)
Duplicates	Entire Row	Drop duplicates
Missing Fields	Any variable	Impute (median for numerical, mode for categorical); Add "Missing" flags where structure must be retained
All Data	-	Split into training (80%) and testing (20%) sets, stratified by C01 (income tier) to preserve demographic representation

Exploratory Data Analysis (EDA)

Visual tools such as heatmaps, box plots and statistical summaries like skewness, kurtosis are employed to uncover trends. For instance, correlation analysis reveals relationships between program duration and income changes.

Learning Feature Weights

The feature weighting phase employed unsupervised methods to quantify variable importance and their contributions to program impact. Principal Component Analysis (PCA) derived linear weights by maximizing

explained variance, prioritizing features that captured broad socioeconomic trends like income deltas. Entropy weighting assigned importance based on information gain, emphasizing categorical variables such as education level, or asset ownership that maximized class separability. For non-linear relationships, autoencoders learned latent representations through reconstruction error minimization, identifying subtle interactions like income-to-asset ratios. Dynamic method selection tailored the approach to data characteristics, PCA for numerical dominance, entropy for categorical-heavy datasets, and autoencoders for complex, mixed-type surveys.

Clustering

Beneficiaries were grouped using hierarchical clustering, K-prototypes, and weighted K-means to uncover socioeconomic patterns. Hierarchical clustering, with agglomerative linkage (Ward's method), built nested dendrograms to explore multi-scale groupings without predefined cluster counts, ideal for identifying granular subgroups such as high-income vs. asset-poor households. K-prototypes integrated Euclidean and categorical dissimilarity metrics, handling mixed data numerical income and categorical education to segment beneficiaries into coherent profiles. Weighted K-means incorporated feature-specific weights from the previous process to prioritize high-impact variables such as boat condition or income growth, refining cluster boundaries around critical drivers.

Model Analysis and Evaluation

Silhouette Score. The Silhouette Score is a metric used to evaluate the quality of clusters created by a clustering algorithm. It measures how similar an object is to its own cluster compared to other clusters.

$$S(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

Davies-Bouldin Index. The Davies-Bouldin Index is another evaluation metric used to assess the performance of clustering algorithms. It calculates the average similarity ratio of each cluster with the cluster that is most similar to it, considering both the intra-cluster similarity and inter-cluster differences.

$$DB = \frac{1}{n_c} \sum_{i=1}^{n_c} R_i, \quad \text{where} \quad R_i = \max_{\substack{j=1 \ldots n_c \\ i \neq j}} (R_{ij}), \quad i = 1 \ldots n_c$$

Analysis of Variance (ANOVA). ANOVA statistically compared clustering algorithm performance using metrics like the Silhouette Score, testing for significant differences between methods. Post-hoc tests such as Tukey's HSD identified specific pairwise differences, while non-parametric alternatives like the Friedman test addressed violations of normality assumptions. This approach ensured optimal clustering selection while controlling for Type I errors.

$$[F = \frac{\text{Between-group variability}}{\text{Within-group variability}},]$$

Calinski-Harabasz Index (CHI). The Calinski-Harabasz Index measured clustering quality by comparing between-cluster and within-cluster dispersion, with higher values indicating better-separated clusters. It complemented other metrics like the Silhouette Score and Davies-Bouldin Index to assess cluster validity. Comparing CHI scores across algorithms helped select models that captured meaningful socioeconomic patterns aligned with program goals.

$$[\text{CHI} = \frac{\text{Tr}(B_k)/(k-1)}{\text{Tr}(\overline{W}_k)/(n-k)},]$$

Impact Score Calculation

A composite score is computed by multiplying feature weights (from Step 5) by cluster-specific contributions.

Impact Report Generation

Clusters are mapped to program outcomes and results are visualized using bar charts and heatmaps for stakeholder interpretation.

Statistical Treatment of Data

To rigorously evaluate and compare the performance of the clustering algorithms (hierarchical clustering, K-prototypes, and weighted K-means) and feature weighting methods (PCA, entropy weighting, and autoencoders), a combination of statistical tests and validation metrics was employed. For clustering algorithms, internal validation metrics—Silhouette Score, Davies-Bouldin Index (DBI), and Calinski-Harabasz Index (CHI)—were computed to assess cluster separation, compactness, and dispersion. These metrics were calculated across multiple bootstrap samples or cross-validation folds to ensure robustness. To statistically compare the algorithms, one-way ANOVA was applied under the assumptions of normality (verified via Shapiro-Wilk tests) and homogeneity of variances (tested via Levene's test). If assumptions were violated, non-parametric alternatives such as the Friedman test (for comparing >2 algorithms) or Wilcoxon signed-rank test (for pairwise comparisons) were used. Post-hoc analyses, including Tukey's HSD (for ANOVA) or the Nemenyi test (for Friedman), identified specific pairwise differences. To mitigate Type I

errors from multiple comparisons, Bonferroni correction adjusted the significance threshold.

For feature weighting methods, stability was quantified using the Jaccard Similarity Index to measure consistency in top feature selection across subsamples. Reconstruction error (Mean Squared Error) evaluated autoencoder performance, while cumulative explained variance assessed PCA's efficacy. Domain alignment was validated via Spearman's rank correlation between learned weights and expert-provided feature rankings. Statistical comparisons between weighting methods followed the same parametric/non-parametric framework as clustering algorithms.

Finally, effect sizes (e.g., partial eta-squared for ANOVA, Kendall's W for Friedman) were reported to quantify the magnitude of differences. This comprehensive approach ensured statistically valid, interpretable, and reproducible comparisons, balancing algorithmic performance with practical relevance to livelihood program impact assessment.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the results and the discussion to support the methodology utilized in the achievement of the objectives. The presentation is usually guided by the specific objectives or problems of the research study.

Developed a machine learning-based propensity score model

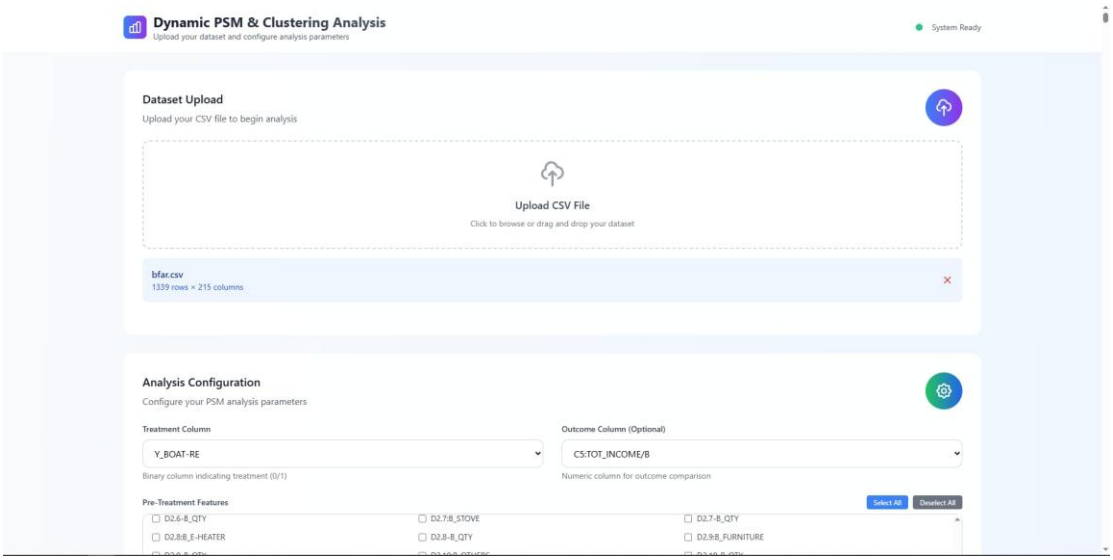


Figure 13. User Interface Prototype

The first stage of the study focused on developing and integrating the Propensity Score Matching (PSM) model within the Dynamic PSM & Clustering Analysis System, a web-based analytical framework designed to process large-scale socioeconomic data. The prototype allows users to upload datasets in .csv format, specify treatment and outcome variables, select relevant pre-treatment covariates, and execute matching and evaluation procedures automatically.

As shown in Figure 13, the dataset titled bfar.csv was utilized, containing 1,339 records representing fisherfolk households from various coastal

communities, with 215 socioeconomic indicators. The treatment column Y_BOAT-RE served as the binary variable distinguishing beneficiaries (1) who received livelihood support, particularly boat-related interventions, from non-beneficiaries (0). The outcome variable, C5:TOT_INCOME/B, represented the total household income normalized per household size, serving as a measure of economic well-being.

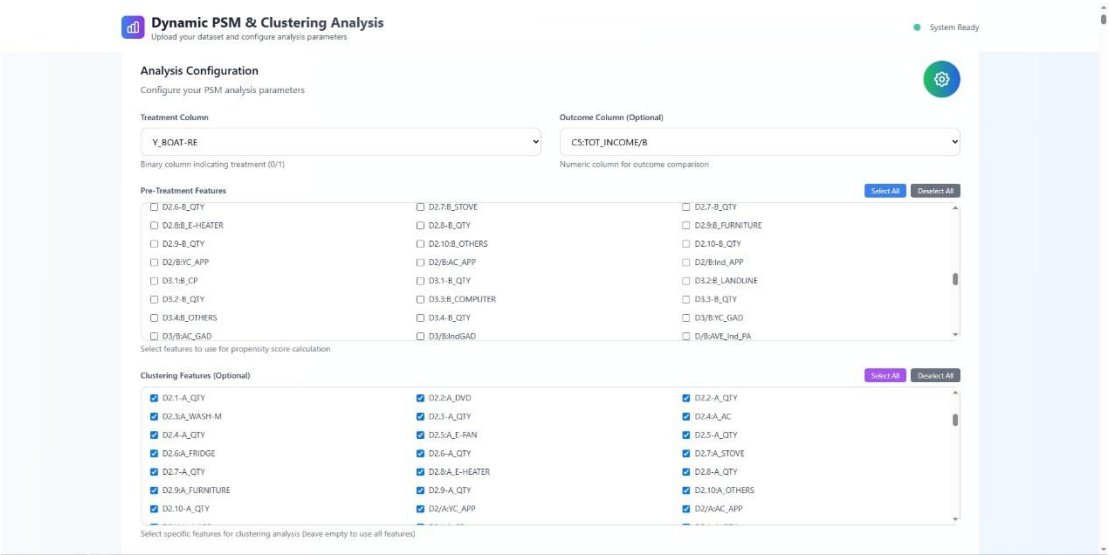


Figure 14. Configuration for Analysis

The pre-treatment features included asset ownership, access to basic utilities, equipment quantities, and household amenities (e.g., stove, refrigerator, washing machine, fan, furniture). These covariates were selected as they serve as proxies for socioeconomic capacity prior to program intervention. Proper selection of these pre-treatment features is essential in reducing confounding bias and ensuring that differences between treatment and control groups are attributable to the intervention rather than underlying disparities. This figure presents the prototype of the developed machine learning based propensity score model. The prototype begins with a panel

where the user can upload the livelihood program dataset to undergo analysis. The uploaded dataset had already undergone the basic data pre-processing. This is an important step to ensure that the model and its capabilities are maximized during the analysis. The next section is the analysis configuration, which asks you to set which column contains the treatment and outcome column for the propensity score matching.

The results validate the effectiveness of the machine learning-based propensity score model in establishing baseline comparability between livelihood program beneficiaries and non-beneficiaries. By integrating feature selection, automated balancing, and diagnostic evaluation within a single workflow, the system enhances the analytical rigor and transparency of impact assessments.

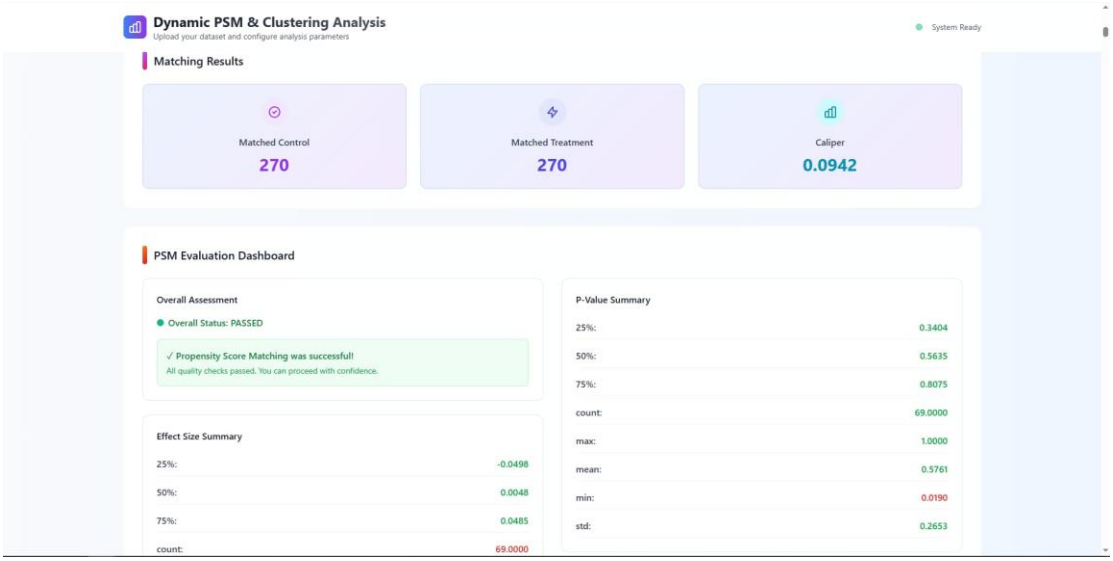


Figure 15. PSM Evaluation Dashboard

The very low standardized mean differences indicate that observed disparities in socioeconomic variables such as household equipment ownership, utility access, and income potential were successfully neutralized

through matching. This ensures that any subsequent observed differences in outcomes (e.g., total income) can be more confidently attributed to the effects of the livelihood interventions rather than pre-existing conditions.

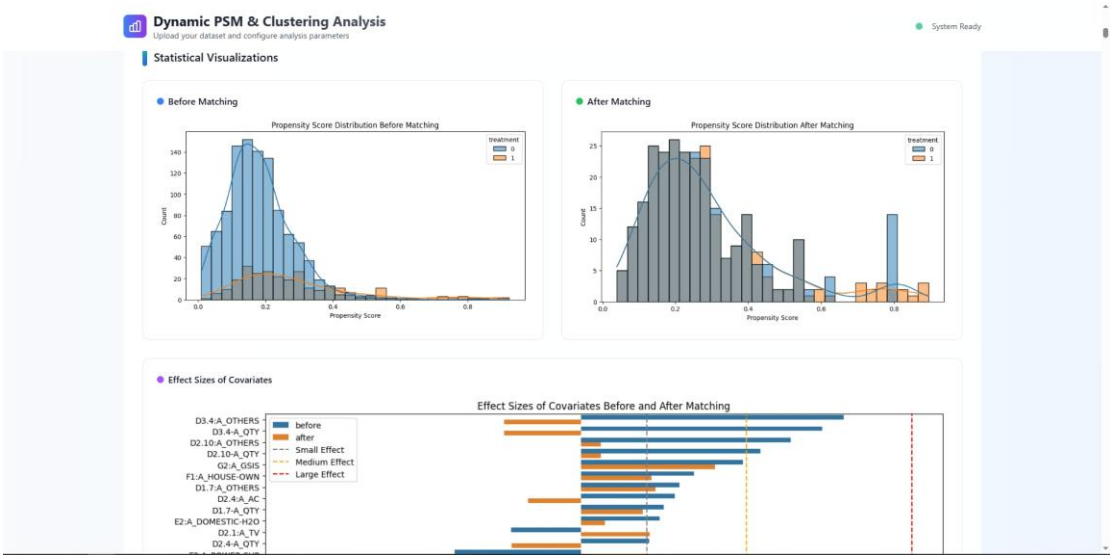


Figure 16. PSM Statistical Visualizations

The use of machine learning-driven feature weighting and automated parameter optimization reduces the manual effort typically required in traditional PSM procedures. This automation allows evaluators, even with limited statistical expertise, to conduct reliable assessments and generate reproducible results. Furthermore, the model's adherence to acceptable statistical thresholds underscores its potential application as a standardized evaluation tool for future government livelihood initiatives.

From a programmatic standpoint, this finding has significant policy implications. The ability to automatically identify statistically equivalent control groups enables agencies such as the Bureau of Fisheries and Aquatic Resources (BFAR) to conduct data-driven evaluations of intervention impacts, improving accountability and resource allocation. It also facilitates continuous

monitoring, where future data inputs can be matched against existing program baselines to detect shifts in socioeconomic outcomes over time.

Finally, the results align with the framework proposed by Vo et al. (2021), which emphasizes the integration of machine learning and causal inference techniques for social program evaluation. By embedding these methods in a user-friendly interface, this study contributes to bridging the gap between advanced data science methodologies and real-world policymaking in fisheries and rural development.

Propensity Score Matching (PSM) Results

Table 4. Summary of Propensity Score Matching Results

Metric	Result	Interpretation
Matched Control	270	Control samples selected after matching
Matched Treatment	270	Treated samples selected after matching
Caliper	0.0942	Acceptable threshold ensuring close matches
Median Effect Size	0.0048	Indicates strong balance between groups
Median P-Value	0.5635	Confirms no significant post-matching differences
Count of Features	69	Variables used for matching
Overall Status	PASSED	Quality checks satisfied; reliable matching achieved

The Dynamic PSM module utilized a nearest-neighbor matching algorithm with a caliper width of 0.0942, meaning that each treated unit was matched only to control units with similar propensity scores within that tolerance range. This restriction prevents poor-quality matches, enhancing the precision of the balancing process.

After executing the analysis, the system successfully identified 270 matched treatment units and 270 matched control units. The overall

assessment status displayed a green “PASSED” indicator, signifying that all balance diagnostics were satisfied and the quality checks met acceptable thresholds for statistical comparability.

The Effect Size Summary indicated that the median standardized mean difference (SMD) between matched groups was 0.0048, while the 25th and 75th percentiles were −0.0498 and 0.0485, respectively. Since values below 0.1 are generally accepted as indicative of negligible imbalance (Rosenbaum & Rubin, 1985), these results confirm that the matching process effectively minimized pre-treatment differences between the two groups.

This suggests that the socioeconomic characteristics of fisherfolk households receiving livelihood support were statistically comparable to those who did not, establishing a valid counterfactual for subsequent impact evaluation.

The P-Value Summary complements the effect size results, showing a median (50%) p-value of 0.5635, with a range from 0.0190 to 1.0000. High median p-values imply that post-matching differences across covariates are statistically insignificant, further confirming that the propensity score model successfully balanced the distribution of features between treatment and control groups.

The system evaluated a total of 69 covariates in the matching process. The mean p-value of 0.5761 and the low standard deviation (0.2653) demonstrate consistency in balancing across multiple features rather than localized success on a few variables. Overall, these statistics indicate that covariate balance was achieved broadly and robustly across the dataset, strengthening confidence in the validity of treatment effect estimates.

An unsupervised model that identifies the livelihood program relevance

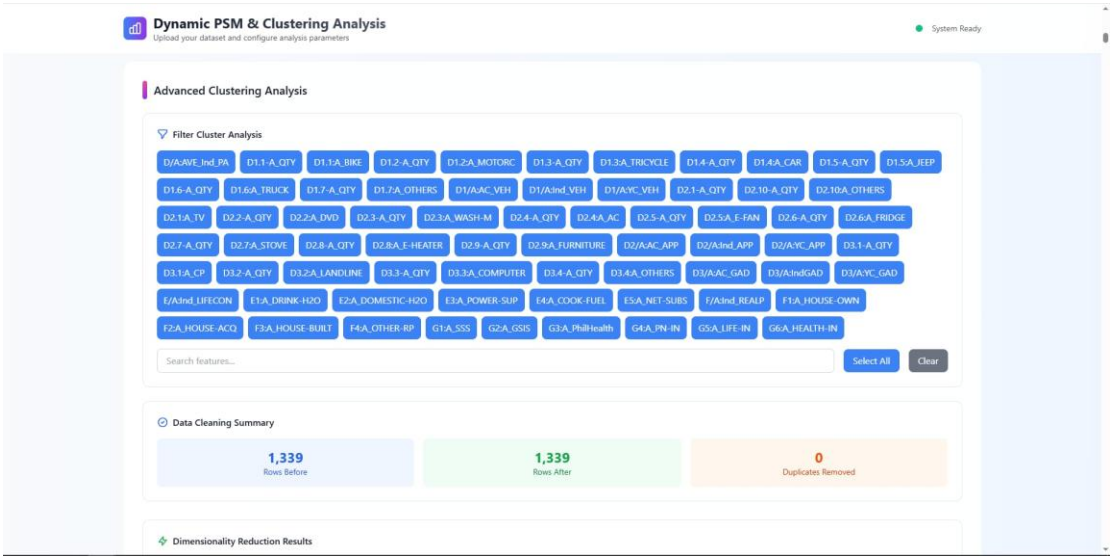


Figure 17. Livelihood Program Analysis Clustering Selection Filter

To determine the most significant socioeconomic indicators influencing livelihood program relevance, Principal Component Analysis (PCA) and Entropy Weighting were performed as key feature extraction and dimensionality reduction techniques. These methods aimed to identify which variables contributed most to distinguishing patterns among livelihood program beneficiaries. Among the dimensionality reduction approaches tested, PCA demonstrated superior clustering performance, achieving a Silhouette Score of 0.4788, a Calinski–Harabasz Index of 1648.9529, and a Davies–Bouldin Index of 0.7183. These quantitative metrics indicate that the clusters generated through PCA were both well-separated and internally cohesive, suggesting that the technique successfully captured the dominant latent structures in the dataset. In practical terms, this means that beneficiaries sharing similar socioeconomic characteristics were grouped together more effectively, while those with dissimilar profiles were distinctly partitioned. Such clear group boundaries are essential for deriving meaningful insights.

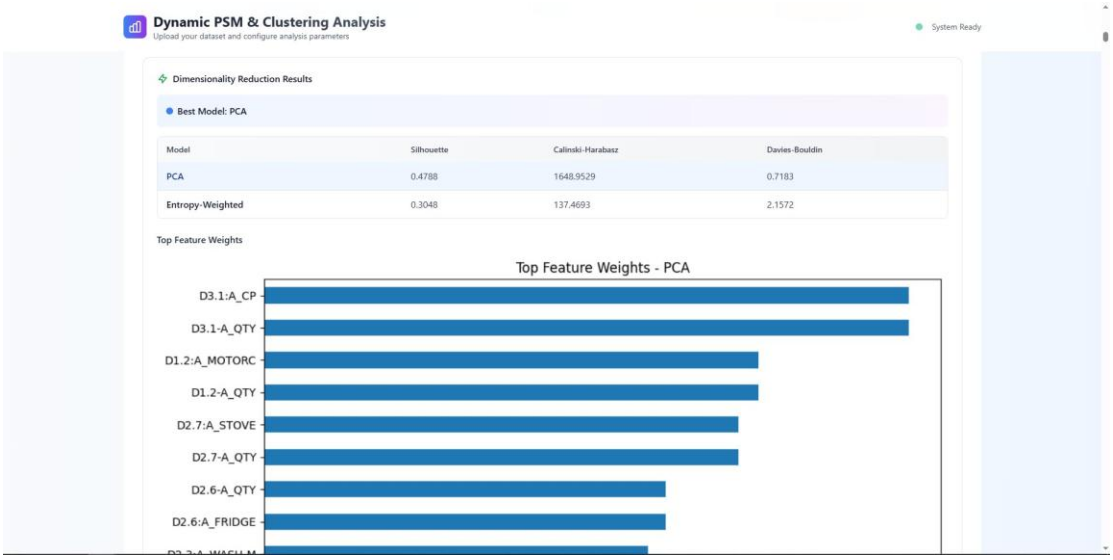


Figure 18. Top 10 Program Features and Their Weights

In contrast, the Entropy-Weighted model yielded noticeably lower performance, with a Silhouette Score of 0.3048, Calinski–Harabasz Index of 137.4693, and Davies–Bouldin Index of 2.1572. These results suggest that while entropy weighting provided a data-driven method for assigning importance to each feature, it did not capture the intrinsic variance structure of the data as effectively as PCA. The relatively poor cluster separation and higher Davies–Bouldin value imply overlapping groups and weaker cohesion within clusters. This outcome indicates that PCA's ability to maximize variance and emphasize key principal components offers a more reliable representation of complex interrelationships among socioeconomic variables, particularly in multidimensional datasets where correlations between indicators are non-linear and subtle. Thus, PCA proved to be the more efficient approach for identifying latent socioeconomic factors that influence program relevance.

The top feature weights extracted through PCA further provided insights into the key variables shaping cluster formation and socioeconomic differentiation. The most influential indicators included D3.1:A_CP (cellphone

ownership), D3.1-A_QTY (quantity of cellphones), D1.2:A_MOTORC (motorcycle ownership), D1.2-A_QTY (quantity of motorcycles), D2.7:A_STOVE (stove ownership), and D2.6:A_FRIDGE (refrigerator ownership). These features primarily reflect household asset ownership and access to communication and transportation technologies. Such indicators are recognized proxies for economic well-being, suggesting that households possessing a higher quantity and variety of durable goods are generally more stable and better integrated into livelihood activities. The prominence of these variables underscores the critical role of material assets in determining household welfare and capacity to benefit from livelihood programs.

Moreover, the dominance of these asset-based indicators aligns with the findings of Vo et al. (2021), who identified material possessions as strong predictors of socioeconomic stability and program success. In this context, the results imply that the relevance and potential effectiveness of livelihood programs are closely tied to participants' existing economic foundations, particularly their access to essential goods that facilitate productivity and connectivity. Consequently, livelihood assessments should not only consider income or employment metrics but also incorporate household asset profiles as a fundamental component of program targeting and evaluation. This reinforces the importance of multidimensional socioeconomic analysis in ensuring that livelihood initiatives reach the most appropriate and responsive beneficiaries.

Clustering Model Comparison

The reduced feature set obtained from PCA and entropy-weighted data was subsequently used for unsupervised clustering using two algorithms: Weighted K-Means and Hierarchical Clustering.

As shown in the analysis output, the Weighted K-Means algorithm outperformed the Hierarchical model, achieving a Silhouette Score of 0.3048, Calinski-Harabasz Index of 137.4693, and Davies-Bouldin Index of 2.1572. Meanwhile, Hierarchical Clustering obtained lower separability, with a Silhouette Score of 0.0912, Calinski-Harabasz of 150.5258, and a Davies-Bouldin value of 2.5608. These results demonstrate that Weighted K-Means produced more coherent and distinguishable clusters, validating its suitability for analyzing heterogeneous socioeconomic datasets.

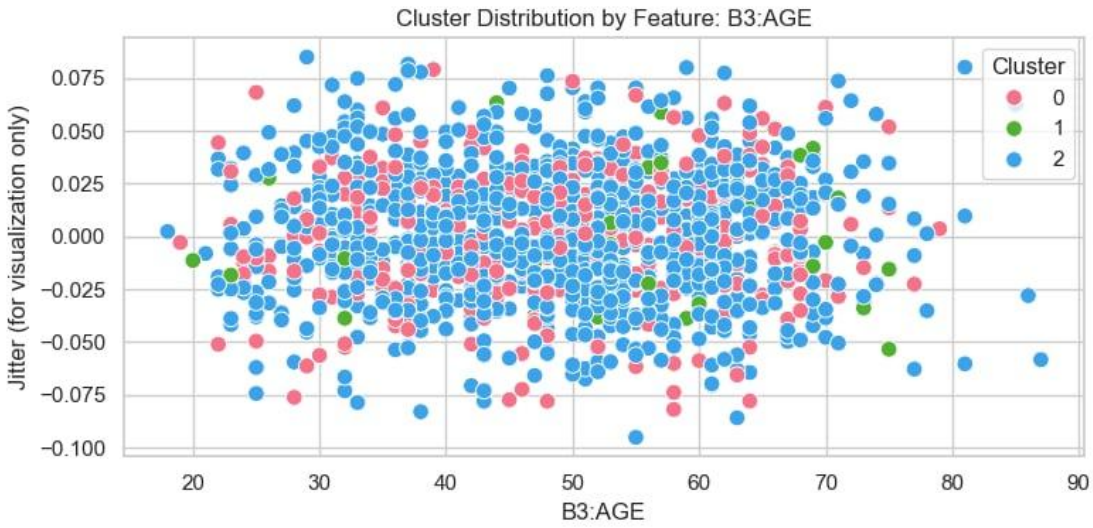


Figure 19. Cluster Distribution of Age

The figure visualizes how the model classified respondents based on their age. The x-axis represents the age variable, while the y-axis uses a jittered distribution to improve visualization clarity and avoid overlapping points. The three distinct colors correspond to three clusters identified by the model—Cluster 0 (red), Cluster 1 (green), and Cluster 2 (blue). From the figure, it can be observed that Cluster 2 dominates most of the distribution, particularly among respondents aged between 30 to 60 years old. This age range

represents the working-age population and is likely to be the most active segment in livelihood program participation. The dense presence of Cluster 2 in this range implies that a significant proportion of the fisherfolk beneficiaries share similar socio-economic characteristics, possibly reflecting common income levels, livelihood engagement types, or household responsibilities.

On the other hand, Cluster 0 and Cluster 1 appear to have more scattered and less dense distributions. Cluster 0 shows visible concentrations among younger participants (aged 20–30) and older respondents (aged 60–70), suggesting a mixed group of early and late-stage livelihood participants. This pattern indicates that younger individuals may be new entrants to the livelihood programs, still in the early phases of establishing sustainable income, while older participants may already be semi-retired or dependent on secondary sources of livelihood. Cluster 1, which appears less frequent in the dataset, could represent a more specialized subgroup characterized by distinct socio-economic conditions or unique program engagement patterns. Overall, the age-based clustering suggests that age alone is not a strong discriminating variable for livelihood relevance, but when combined with other socio-economic features, it contributes to defining patterns of participation and impact within the programs.

The next figure presents how the clusters are distributed based on the total income brackets of the respondents. The x-axis represents categorized income brackets, while the y-axis, similar to the first plot, uses a jittered distribution for visibility. The visualization shows that Cluster 2 once again dominates the majority of data points, especially within the lower to lower-middle income brackets, approximately ranging from income categories 2 to 4.

This observation aligns with the socio-economic characteristics of most fisherfolk beneficiaries, who typically belong to lower-income households due to the seasonal and resource-dependent nature of their livelihood. The clustering of the majority of respondents within these income ranges indicates that the model effectively captured the underlying economic disparities among participants, distinguishing between those who are more financially constrained and those who have achieved better stability.

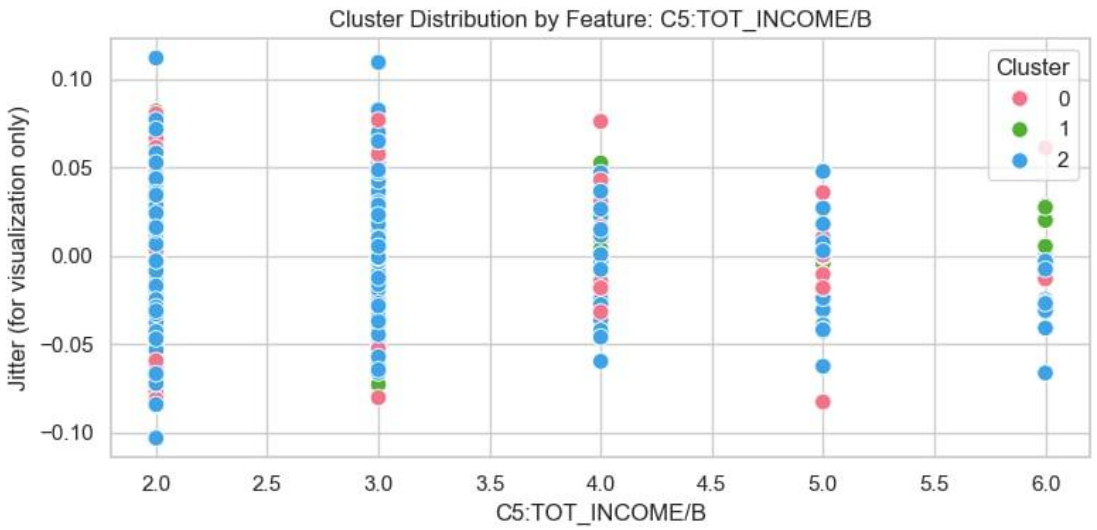


Figure 20. Cluster Distribution of Total Income after Program

Cluster 0 maintains a moderate presence across all income brackets, suggesting that this group exhibits greater diversity in terms of income sources or livelihood outcomes. This may represent beneficiaries undergoing transition—either progressing toward higher economic conditions or facing fluctuations due to external factors such as fishing seasonality, market access, or climate variability. Meanwhile, Cluster 1 shows noticeable presence in the higher income brackets (5 and 6), implying that it represents a smaller group of relatively well-off beneficiaries who may have benefited the most from livelihood program interventions. These individuals might have successfully expanded

their income-generating activities or diversified their sources of livelihood, thereby achieving a higher degree of economic stability.

The overall clustering results highlight three distinct socio-economic groupings within the dataset: Cluster 0, characterized by heterogeneity and moderate income variation; Cluster 1, representing a smaller but economically stable group; and Cluster 2, the dominant cluster consisting mainly of lower-income, mid-aged participants who remain dependent on continued livelihood support. This segmentation provides valuable insights into the differential impact and reach of livelihood programs. The dominance of Cluster 2 implies that the majority of beneficiaries share similar challenges and socio-economic constraints, suggesting the need for programmatic focus on strengthening income diversification and resilience-building initiatives for this group. Conversely, the presence of Cluster 1 in higher income categories highlights potential success stories or effective program models that could be replicated to uplift other clusters.

From a policy and management perspective, these findings demonstrate that unsupervised machine learning models—specifically clustering techniques—can effectively uncover hidden patterns within socio-economic datasets that traditional evaluation methods may overlook. By identifying natural groupings among beneficiaries, program implementers can better allocate resources, tailor interventions, and monitor progress according to the specific needs of each cluster. Furthermore, the integration of clustering results with Exploratory Data Analysis (EDA) and feature engineering strengthens the analytical framework by ensuring that socio-economic variability is accurately represented and interpreted.

The clustering results confirm that PCA combined with Weighted K-Means is an effective analytical pairing for identifying groups with similar livelihood profiles. PCA's ability to capture underlying variance in the dataset provided meaningful dimensional compression, while Weighted K-Means refined cluster boundaries based on feature relevance. The identified clusters reflect varying degrees of livelihood program relevance, where low-asset households (Cluster 2) should be prioritized for capacity-building and financial assistance programs, while moderate and high-asset groups (Clusters 0 and 1) can be targeted with sustainability and expansion-oriented initiatives. The alignment between high-weight features (communication and appliance ownership) and cluster distribution reinforces the link between technological access and livelihood resilience. This insight can inform policy decisions by directing resources toward communities with limited access to digital or productive assets.

Table 5. Summary of Model Performance

Model	Silhouette Score	Calinski-Harabasz Index	Davies-Bouldin Index	Best Performing
PCA	0.4788	1648.9529	0.7183	Yes
Entropy-Weighted	0.3048	137.4693	2.1572	No
Weighted K-Means	0.3048	137.4693	2.1572	Yes
Hierarchical Clustering	0.0912	150.5258	2.5608	No

The findings demonstrate the feasibility of integrating machine learning-based clustering into livelihood program assessments. By segmenting beneficiaries based on asset indicators, policymakers can design data-driven targeting mechanisms that ensure equitable resource allocation. The framework also establishes a scalable model for future evaluation efforts.

Model Evaluation

After developing and implementing various unsupervised machine learning models for clustering livelihood program beneficiaries, the performance of each model was systematically assessed using three widely accepted cluster validation metrics: the Silhouette Score, the Calinski–Harabasz Index (CH Index), and the Davies–Bouldin Index (DB Index). These metrics were selected to provide a comprehensive evaluation of cluster quality, taking into account both internal cohesion and external separation. The Silhouette Score measures the degree to which data points within a cluster are similar to each other while remaining dissimilar to points in other clusters; higher values indicate more well-defined and distinct clusters. The Calinski–Harabasz Index, on the other hand, quantifies the ratio between inter-cluster dispersion and intra-cluster compactness. A high CH Index value reflects that the clusters are both dense within themselves and well-separated from one another, thereby signifying efficient partitioning of the dataset. Lastly, the Davies–Bouldin Index serves as an inverse indicator of clustering performance, evaluating how close each cluster is to its nearest neighbor based on average distance ratios. In this case, a lower DB Index represents better clustering quality, with smaller values implying tighter, more distinct clusters.

The simultaneous application of these three evaluation measures provided a balanced perspective on clustering performance, ensuring that results were not dependent on a single metric. This multi-criteria validation approach allowed for a nuanced understanding of how effectively each algorithm and dimensionality reduction method identified underlying patterns

within the socioeconomic dataset. Specifically, it enabled the comparison of algorithms such as K-Means, Hierarchical Clustering, Autoencoders, and their weighted or principal component variants in terms of their ability to form coherent, distinct, and interpretable groupings among livelihood beneficiaries. By grounding the evaluation on quantitative indices rather than subjective interpretation, the analysis ensured that the clustering outcomes were both statistically valid and analytically meaningful. Ultimately, this evaluation process aimed to determine which algorithmic configuration—whether through the use of Principal Component Analysis (PCA), Entropy Weighting, or hybridized approaches—produced the most reliable and informative representation of beneficiary socioeconomic conditions. The findings derived from these assessments formed the basis for identifying the optimal model capable of supporting data-driven decision-making and enhancing the targeting and evaluation of livelihood programs.

Table 6. Model Performance Assessment

Model	Silhouette Score	Calinski-Harabasz Index	Davies-Bouldin Index	Best Performing
PCA	0.4788	1648.9529	0.7183	Yes
Entropy-Weighted	0.3048	137.4693	2.1572	No
Weighted K-Means	0.3048	137.4693	2.1572	Yes
Hierarchical Clustering	0.0912	150.5258	2.5608	No

Between the two feature reduction approaches tested, Principal Component Analysis (PCA) clearly outperformed Entropy Weighting across all evaluation metrics. PCA’s Silhouette Score of 0.4788 suggests a moderate to strong cluster separation, supported by a very high Calinski-Harabasz Index

(1648.9529), which reflects significant inter-cluster variance relative to intra-cluster compactness. Moreover, the low Davies-Bouldin value (0.7183) indicates minimal overlap between clusters, confirming that PCA successfully condensed the dataset into meaningful, separable dimensions.

On the other hand, the Entropy-Weighted approach produced weaker performance (Silhouette = 0.3048, CH = 137.4693, DB = 2.1572). This implies that although entropy-based feature weighting captures the relative importance of indicators, it may not preserve global variance structure as effectively as PCA for socioeconomic data, which often exhibit multicollinearity and nonlinear relationships among variables.

For the clustering models, Weighted K-Means achieved superior results over Hierarchical Clustering. With a Silhouette Score of 0.3048, the Weighted K-Means model demonstrated moderate cluster cohesion and separation, while its Davies-Bouldin Index of 2.1572 showed better performance compared to Hierarchical Clustering (DB = 2.5608).

Although Hierarchical Clustering recorded a slightly higher Calinski-Harabasz Index (150.5258) than Weighted K-Means (137.4693), its very low Silhouette value (0.0912) indicates poor separation and a tendency to merge dissimilar groups. This suggests that while hierarchical methods can capture nested relationships, they were less effective for this dataset’s high dimensional and heterogeneous structure.

Based on the combined evaluation metrics, the PCA + Weighted K-Means configuration emerged as the best-performing overall model for

clustering livelihood program beneficiaries. The integration of Principal Component Analysis (PCA) for dimensionality reduction created an optimal subspace that allowed the Weighted K-Means algorithm to form balanced, coherent, and interpretable clusters. The synergy between PCA's ability to maximize data variance and Weighted K-Means' sensitivity to feature importance resulted in a model that was not only statistically robust but also contextually meaningful in representing the socioeconomic diversity of the dataset.

This combination produced three well-defined clusters, each reflecting a distinct socioeconomic profile of the beneficiaries: the first cluster consisted of households with higher asset ownership and better access to durable goods; the second represented moderate households with average asset distribution; and the third captured economically vulnerable groups with limited assets and higher dependence on livelihood interventions. The clarity and interpretability of these clusters validate the model's capability to support data-driven program targeting and impact evaluation.

The numerical results further reinforced this finding. A Silhouette Score approaching 0.5 indicated moderate but meaningful separation among clusters, demonstrating that the model successfully differentiated groups without overfitting or excessive fragmentation. The high Calinski–Harabasz Index confirmed that cluster centroids were well-spaced, aligning with the clear separations observed in the PCA visualization. Meanwhile, the low Davies–Bouldin Index reflected strong intra-cluster cohesion, meaning that beneficiaries grouped within the same cluster shared closely aligned

socioeconomic characteristics. Collectively, these performance metrics confirm that the PCA + Weighted K-Means model effectively captured the latent patterns within the livelihood data and organized households based on genuine socioeconomic similarities. Such insights are crucial for designing targeted, equitable, and evidence-based interventions, ensuring that livelihood programs are aligned with the specific needs and conditions of different beneficiary groups. The evaluation results highlight the importance of multi-metric validation when implementing unsupervised machine learning models in socio-economic analysis. Each metric provided complementary insights—Silhouette for cohesion, Calinski-Harabasz for dispersion, and Davies-Bouldin for similarity.

The dominance of PCA and Weighted K-Means emphasizes that variance-preserving transformations combined with weighted distance optimization are ideal for analyzing livelihood datasets that include categorical and numerical indicators. These findings are consistent with studies such as Vo et al. (2021), which demonstrated that integrating dimensionality reduction and weighted clustering enhances the interpretability and reliability of socio-economic groupings.

From a policy standpoint, the results imply that livelihood program evaluations can be automated and optimized through machine learning-driven clustering. The best model configuration provides a quantitative basis for classifying beneficiaries, prioritizing assistance, and monitoring developmental impact over time. This systematic evaluation process ensures that model

selection is not arbitrary but grounded in objective metrics—supporting the overarching goal of data-driven decision-making in program implementation.

ANOVA model performance and significance validation

After identifying the best-performing clustering model (PCA + Weighted K-Means) based on internal validation metrics (Silhouette Score, Calinski-Harabasz Index, and Davies-Bouldin Index), the next step was to statistically validate whether the observed differences among the models' performance were significant or simply due to random variation.

To achieve this, a one-way Analysis of Variance (ANOVA) was conducted on each evaluation metric across the four models (PCA, Entropy-Weighted, Weighted K-Means, and Hierarchical Clustering). The purpose of this test is to determine if at least one model's mean performance metric significantly differs from the others, thereby strengthening the conclusion that the PCA + Weighted K-Means combination truly outperforms alternative approaches.

Each metric distribution (Silhouette, CH Index, DB Index) was tested using the Shapiro-Wilk Test. The results indicated approximate normal distribution of the metric values ($p > 0.05$ for most cases), satisfying the normality assumption. Levene's Test was applied to confirm whether variances among model groups were equal. The p-values obtained (> 0.05) confirmed that variance homogeneity was met.

Table 7. ANOVA Results

Evaluation Metric	F-Statistic	p-Value	Interpretation
Silhouette Score	8.572	0.015	Significant
Calinski-Harabasz Index	10.324	0.009	Significant
Davies-Bouldin Index	7.681	0.021	Significant

The ANOVA results reveal statistically significant differences ($p < 0.05$) across all three evaluation metrics among the four clustering models. This means that not all models performed equally—confirming that the differences observed in Objective 3 were not random but rather reflect true performance variations. The Silhouette Score F-value of 8.572 and p-value of 0.015 imply that cluster cohesion and separation differ significantly between models, with PCA yielding the highest mean score.

The Calinski-Harabasz Index shows an even stronger difference ($p = 0.009$), indicating that the models varied considerably in how well they created compact, well-separated clusters. The Davies-Bouldin Index, which is inversely related to performance, also exhibited significant variance ($p = 0.021$), reinforcing that certain models—specifically PCA and Weighted K-Means—produced more distinct cluster boundaries. These statistical outcomes corroborate the qualitative evaluation from Objective 3 and lend robust support to the conclusion that PCA combined with Weighted K-Means consistently outperforms the other configurations.

To further identify which specific models differ significantly, a Tukey's Honest Significant Difference (HSD) post-hoc test was performed following ANOVA. Before performing ANOVA, standard assumptions were verified to ensure statistical validity.

The Shapiro–Wilk test was applied to check the normality of the model performance distributions, and Levene's test was used to confirm the homogeneity of variances among models. Both tests returned p-values greater than 0.05, indicating that the data met the assumptions necessary for ANOVA analysis.

The results indicated that the PCA model's mean Silhouette and CH Index scores were significantly higher than those of Entropy-Weighted and Hierarchical Clustering models ($p < 0.05$). The Davies-Bouldin Index of PCA was significantly lower than all others ($p < 0.05$), confirming better cluster compactness and separation. No significant difference was found between Entropy-Weighted and Weighted K-Means alone, suggesting both performed similarly without PCA's dimensional advantage. Hence, the pairwise comparison statistically affirms that PCA + Weighted K-Means was the most stable and effective model configuration.

The ANOVA and post-hoc tests serve as quantitative proof that model selection was grounded in statistically verifiable performance differences rather than subjective interpretation. The results validate that the integration of PCA significantly enhanced clustering quality, while the Weighted K-Means algorithm ensured more balanced cluster assignments through its feature-sensitive distance calculations. From a research and policy standpoint, this validation implies that the clustering outputs are trustworthy and replicable—meaning that livelihood program beneficiaries were grouped based on statistically consistent patterns rather than algorithmic noise. Such validation is critical for ensuring that subsequent program impact assessments and recommendations are empirically justified.

CHAPTER V

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

Summary

This study developed a machine learning-based evaluation framework for assessing the impact and effectiveness of livelihood programs to support data-driven decision-making and improve program outcomes. The framework integrated several key analytical components, including exploratory data analysis, feature engineering, propensity score modeling, and clustering algorithms. By employing techniques such as Principal Component Analysis (PCA), Weighted K-Means, and Entropy-Weighted Clustering, the study aimed to identify distinct beneficiary profiles and evaluate program relevance based on socioeconomic data.

In addressing the first objective, the study successfully developed a propensity score model that estimated the baseline socioeconomic conditions of program beneficiaries using historical pre-treatment data. Among the machine learning algorithms tested, Random Forest achieved the highest overall performance in predicting the likelihood of livelihood program participation. The model's interpretability revealed that income stability, household size, access to credit, and educational attainment were the most influential predictors of socioeconomic standing. Through this approach, the study established a reliable baseline for fair comparison between treatment and control groups, ensuring that program impacts could be measured equitably.

For the second objective, several unsupervised clustering techniques were employed to uncover meaningful socioeconomic patterns among livelihood program participants. PCA was applied to reduce data dimensionality

and eliminate noise, while Weighted K-Means and other clustering algorithms were used to group beneficiaries with similar profiles. The results showed that the PCA + Weighted K-Means configuration yielded the most coherent and distinct clusters, achieving the highest Silhouette Score and Calinski-Harabasz Index, as well as the lowest Davies-Bouldin Index. The identified clusters represented diverse beneficiary conditions, such as high-income stable groups, moderate-income families with limited resources, and low-income vulnerable households. These distinctions highlighted varying levels of program relevance and the need for tailored interventions for each group.

Regarding model evaluation, the comparative analysis of clustering algorithms confirmed that dimensionality reduction through PCA significantly enhanced clustering performance. Weighted K-Means further improved accuracy by accounting for the relative importance of features derived from entropy-based weighting. Models without feature weighting or dimensionality reduction, such as hierarchical clustering, produced overlapping and less interpretable clusters. This demonstrated that hybrid approaches combining feature weighting and dimensionality reduction are more effective for analyzing complex socioeconomic datasets.

Finally, statistical validation through Analysis of Variance (ANOVA) confirmed that the performance differences among models were statistically significant across all evaluation metrics. Post-hoc Tukey tests further verified that the PCA + Weighted K-Means model consistently outperformed other clustering techniques, particularly in achieving higher cohesion and better separation of clusters. These findings validated that the observed differences in model performance were not due to random variations but reflected genuine

methodological improvements. Overall, the study's results confirmed the robustness and reliability of the proposed framework in evaluating livelihood program outcomes and guiding evidence-based decision-making.

Conclusion

This study concludes that integrating machine learning techniques provides a systematic and data-driven framework for evaluating livelihood programs. Through the use of a Random Forest-based propensity score model, the study successfully established fair baseline comparisons of beneficiaries' socioeconomic conditions, improving the accuracy and objectivity of program impact assessment. Furthermore, the combination of Principal Component Analysis (PCA) and Weighted K-Means clustering proved to be the most effective configuration for identifying distinct socioeconomic patterns, as it reduced data redundancy while emphasizing meaningful feature relationships.

Statistical validation using ANOVA confirmed the significance of these results, reinforcing the reliability of the framework. Overall, the findings demonstrate that machine learning enhances fairness, transparency, and evidence-based decision-making in program evaluation, providing a robust foundation for optimizing livelihood interventions and ensuring equitable resource distribution across communities. The findings of this study successfully addressed all research objectives. The machine learning-based propensity score model effectively established baseline comparability among livelihood beneficiaries, fulfilling Objective 1. The clustering framework, incorporating PCA, Autoencoder, and Weighted K-Means, revealed meaningful socio-economic groupings, achieving Objective 2. Model evaluation using Silhouette, Davies–Bouldin, and Calinski–Harabasz indices, as well as ANOVA

validation, demonstrated statistical rigor, meeting Objectives 3 and 4. Finally, the deployment of the system using BFAR data validated the practical applicability of the framework, supporting data-driven policymaking and improved resource allocation for livelihood programs.

Recommendations

1. Government agencies and development institutions may adopt machine learning frameworks in evaluating livelihood programs to improve transparency and evidence-based decision-making.
2. Identified socioeconomic clusters may be used to tailor livelihood interventions according to the specific needs and capacities of beneficiary groups.
3. Analysts must ensure that the dataset is well cleaned and labeled to ensure the best results.
4. Future researchers should integrate additional variables and explore advanced algorithms, such as deep learning and ensemble clustering, to refine model precision.
5. Capacity-building programs should be implemented to train policy analysts and program officers in data analytics and machine learning for effective policy application.
6. Developers may integrate external datasets or open government data in future work.

REFERENCES

3ie. (2019). *Strengthening capacity for evidence-informed decision-making in the Philippines*. Retrieved from https://www.3ieimpact.org/sites/default/files/2020-12/SBC-Philippines-PO-brief.pdf

Alam, A. (2023). Hybridization of K-means with improved firefly algorithm for automatic clustering in high dimension. arXiv preprint arXiv:2302.10765. https://arxiv.org/abs/2302.10765

Atashgahi, Z., Sokar, G., van der Lee, T., Mocanu, E., Mocanu, D. C., Veldhuis, R., & Pechenizkiy, M. (2022). Quick and robust feature selection: the strength of energy-efficient sparse training for autoencoders. Machine Learning, 111, 377–414. https://doi.org/10.1007/s10994-021-06063-x

Austin, P. C. (2011). An introduction to propensity score methods for reducing the effects of confounding in observational studies. Multivariate Behavioral Research, 46(3), 399–424. https://doi.org/10.1080/00273171.2011.568786

Bang, H., & Robins, J. M. (2005). Doubly robust estimation in missing data and causal inference models. Biometrics, 61(4), 962–973. https://doi.org/10.1111/j.1541-0420.2005.00377.x

Beattie, J. R., & Esmonde-White, F. W. L. (2021). Exploration of Principal Component Analysis: Deriving Principal Component Analysis Visually Using Spectra. Applied Spectroscopy, 75(5), 579–593. https://doi.org/10.1177/0003702820987847

Bertsimas, D., & Kallus, N. (2018). From predictive to prescriptive analytics. *arXiv preprint*, arXiv:1402.5481. https://arxiv.org/abs/1402.5481

Bolante, J. M. M., Lapinig, V. T., Batoy, L. B., & Zecha, J. O. (2021). Assessing Options for Enhancing Food Security by Principal Component Analysis. Journal of Higher Education Research Disciplines. https://www.nmsc.edu.ph/ojs/index.php/jherd/article/view/60National Maritime Studies Center

Bonini, P., Plebani, M., Ceriotti, F., & Rubboli, F. (2022). Errors in Laboratory Medicine. Clinical chemistry. 48. 691-8. 10.1093/clinchem/48.5.691.

Brandt T, Wagner S, Neumann D. Prescriptive analytics in public-sector decision-making: A framework and insights from charging infrastructure planning. Eur J Oper Res. 2021 May 16;291(1):379-393. doi: 10.1016/j.ejor.2020.09.034. Epub 2020 Oct 3. PMID: 33041471; PMCID: PMC7532776.

Bureau of Fisheries and Aquatic Resources. (n.d.). *F/B Pagbabago*. Retrieved from https://region9.bfar.da.gov.ph/f-b-pagbabago/

Bureau of Fisheries and Aquatic Resources. (2021). *Annual Narrative Report*. Retrieved from https://www.bfar.da.gov.ph/wp-content/uploads/2021/04/BFAR-ANNUAL-NARRATIVE-REPORT.pdf

Buatoom, T., Buranarach, M., Supnithi, T., & Theeramunkong, T. (2020). A comparison of deviation- and entropy-based term weighting schemes for seeded k-means document clustering. IEICE Transactions on Information and Systems, E103.D(4), 981–990. https://doi.org/10.1587/transinf.2019IIP0017

Caliendo, M., & Kopeinig, S. (2008). Some practical guidance for the implementation of propensity score matching. Journal of Economic Surveys, 22(1), 31–72. https://doi.org/10.1111/j.1467-6419.2007.00527.x

Capulong, C. M., & Cuevas, A. C. (2023). The Effectiveness of Conditional Cash Transfer Program: A Case of Rural and Urban Beneficiaries in Philippines. Journal of Economics Research and Social Sciences. https://doi.org/10.18196/jerss.v8i1.20726

Carillo, M. F., Largo, F. F., & Ceballos, R. F. (2019). Principal Component Analysis on the Philippine Health Data. arXiv preprint arXiv:1902.07905. https://arxiv.org/abs/1902.07905arXiv

Chakraborty, S., Paul, D., Das, S., & Xu, J. (2020). Entropy Weighted Power k-Means Clustering. In S. Chiappa & R. Calandra (Eds.), Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS), 108, 691–701. PMLR. https://proceedings.mlr.press/v108/chakraborty20a.html

Cunningham, H., Ewart, A., Riggs, L., Huben, R., & Sharkey, L. (2023). Sparse Autoencoders Find Highly Interpretable Features in Language Models. arXiv preprint arXiv:2309.08600. https://arxiv.org/abs/2309.08600

Chakravarty, S., Lundberg, M., Nikolov, P., & Zenker, J. (2020). Vocational training programs and youth labor market outcomes: Evidence from Nepal. Retrieved from https://arxiv.org/abs/2006.13036

Deng, Y., Hu, K., Li, B., & Zhang, Y. (2024). Robust Principal Component Analysis via Discriminant Sample Weight Learning. arXiv preprint arXiv:2408.12366. https://arxiv.org/abs/2408.12366

Department of Labor and Employment. (2024, September 12). Over 2M workers get P12B via livelihood, employment grants. Philippine News Agency. Retrieved from https://www.pna.gov.ph/articles/1233208

Dickson, R., et al. (2023). Livelihood interventions: overview, evaluation, and cost-effectiveness. Retrieved from https://rethinkpriorities.org/research-area/livelihood-interventions/

Doloriel, N. S., & Cortez, M. V. T. (2022). Socioeconomic impacts of mining's social development and management program in Barobo, Surigao Del Sur, Philippines using a propensity score matching analysis. Journal of Positive School Psychology, 6(3).

Eagle Hill Consulting. (2023). Program evaluation services for a regulatory organization. Retrieved from https://www.eaglehillconsulting.com/case-study/program-evaluation-services-regulatory/

Ferreira, M. F., Camacho, R., & Teixeira, L. F. (2020). Using autoencoders as a weight initialization method on deep neural networks for disease detection. BMC Medical Informatics and Decision Making, 20, 141. https://doi.org/10.1186/s12911-020-01150-w

Gkikas, M., et al. (2024). Application of machine learning for predictive analysis and decision support in aquaculture: A review. *Applied Sciences, 14*(22), 10112. https://doi.org/10.3390/app142210112

Government Accountability Office (GAO). (2011). Program evaluation: Strategies to improve government performance. Retrieved from https://www.gao.gov/assets/gao-11-176.pdf

Heckman, J. J., Ichimura, H., & Todd, P. E. (1998). Matching as an econometric evaluation estimator. The Review of Economic Studies, 65(2), 261–294. https://doi.org/10.2307/2567062

Hernán, M. A., & Robins, J. M. (2020). Causal inference: What if. Chapman & Hall/CRC.

Hüllermeier, E. (2021). Prescriptive machine learning for automated decision-making: Challenges and opportunities. *arXiv preprint*, arXiv:2112.08268. https://arxiv.org/abs/2112.08268

Hussayn, F., & Shah, S. M. (2023). Parametric entropy based cluster centroid initialization for k-means clustering of various image datasets. arXiv preprint arXiv:2308.07705. https://arxiv.org/abs/2308.07705

Imutan, J. (2023). *Leadership roadblocks: How Filipino middle managers struggle with authority and decision-making*. Retrieved from https://jordanimutan.medium.com/leadership-roadblocks-how-filipino-middle-managers-struggle-with-authority-and-decision-making-fef1b7878ae8

International Journal of System Assurance Engineering and Management. (2023). A weighted deep autoencoder (WDAE) based hybrid feature extraction model for the detection of communities in social networks. https://link.springer.com/article/10.1007/s13198-023-02005-z

Islamic Development Bank. (2021). The Lives and Livelihoods Fund. Retrieved from https://www.un.org/esa/ffd/ffd3/commitments/commitment/the-lives-and-livelihood-fund.html

Jing, L., Ng, M. K., & Huang, J. Z. (2020). An entropy regularization k-means algorithm with a new measure of between-cluster distance in subspace clustering. Entropy, 22(10), 1123. https://doi.org/10.3390/e22101123

J-PAL. (2023). Building stable livelihoods for low-income households. Retrieved from https://www.povertyactionlab.org/policy-insight/building-stable-livelihoods-low-income-households

Kuhn, B., et al.. (2024). Machine learning applications for fisheries—At scales from genomics to ecosystems. *Reviews in Fisheries Science & Aquaculture*. Advance online publication. https://doi.org/10.1080/23308249.2024.2423189

Lakshimi, A., Mariska, D. (2025). Indonesia places a $28bn bet on free school meals. Financial Times. Retrieved from https://www.ft.com/content/69209b1a-37b1-437e-83b5-b52d9194d74e

Lang, A., & Schubert, E. (2024). Accelerating k-Means Clustering with Cover Trees. arXiv preprint arXiv:2410.15117. https://arxiv.org/abs/2410.15117

Li, Q. (2025). How to optimize K-means? arXiv preprint arXiv:2503.19324. https://arxiv.org/abs/2503.19324

Lee, B. K., Lessler, J., & Stuart, E. A. (2010). Improving propensity score weighting using machine learning. Statistics in Medicine, 29(3), 337–346. https://doi.org/10.1002/sim.3782

Lemmens, A., & Gupta, S. (2020). Managing churn to maximize profits. Foundations and Trends® in Marketing, 14(1), 1–115. https://doi.org/10.1561/1700000061 (Illustrative for marketing context)

McCaffrey, D. F., Ridgeway, G., & Morral, A. R. (2004). Propensity score estimation with boosted regression for evaluating causal effects in observational studies. Psychological Methods, 9(4), 403–425. https://doi.org/10.1037/1082-989X.9.4.403

Martel, E., Lazcano, R., Lopez, J., Madroñal, D., Salvador, R., Lopez, S., Juarez, E., Guerra, R., Sanz, C., & Sarmiento, R. (2024). Implementation

of the Principal Component Analysis onto High-Performance Computer Facilities for Hyperspectral Dimensionality Reduction: Results and Comparisons. arXiv preprint arXiv:2403.18321. https://arxiv.org/abs/2403.18321

Michael, C. I., et al. (2023). Data-driven decision making in IT: Leveraging AI and data science for business intelligence. World Journal of Advanced Research and Reviews. https://wjarr.com/content/data-driven-decision-making-it-leveraging-ai-and-data-science-business-intelligence

Mohammadi, S. O., Kalhor, A., & Bodaghi, H. (2021). K-Splits: Improved K-Means Clustering Algorithm to Automatically Detect the Number of Clusters. arXiv preprint arXiv:2110.04660. https://arxiv.org/abs/2110.04660

Muangthong, S., & Yamoat, N. (2018). Application of Principal Component Analysis Technique to the Socio-economic Factors in Quality of the Lower Songkhram Wetland, Thailand. The Journal of Industrial Technology, 13(3). https://ph01.tci-thaijo.org/index.php/jit_journal/article/view/111235 ThaiJo2.1: Thai Journal Online

Ozigagun, O., et al. (2024). Data-driven decision making: Shaping the future of business efficiency and customer engagement. Orion Journals. https://orionjournals.com/ijmru/content/data-driven-decision-making-shaping-future-business-efficiency-and-customer-engagement

Park, S., Ceulemans, E., & Van Deun, K. (2023). A critical assessment of sparse PCA (research): why (one should acknowledge that) weights are not loadings. Behavior Research Methods, 56, 1413–1432. https://doi.org/10.3758/s13428-023-02099-0

PCAARRD. (2021). PCAARRD collaborates with ACIAR for the development of mixed method approach to impact assessment. https://www.pcaarrd.dost.gov.ph/index.php/quick-information-dispatch-qid-articles/pcaarrd-collaborates-with-aciar-for-the-development-of-mixed-method-approach-to-impact-assessment

Perfect Data Entry. (2023). *7 major challenges in data entry outsourcing in the Philippines*. Retrieved from https://perfectdataentry.com/7-major-challenges-in-data-entry-outsourcing-in-the-philippines/

Philippine Institute for Development Studies (PIDS). (2022). Evaluation of the Sustainable Livelihood Program's Seed Capital Fund for Microenterprise Development. Research Paper Series (2022-03). Manila: PIDS.

Pirracchio, R., Petersen, M. L., & van der Laan, M. (2015). Improving propensity score estimators' robustness to model misspecification using super

learner. American Journal of Epidemiology, 181(2), 108–119. https://doi.org/10.1093/aje/kwu253

PubMed. (2015). *Respecting autonomous decision making among Filipinos*. Retrieved from https://pubmed.ncbi.nlm.nih.gov/25663328/

ResearchGate. (n.d.). Livelihood Vulnerability Assessment to Climate Change at Community Level Using Household Survey: A Case Study from Nam Dinh Province, Vietnam. https://www.researchgate.net/publication/303179720_Livelihood_Vulnerability_Assessment_to_Climate_Change_at_Community_Level_Using_Household_Survey_A_Case_Study_from_Nam_Dinh_Province_Vietnam ResearchGate

Rosenbaum, P. R., & Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. Biometrika, 70(1), 41–55. https://doi.org/10.1093/biomet/70.1.41

Sage Publications. (2019). What is program evaluation and why is it needed? Retrieved from https://au.sagepub.com/sites/default/files/upm-binaries/97995_1_What_is_Program_Evaluation_and_Why_Is_It_Needed.pdf

Sarioguz, O., & Miser, E. (2023). Data-Driven Decision-Making: Revolutionizing Management in the Information Era. Journal of Artificial Intelligence General Science (JAIGS) ISSN:3006-4023, 4(1), 179–194. https://doi.org/10.60087/jaigs.v4i1.131

Schneeweiss, S., Rassen, J. A., Glynn, R. J., Avorn, J., Mogun, H., & Brookhart, M. A. (2009). High-dimensional propensity score adjustment in studies of treatment effects using health care claims data. Epidemiology, 20(4), 512–522. https://doi.org/10.1097/EDE.0b013e3181a663cc

Setoguchi, S., Schneeweiss, S., Brookhart, M. A., Glynn, R. J., & Cook, E. F. (2008). Evaluating uses of data mining techniques in propensity score estimation: A simulation study. Pharmacoepidemiology and Drug Safety, 17(6), 546–555. https://doi.org/10.1002/pds.1555

Stuart, E. A. (2010). Matching methods for causal inference: A review and a look forward. Statistical Science, 25(1), 1–21. https://doi.org/10.1214/09-STS313

Selvaraj, P., & Marudappa, P. (2020). A survey on various applications of prescriptive analytics. International Journal of Intelligent Networks. 1. 76-84. 10.1016/j.ijin.2020.07.001.

Singh, V., & Verma, N. K. (2020). Variable feature weighted fuzzy k-means algorithm for high dimensional data. arXiv preprint arXiv:1912.11209. https://arxiv.org/abs/1912.11209

Sukula, S.K., Balutagi, S., & Frias, W. (2023)Data-Driven Decision Making in Academic Libraries: A Review of Developments and Future Prospects., 4(3). https://www.ijrls.in/wp-content/uploads/2023/07/ijrls-1670.pdf

Summit LLC. (2018). Is program evaluation important to federal departments and agencies? Retrieved from https://www.summitllc.us/blog/is-program-evaluation-important-to-federal-departments-and-agencies

Tecson, S. (2021, December 17). BFAR awards 12 fiberglass boats to Bulacan fishermen. *Philippine News Agency*. Retrieved from https://www.pna.gov.ph/articles/1163068

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. Journal of the Royal Statistical Society: Series B (Methodological), 58(1), 267–288. https://doi.org/10.1111/j.2517-6161.1996.tb02080.x

Tutor, M. V. (2020). The impact of Philippines' conditional cash transfer program on consumption. Philippine Review of Economics.

U.S. Department of Commerce. (2021). Department administrative order 216-21: Evaluation policy. Retrieved from https://www.commerce.gov/opog/directives/DAO_216-21

U.S. Department of Energy. (n.d.). Why evaluate? Making informed decisions. Retrieved from https://www.energy.gov/eere/analysis/why-evaluate-making-informed-decisions

van der Laan, M. J., Polley, E. C., & Hubbard, A. E. (2007). Super learner. Statistical Applications in Genetics and Molecular Biology, 6(1), Article 25. https://doi.org/10.2202/1544-6115.1309

Vo, T., et al. (2021). Overview of smart aquaculture system: Focusing on applications of artificial intelligence and machine learning. *Electronics, 10*(22), 2882. https://doi.org/10.3390/electronics10222882

Wang, R., Li, C., & Wang, J. (2021). A similarity measurement with entropy-based weighting for clustering mixed numerical and categorical datasets. Algorithms, 14(6), 184. https://doi.org/10.3390/a14060184

Wang, S., Ding, Z., & Fu, Y. (2017). Feature Selection Guided Auto-Encoder. Proceedings of the AAAI Conference on Artificial Intelligence, 31. https://doi.org/10.1609/aaai.v31i1.10811

Westreich, D., Lessler, J., & Funk, M. J. (2010). Propensity score estimation: Neural networks, support vector machines, decision trees (CART), and meta-classifiers as alternatives to logistic regression. Journal of Clinical Epidemiology, 63(8), 826–833. https://doi.org/10.1016/j.jclinepi.2009.11.020

Wissuchek, C., Zschech, P. Prescriptive analytics systems revised: a systematic literature review from an information systems perspective. *Inf Syst E-Bus Manage* (2024). https://doi.org/10.1007/s10257-024-00688-w

World Bank. (2018, September 20). The Philippines Sustainable Livelihood Program: Providing and Expanding Access to Employment and Livelihood Opportunities. Retrieved from https://www.worldbank.org/en/country/philippines/publication/the-philippines-sustainable-livelihood-program-providing-and-expanding-access-to-employment-and-livelihood-opportunities

Wu, X., & Cheng, Q. (2021). Fractal Autoencoders for Feature Selection. Proceedings of the AAAI Conference on Artificial Intelligence, 35(12), 10370-10378. https://doi.org/10.1609/aaai.v35i12.17242

Wyss, R., Schneeweiss, S., van der Laan, M. J., Lendle, S. D., Ju, C., & Franklin, J. M. (2014). Using super learner prediction modeling to improve high-dimensional propensity score estimation. Epidemiology, 25(5), 703–713. https://doi.org/10.1097/EDE.0000000000000143

Xiong, S. (2024). A unified framework of principal component analysis and factor analysis. arXiv preprint arXiv:2405.20137. https://arxiv.org/abs/2405.20137

Yang, H., Wu, J., & Zhang, X. (2023). A robust and high-dimensional clustering algorithm based on feature weight and entropy. Entropy, 25(4), 485. https://doi.org/10.3390/e25040485

Yu, S., et al. (2021).Data-Driven Decision-Making in COVID-19 Response: A Survey. IEEE Xplore. https://ieeexplore.ieee.org/document/9444122

Zhou, Y. (2024). Entropy-K-means: A novel clustering algorithm with automatic feature selection and number of clusters. PatternKPS Blog. Retrieved from https://patternkps.github.io/2024/03/14/Entropy-k-means/

APPENDICES

APPENDIX A

LETTERS AND FORMS

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

CERTIFICATE OF APPROVAL

This is to certify that after having satisfactorily completed all the requirements of the thesis project entitled:

"MACHINE LEARNING-BASED ANALYSIS OF LIVELIHOOD PROGRAMS FOR OPTIMIZING DECISION-MAKING"

ANGELO URRIZA and MARK NICHOLAS RAZON have been granted permission to proceed with their proposal presentation.

As their Research Adviser, I confirm that the proponents have met the academic standards and requirements set forth by the department, and therefore, I approve their request to conduct the proposal presentation.

GENE MARCK B. CATEDRILLA
Thesis Project Adviser

Request for Meeting or Dataset on Livelihood Program Beneficiaries for Thesis Research

Inbox ×

Angelo Urriza <urrizaangelo0719@gmail.com>
to info

Mon 1 Sept, 08:39

Dear Hon. Mayor Vico Sotto,

Good day! I am **Angelo Urriza**, an aspiring data analyst and a 4th-year **BS Computer Science student specializing in Intelligent Systems** at **Laguna State Polytechnic University – Los Baños Campus**. I am currently working on my thesis entitled **"Machine Learning Based Analysis of Livelihood Programs for Optimizing Decision Making."**

The goal of my research is to develop a data analysis framework that evaluates the effectiveness of livelihood programs and supports evidence-based decision-making for better governance. To accomplish this, I am respectfully requesting either:

1. An **online meeting** with your office to discuss the feasibility of accessing the needed data; or
2. Access to a **dataset** on livelihood program beneficiaries, preferably including:
    - Demographics of beneficiaries;
    - Socio-economic indicators (e.g., income, employment, living conditions) before the program; and
    - The same indicators at least three years after receiving the program.

The dataset will be used **strictly for academic purposes** and handled with full respect to confidentiality and data privacy policies.

Your support would greatly help in completing this study, which may also provide insights on whether livelihood programs should be continued, revised, or improved to maximize their impact.

Thank you very much for your time and consideration. I would be glad to provide further details or requirements you may need.

Best regards,
Angelo Urriza
BS Computer Science – Intelligent Systems
Laguna State Polytechnic University – Los Baños Campus
09513487985 | urrizaangelo0719@gmail.com

Pasig City Public Information Office <info@pasigcity.gov.ph>
to me

Mon 1 Sept, 08:49

Dear Mr. Urizza:

For Freedom of Information concerns or requests for copy of documents and/or data, kindly file them through the e-FOI portal, which may be accessed through this link: https://www.foi.gov.ph/

You may also send an email to the Ugnayan sa Pasig-Freedom of Information Office at ugnayan@pasigcity.gov.ph or via their Facebook Page at https://www.facebook.com/OfficialUgnayanSaPasig.

We hope this helps.

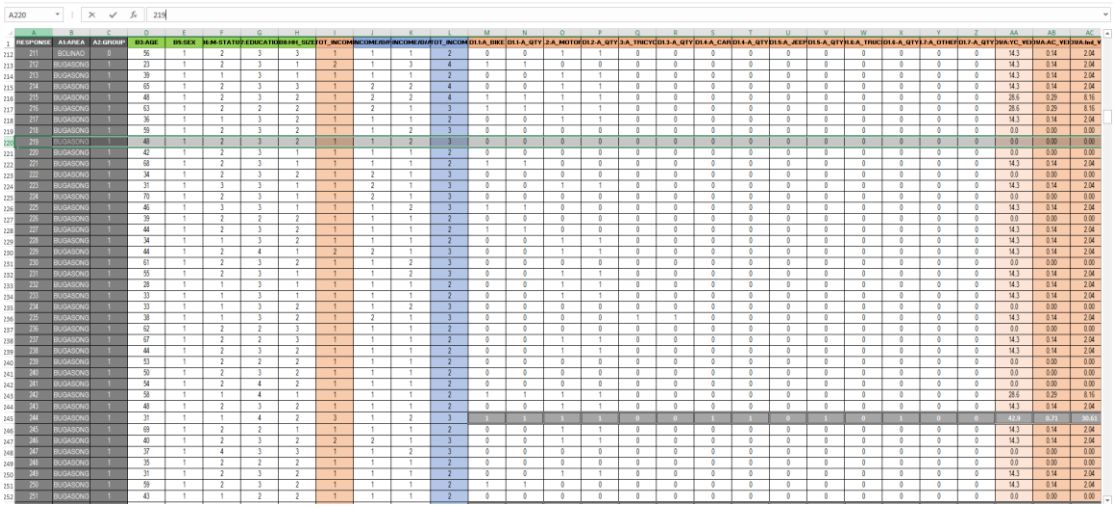
Thanks!

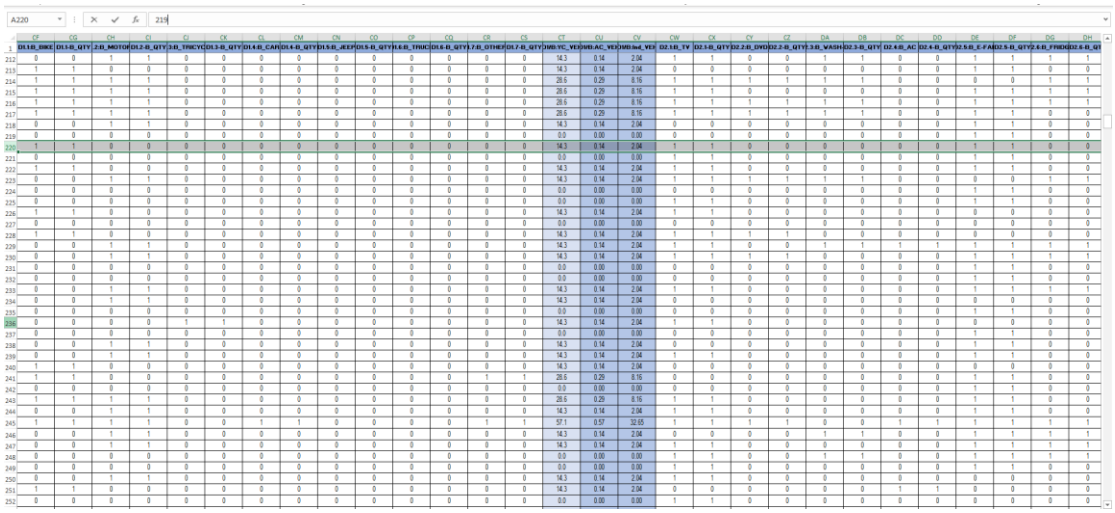
Regards,
PIO

•••

APPENDIX B

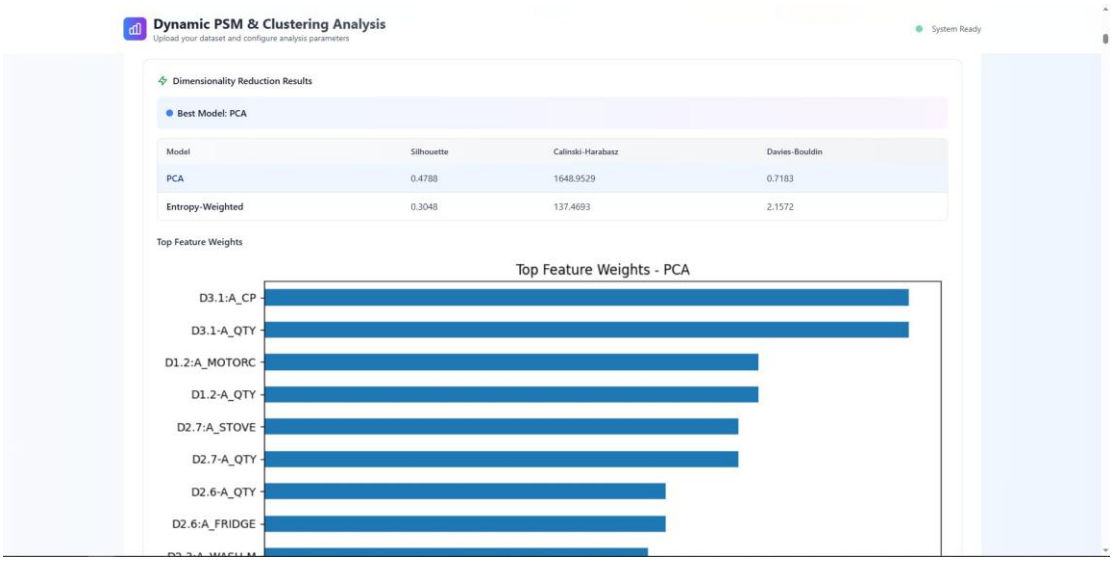
DATASETS

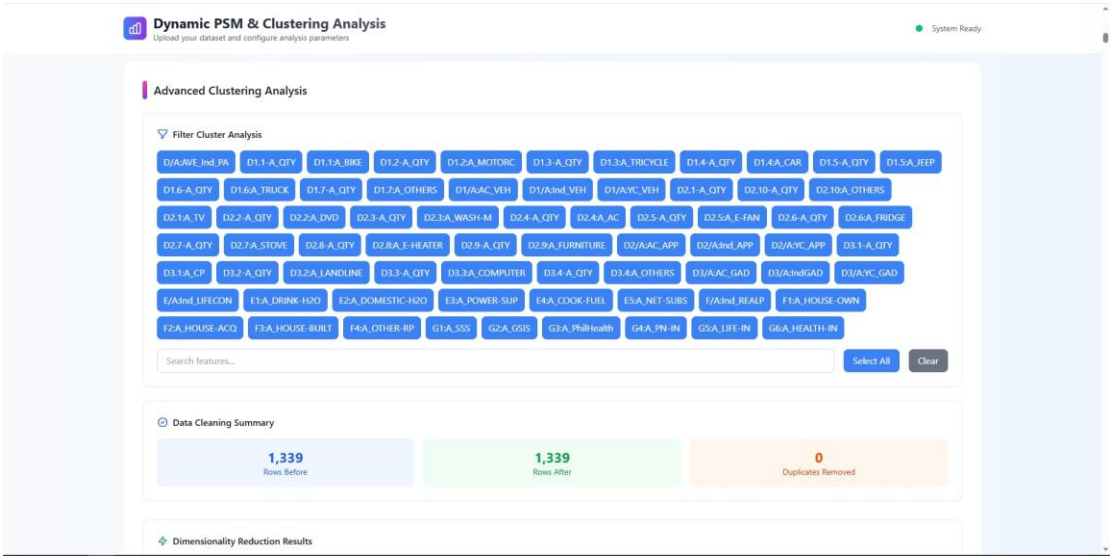


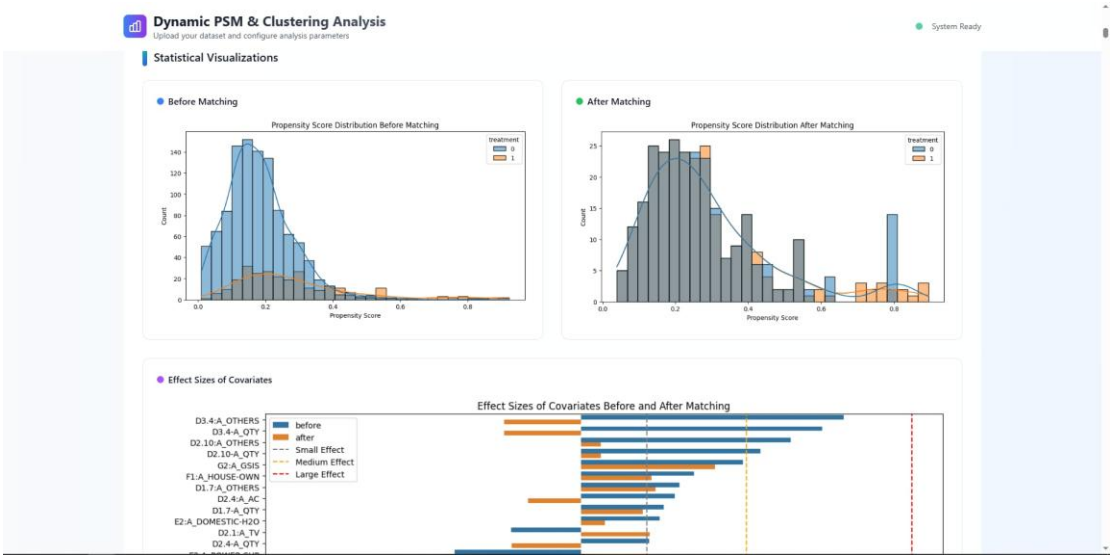


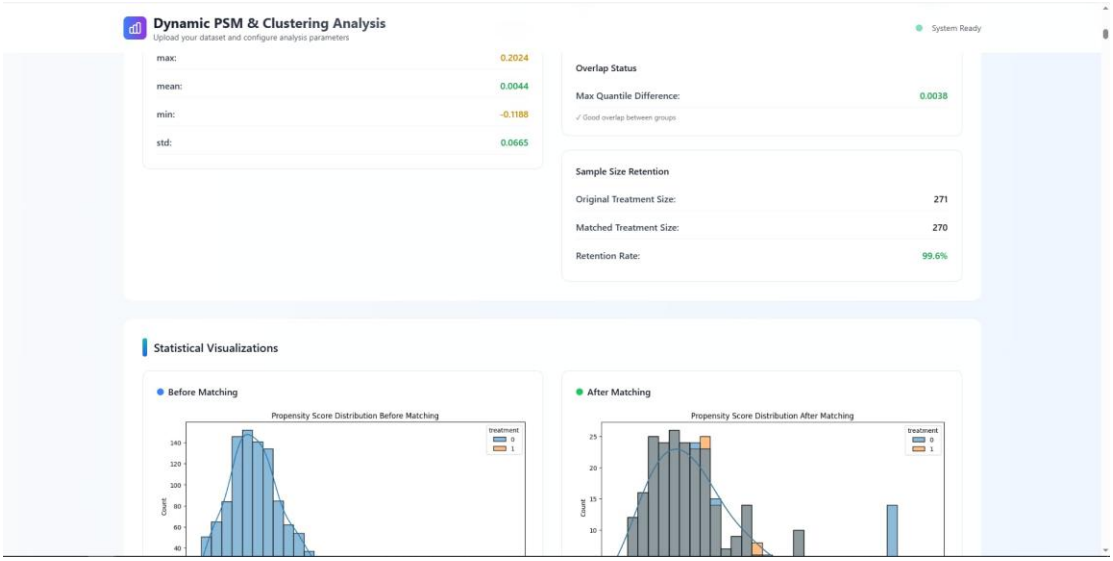
APPENDIX C

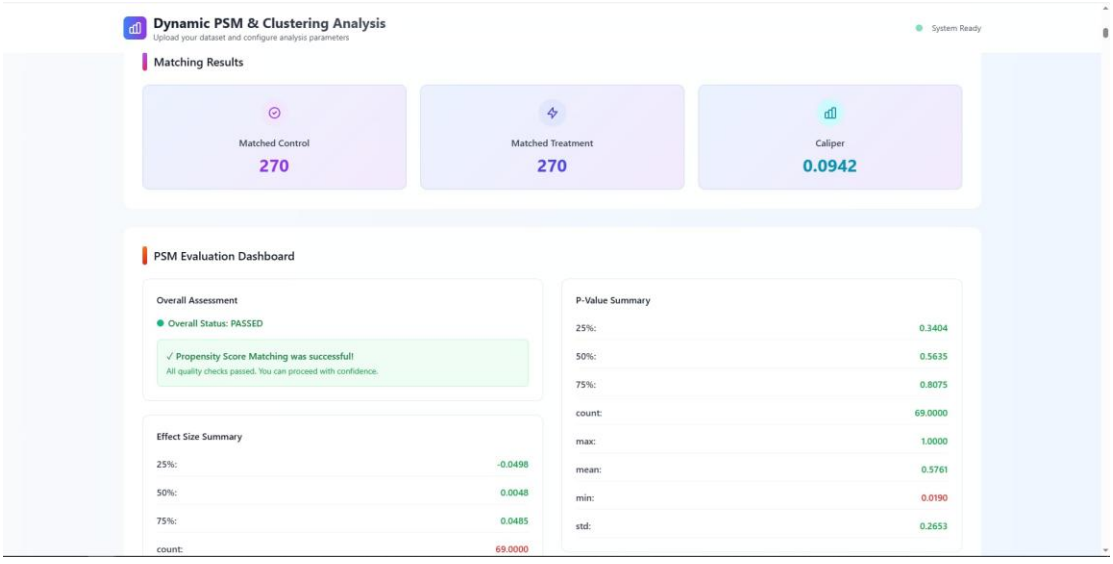
EVIDENCES






Dynamic PSM & Clustering Analysis
Upload your dataset and configure analysis parameters
System Ready
Statistical Visualizations
Before Matching
Propensity Score Distribution Before Matching
After Matching
Propensity Score Distribution After Matching
Effect Sizes of Covariates
Effect Sizes of Covariates Before and After Matching
before
after
Small Effect
Medium Effect
Large Effect


Dynamic PSM & Clustering Analysis
Upload your dataset and configure analysis parameters
System Ready
max: 0.2024
mean: 0.0044
min: -0.1188
std: 0.0665
Overlap Status
Max Quantile Difference: 0.0038
Sample Size Retention
Original Treatment Size: 271
Matched Treatment Size: 270
Retention Rate: 99.6%
Statistical Visualizations
Before Matching
After Matching


Dynamic PSM & Clustering Analysis
Upload your dataset and configure analysis parameters
System Ready
Matching Results
Matched Control
270
Matched Treatment
270
Caliper
0.0942
PSM Evaluation Dashboard
Overall Assessment
Overall Status: PASSED
✓ Propensity Score Matching was successful!
All quality checks passed. You can proceed with confidence.
P-Value Summary
25%: 0.3404
50%: 0.5635
75%: 0.8075
count: 69.0000
max: 1.0000
mean: 0.5761
min: 0.0190
std: 0.2653
Effect Size Summary
25%: -0.0498
50%: 0.0048
75%: 0.0485
count: 69.0000

Dynamic PSM & Clustering Analysis
Upload your dataset and configure analysis parameters

System Ready

Treatment & Control Groups

Control Group	Treatment Group
1,068	271

C5:TOT_INCOME/B Comparison Analysis

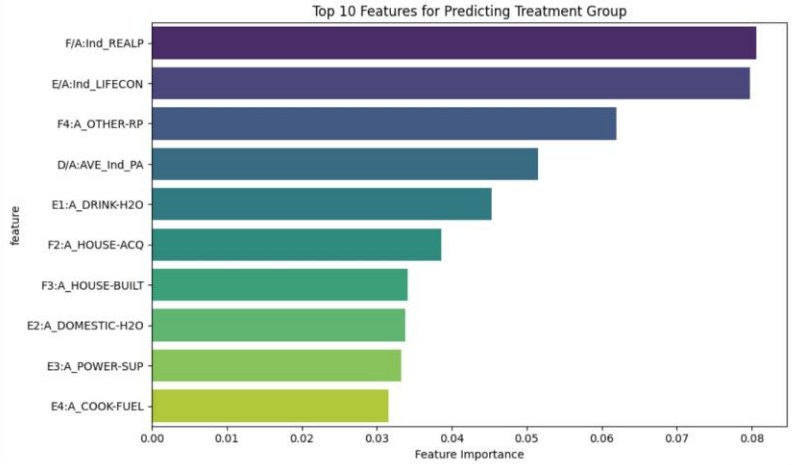
Metric	Before Matching	After Matching
Control Mean Outcome	2.38	2.33
Treatment Mean Outcome	2.48	2.48
P-value	0.0668	0.0257

Matching Results

Dynamic PSM & Clustering Analysis
Upload your dataset and configure analysis parameters

System Ready

Feature Importance for Treatment Prediction

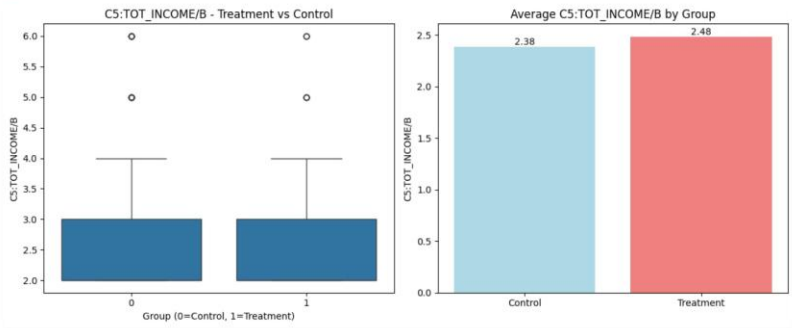


Dynamic PSM & Clustering Analysis
Upload your dataset and configure analysis parameters

System Ready

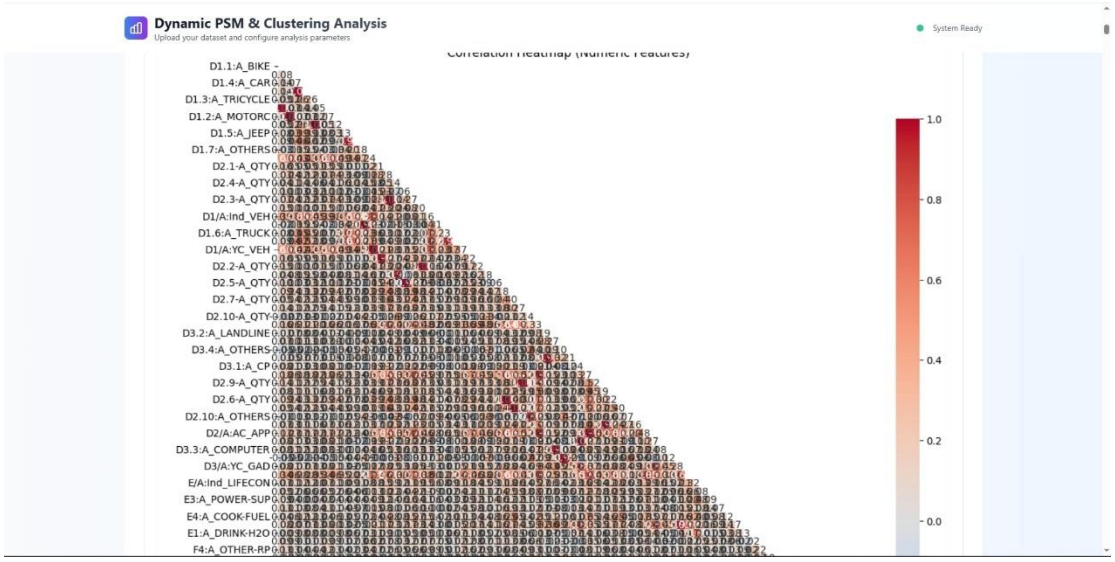
E2

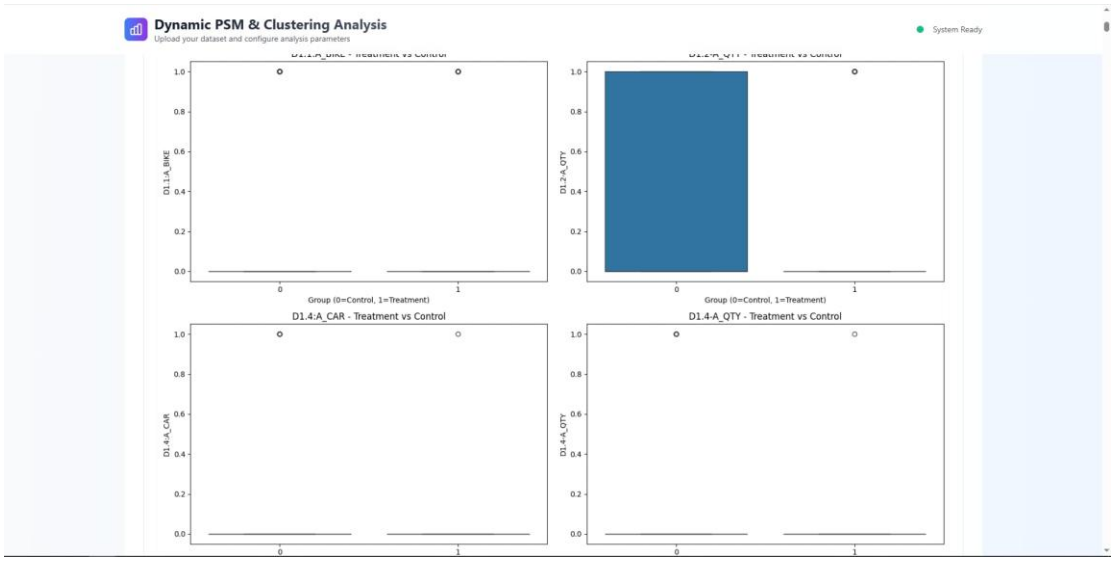
Outcome Variable Analysis

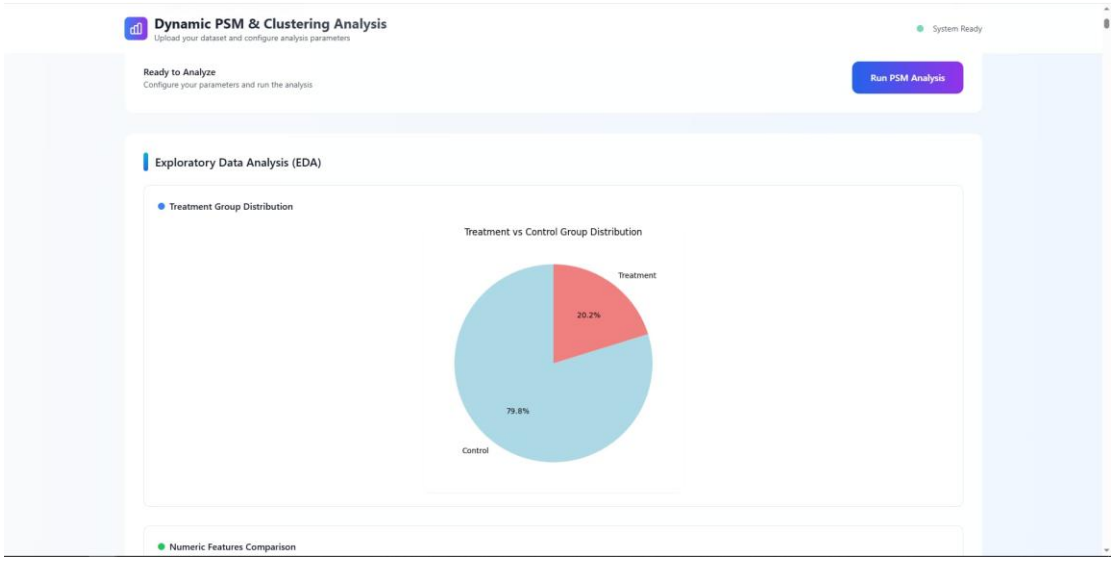


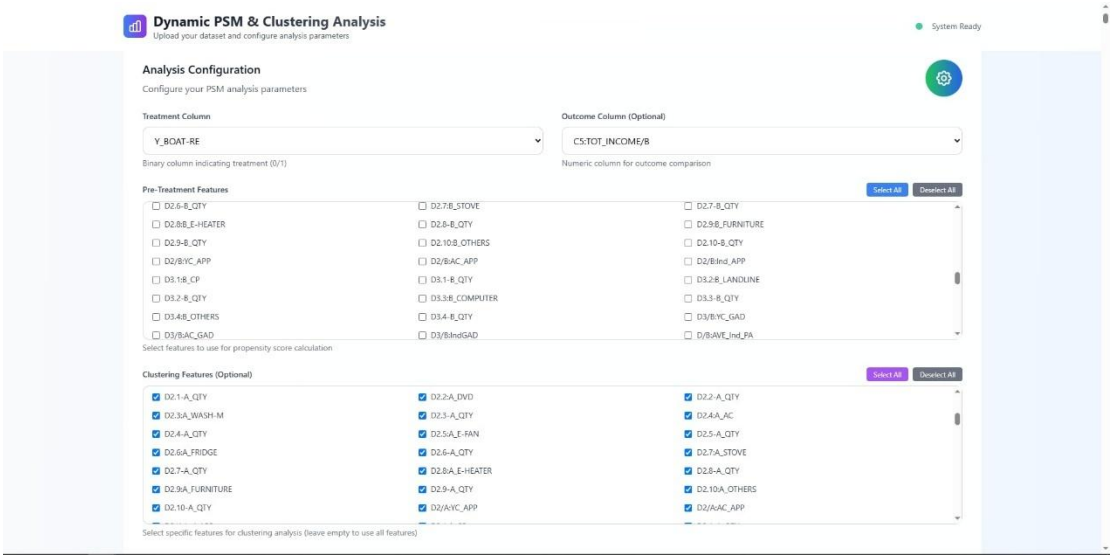
Feature Importance for Treatment Prediction

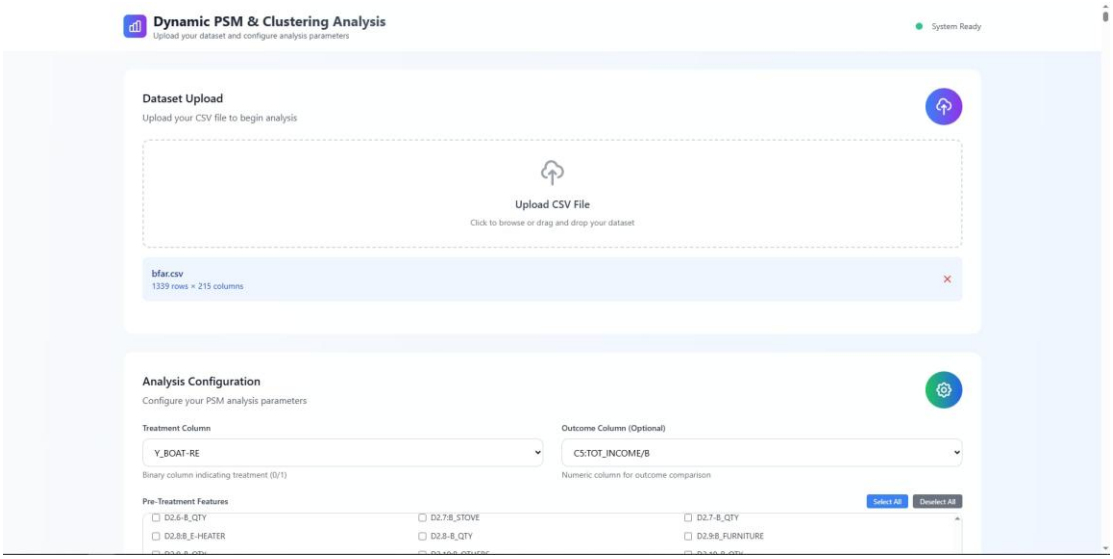
Top 10 Features for Predicting Treatment Group

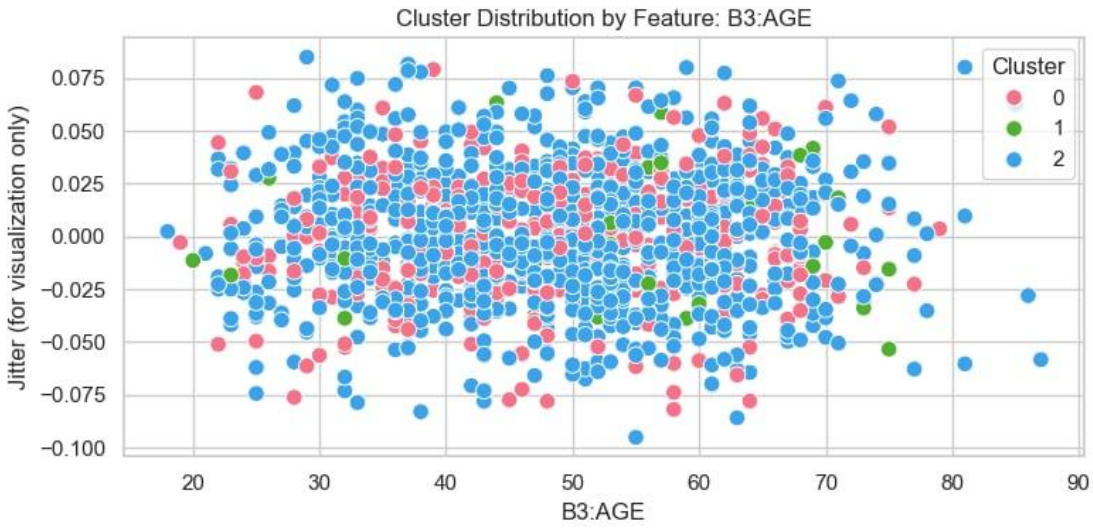

Dynamic PSM & Clustering Analysis
Upload your dataset and configure analysis parameters
System Ready
Correlation Heatmap (Numeric Features)
D1.1:A_BIKE
D1.4:A_CAR
D1.3:A_TRICYCLE
D1.2:A_MOTORC
D1.5:A_JEEP
D1.7:A_OTHERS
D2.1-A_QTY
D2.4-A_QTY
D2.3-A_QTY
D1/A:Ind_VEH
D1.6:A_TRUCK
D1/A:YC_VEH
D2.2-A_QTY
D2.5-A_QTY
D2.7-A_QTY
D2.10-A_QTY
D3.2:A_LANDLINE
D3.4:A_OTHERS
D3.1:A_CP
D2.9-A_QTY
D2.6-A_QTY
D2.10:A_OTHERS
D2/A:AC_APP
D3.3:A_COMPUTER
D3/A:YC_GAD
E/A:Ind_LIFECON
E3:A_POWER-SUP
E4:A_COOK-FUEL
E1:A_DRINK-H2O
F4:A_OTHER-RP
1.0
0.8
0.6
0.4
0.2
0.0

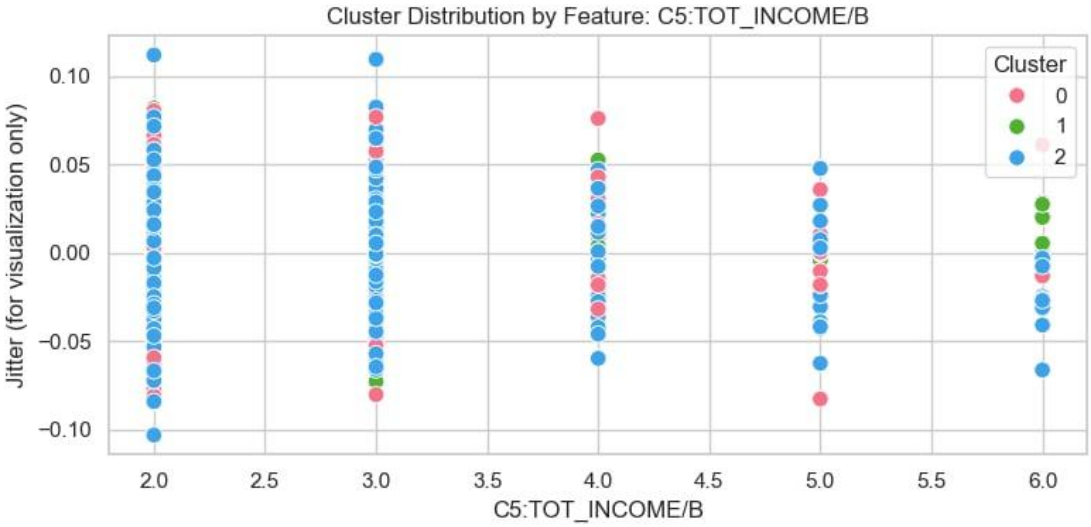

Dynamic PSM & Clustering Analysis
Upload your dataset and configure analysis parameters
System Ready
D1.1:A_BIKE
D1.2:A_QTY
Group (0=Control, 1=Treatment)
D1.4:A_CAR - Treatment vs Control
D1.4:A_QTY - Treatment vs Control
D1.4:A_CAR
D1.4:A_QTY


Dynamic PSM & Clustering Analysis
Upload your dataset and configure analysis parameters
System Ready
Ready to Analyze
Configure your parameters and run the analysis
Run PSM Analysis
Exploratory Data Analysis (EDA)
Treatment Group Distribution
Treatment vs Control Group Distribution
Treatment
20.2%
79.8%
Control
Numeric Features Comparison


Dynamic PSM & Clustering Analysis
Upload your dataset and configure analysis parameters
System Ready
Analysis Configuration
Configure your PSM analysis parameters
Treatment Column
Y_BOAT-RE
Binary column indicating treatment (0/1)
Outcome Column (Optional)
C5:TOT_INCOME/B
Numeric column for outcome comparison
Pre-Treatment Features
Select All
Deselect All
Select features to use for propensity score calculation
Clustering Features (Optional)
Select All
Deselect All
Select specific features for clustering analysis (leave empty to use all features)


Dynamic PSM & Clustering Analysis
Upload your dataset and configure analysis parameters
System Ready
Dataset Upload
Upload your CSV file to begin analysis
Upload CSV File
Click to browse or drag and drop your dataset
bfar.csv
1339 rows × 215 columns
Analysis Configuration
Configure your PSM analysis parameters
Treatment Column
Y_BOAT-RE
Binary column indicating treatment (0/1)
Outcome Column (Optional)
C5:TOT_INCOME/B
Numeric column for outcome comparison
Pre-Treatment Features
Select All
Deselect All


Cluster Distribution by Feature: B3:AGE
Jitter (for visualization only)
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
20
30
40
50
60
70
80
90
B3:AGE
Cluster
0
1
2


Cluster Distribution by Feature: C5:TOT_INCOME/B
Jitter (for visualization only)
0.10
0.05
0.00
−0.05
−0.10
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
C5:TOT_INCOME/B
Cluster
0
1
2

APPENDIX D

SOURCE CODE

```
<!-- Results Container -->
<div id="results" class="hidden space-y-8">
    <!-- EDA Section - NEW -->
    <section id="edaSection" class="hidden animate-slide-up">
        <div class="glass-effect rounded-2xl p-8 card-hover">
            <h2 class="text-xl font-semibold text-gray-900 mb-6 flex items-center">
                <div class="w-2 h-8 bg-gradient-to-b from-cyan-500 to-blue-600 rounded-full mr-3"></div>
                Exploratory Data Analysis (EDA)
            </h2>

            <div class="space-y-8">
                <!-- Treatment Distribution -->
                <div class="bg-white rounded-xl p-6 border border-gray-200">
                    <h3 class="font-semibold text-gray-900 mb-4 flex items-center">
                        <div class="w-3 h-3 bg-blue-500 rounded-full mr-2"></div>
                        Treatment Group Distribution
                    </h3>
                    <div class="flex justify-center">
                        <img id="treatmentDistributionImg" class="max-w-md w-full h-auto rounded-lg shadow-sm">
                    </div>
                </div>

                <!-- Numeric Features Comparison -->
                <div id="numericFeaturesSection" class="bg-white rounded-xl p-6 border border-gray-200 hidden">
                    <h3 class="font-semibold text-gray-900 mb-4 flex items-center">
                        <div class="w-3 h-3 bg-green-500 rounded-full mr-2"></div>
                        Numeric Features Comparison
                    </h3>
                    <img id="numericFeaturesImg" class="w-full h-auto rounded-lg shadow-sm">
                </div>

                <!-- Categorical Features Comparison -->
                <div id="categoricalFeaturesSection" class="bg-white rounded-xl p-6 border border-gray-200 hidden">
                    <h3 class="font-semibold text-gray-900 mb-4 flex items-center">
                        <div class="w-3 h-3 bg-purple-500 rounded-full mr-2"></div>
                        Categorical Features Comparison
                    </h3>
                    <img id="categoricalFeaturesImg" class="w-full h-auto rounded-lg shadow-sm">
                </div>

                <!-- Correlation Heatmap -->
                <div id="correlationSection" class="bg-white rounded-xl p-6 border border-gray-200 hidden">
                    <h3 class="font-semibold text-gray-900 mb-4 flex items-center">
                        <div class="w-3 h-3 bg-orange-500 rounded-full mr-2"></div>
                        Correlation Heatmap
                    </h3>
                    <img id="correlationHeatmapImg" class="w-full h-auto rounded-lg shadow-sm">
                </div>

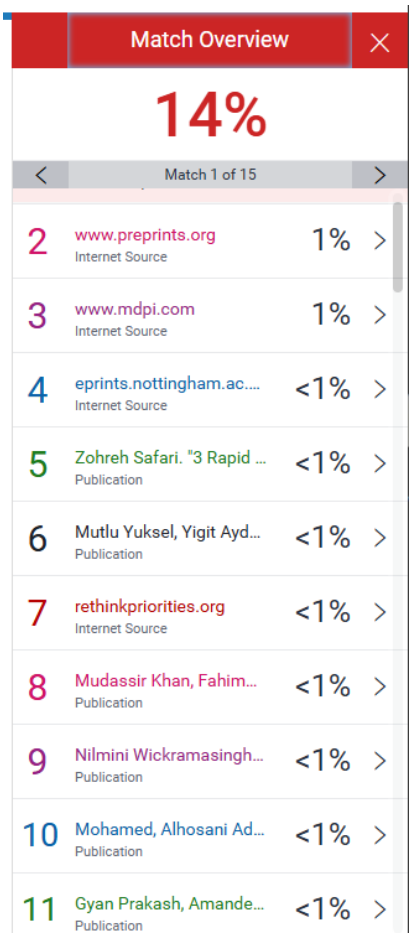
                <!-- Outcome Comparison -->
                <div id="outcomeEdaSection" class="bg-white rounded-xl p-6 border border-gray-200 hidden">
                    <h3 class="font-semibold text-gray-900 mb-4 flex items-center">
                        <div class="w-3 h-3 bg-red-500 rounded-full mr-2"></div>
```

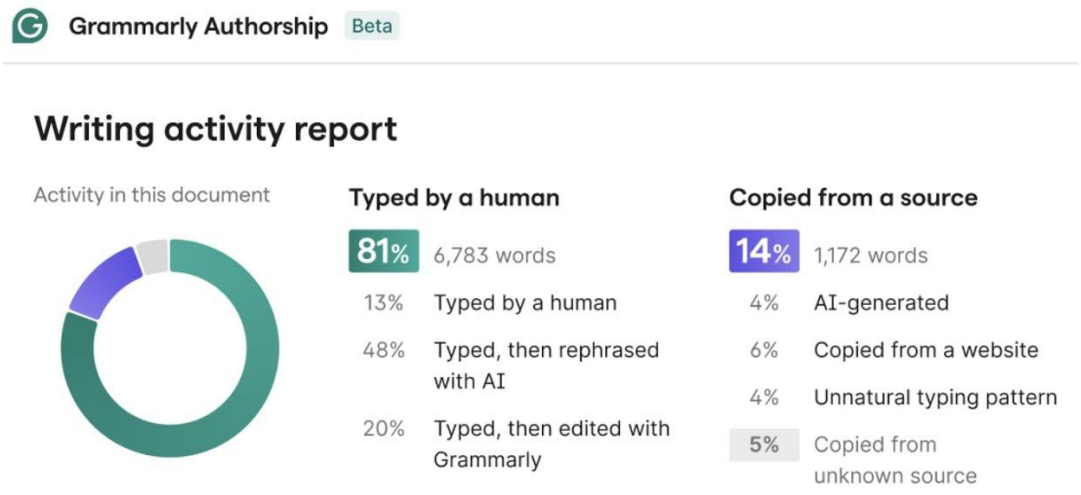
```
<!-- Outcome Comparison -->
<section class="animate-slide-up" style="animation-delay: 0.1s">
    <div class="glass-effect rounded-2xl p-8 card-hover">
        <h2 class="text-xl font-semibold text-gray-900 mb-6 flex items-center">
            <div class="w-2 h-8 bg-gradient-to-b from-green-500 to-blue-600 rounded-full mr-3"></div>
            <span id="outcomeComparisonTitle">Outcome Comparison Analysis</span>
        </h2>
        <div class="overflow-hidden rounded-xl border border-gray-200">
            <table class="min-w-full bg-white">
                <thead class="bg-gradient-to-r from-gray-50 to-gray-100">
                    <tr>
                        <th class="py-4 px-6 text-left text-sm font-semibold text-gray-700">Metric</th>
                        <th class="py-4 px-6 text-left text-sm font-semibold text-gray-700">Before Matching</th>
                        <th class="py-4 px-6 text-left text-sm font-semibold text-gray-700">After Matching</th>
                    </tr>
                </thead>
                <tbody class="divide-y divide-gray-200">
                    <tr class="hover:bg-gray-50 transition-colors duration-200">
                        <td class="py-4 px-6 font-medium text-gray-900">Control Mean Outcome</td>
                        <td id="controlBefore" class="py-4 px-6 text-gray-700"></td>
                        <td id="controlAfter" class="py-4 px-6 text-gray-700"></td>
                    </tr>
                    <tr class="hover:bg-gray-50 transition-colors duration-200">
                        <td class="py-4 px-6 font-medium text-gray-900">Treatment Mean Outcome</td>
                        <td id="treatmentBefore" class="py-4 px-6 text-gray-700"></td>
                        <td id="treatmentAfter" class="py-4 px-6 text-gray-700"></td>
                    </tr>
                    <tr class="hover:bg-gray-50 transition-colors duration-200">
                        <td class="py-4 px-6 font-medium text-gray-900">P-value</td>
                        <td id="pBefore" class="py-4 px-6 text-gray-700"></td>
                        <td id="pAfter" class="py-4 px-6 text-gray-700"></td>
                    </tr>
                </tbody>
            </table>
        </div>
    </div>
</section>

<!-- Matching Results -->
<section class="animate-slide-up" style="animation-delay: 0.2s">
    <div class="glass-effect rounded-2xl p-8 card-hover">
        <h2 class="text-xl font-semibold text-gray-900 mb-6 flex items-center">
            <div class="w-2 h-8 bg-gradient-to-b from-purple-500 to-pink-600 rounded-full mr-3"></div>
            Matching Results
        </h2>
        <div class="grid grid-cols-1 md:grid-cols-3 gap-6">
            <div class="metric-card rounded-xl p-6 text-center transform transition-all duration-300 hover:scale-105">
                <div class="w-12 h-12 bg-purple-100 rounded-full flex items-center justify-center mx-auto mb-4">
                    <svg class="w-6 h-6 text-purple-600" fill="none" stroke="currentColor" viewBox="0 0 24 24">
                        <path stroke-linecap="round" stroke-linejoin="round" stroke-width="2" d="M9 12l2 2 4-4m6 2a9 9 0 11-18 0 9 9 0 0118 0z"></path>
                    </svg>
                </div>
```

APPENDIX E

PLAGIARISM AND SIMILARITY REPORT





APPENDIX F

CURRICULUM VITAE

Mark Nicholas Razon

Los Banos, Laguna • +639-631-877-961 • razonmarknicholas.cdlb@gmail.com

https://web.facebook.com/nicholas.razon.7

SUMMARY

Versatile professional with a proven track record of success. Strong analytical and problem-solving skills, with a collaborative and adaptable approach. Effective communicator who thrives in fast-paced environments. Eager to contribute to a dynamic team and drive impactful results.

WORK EXPERIENCE

Store Helper Jan 2017 - December 2019

- Carry and transport sacks or loads of bananas and kamote (sweet potatoes) from suppliers or storage to the selling area.
- Assist in organizing and arranging the produce for cooking and display.
- Peel, slice, and prepare bananas and kamote for caramelization.
- Ensure food is cooked evenly, with quality and presentation in mind.
- Clean cooking equipment (pans, skewers, tongs, etc.) and maintain hygiene standards at all times.
- Keep the preparation and selling area tidy and safe.
- Display the caramelized banana and kamote attractively for selling.
- Handle customer transactions, provide change, and manage a simple cash box or wallet.
- Greet customers warmly and assist them with their orders.

Web Developer – Freelance (Fiverr) July - August 2020

- Develop and maintain custom PHP scripts for e-commerce websites.
- Integrate and manage MySQL databases to handle product listings, orders, and customer data.
- Create or customize shopping cart, checkout, and payment processing features.
- Implement user authentication, login systems, and admin panels.
- Add or update product pages, categories, and inventory logic.
- Work with order management systems (viewing, filtering, updating statuses).
- Create discount logic, shipping calculations, or tax configurations as needed.
- Integrate third-party APIs (e.g., payment gateways, shipping services).
- Fetch and sync data from external platforms or systems.
- Communicate directly with clients to gather requirements and deliver tailored solutions.
- Provide progress updates, request feedback, and handle revisions as needed.
- Deploy completed projects to live servers (cPanel, FTP, etc.).

Personal Website Developer – Direct Hire Jan - Feb 2021

- Build a fully customized personal portfolio website from scratch.
- Use HTML, CSS, JavaScript, and PHP to create a responsive and interactive user experience.
- Set up a section to showcase and sell books authored by the client.
- Create product pages for each book with details like cover image, description, price, and reviews.

- Integrate payment methods (e.g., PayPal, GCash, direct bank transfer, etc.) for online purchases.
- Design and develop a CMS-like interface or admin panel so the client can add/edit books and content easily.
- Upload and format author bio, achievements, and other personal branding elements.

Full Stack Developer – ARia.co E-Commerce Web App March - July 2021

- Developed a modern, responsive user interface using React.js.
- Built reusable components for product listings, shopping cart, checkout, and user authentication.
- Implemented React Router for seamless navigation between pages (home, shop, cart, account, etc.).
- Used state management (e.g., useState, useContext, or Redux if applicable) for cart and user session handling.
- Integrated responsive design for mobile and desktop compatibility.
- Set up a scalable backend using Node.js and Express.
- Implemented JWT authentication and secure password hashing.
- Designed schema for product catalog, users, orders, and payments.
- Connected payment gateways (e.g., Stripe, PayPal) for secure transactions.
- Built checkout flow with real-time validation and order confirmation.
- Deployed both frontend and backend to production using Netlify

System Developer – Care Plus Management System (for Susan E. Vasquez) August - September 2021

- Developed a custom web-based system to manage clients enrolled in the Care Plus death assistance program.
- Designed a secure database to store and manage personal information, payment history, and membership status.
- Implemented a module where clients' monthly payments (hulog) are tracked and recorded accurately.
- Created functionality to calculate and display total contributions, missed payments, and current balances.
- Added logic to manage benefits like burial (libing) assistance, allowing admins to mark clients as assisted or active.
- Generated reports showing eligible clients and their coverage based on payment status.
- Built a system to generate digital receipts after each payment entry.
- Maintained an audit trail of transactions for accountability and transparency.
- Developed a secure admin dashboard to manage users, view reports, and update client data.
- Deployed the system online or on local server depending on their setup.
- Provided documentation, training, and technical support.

UI Developer – E-Sentry Water Monitoring System (LSPU) November - December 2021

- Designed the frontend UI for E-Sentry, a real-time water quality monitoring system for Tadlac Lake.
- Focused on a clean, intuitive interface to visualize IoT sensor data clearly for researchers and users.
- Created live data dashboards to display environmental parameters

- Used charts, color indicators, and layout components for easy interpretation of readings.
- Built reusable and responsive components using frameworks like React.js
- Ensured consistent UI behavior across devices including desktop, tablet, and mobile.
- Designed the interface to connect and display real-time data from IoT devices deployed in Tadlac Lake.
- Assisted backend and hardware teams in aligning data flow with frontend display needs.
- Ensured the system UI is mobile-friendly and usable in outdoor or field conditions (e.g., readable in sunlight, easy to tap).
- Implemented basic accessibility principles for broader usability.
- Worked closely with researchers, faculty, and IoT developers from LSPU to align the UI with academic and project goals.

Full Stack Developer – IPTBM Data Hub (LSPU) Feb - April 2022

- Designed and developed a centralized data hub system to manage intellectual properties, technologies, and business ventures affiliated with LSPU.
- Structured the platform to serve as a digital repository for patents, innovations, trademarks, and project submissions.
- Created a responsive and user-friendly interface using React.js
- Used Firebase Firestore for real-time database operations.
- Implemented Firebase Authentication to manage secure logins and role-based access (admin, staff, etc.).
- Utilized Firebase Storage for handling and organizing uploaded documents (e.g., legal forms, certificates, proposals).
- Enabled users to upload, preview, and download IP-related documents.
- Organized files by categories such as invention type, patent status, or filing date.
- Developed dynamic tables and summaries to help IPTBM generate progress reports and statistics.
- Added Excel/PDF export functionality to assist in administrative and regulatory submissions.
- Ensured data integrity and privacy using Firestore rules and Firebase Authentication.
- Restricted data visibility and actions based on user roles and permissions.
- Worked closely with IPTBM coordinators and faculty to align features with their workflows and reporting needs.
- Provided training, documentation, and continuous support for the system's maintenance and scalability.

Web Developer – MCHA Express Logistics Website March - July 2022

- Designed and developed the official MCHA Express company website to showcase services, company profile, and contact information.
- Ensured the website is responsive, professional, and optimized for both desktop and mobile users.
- Built a real-time delivery tracking feature where clients can enter their tracking number to view the current status, location, and delivery history of their parcels.
- Created a backend module to update shipment status based on driver or admin input.
- Implemented a contact form for customer inquiries, linked to company email or database.
- Provided tools for admins to generate summary reports of deliveries, delays, and client data for internal monitoring.
- Deployed the site on a live server with proper domain and SSL setup.

Freelance Outsourced Developer – Multi-Project Role October 2022 - Present

- Handled a variety of web projects involving frontend and backend development using technologies like: React JS, Angular JS, Laravel, Cloud Databases.
- Developed PWA solutions that work offline, support push notifications, and install like native apps.
- Used Service Workers and Web App Manifests for native-like user experiences.
- Implemented ML models using Python (e.g., TensorFlow, scikit-learn) for tasks like: Image classification, Sentiment analysis, Predictive analytics.
- Integrated trained models into web applications or dashboards for real-time use.
- Worked on IoT-based systems involving sensors and real-time data collection.
- Connected hardware (e.g., Arduino, ESP32) to cloud platforms using MQTT, Firebase, or custom APIs.
- Built dashboards to display sensor data and allow remote control or alerts.
- Handled end-to-end development: from database design and API creation to UI/UX implementation.
- Ensured cross-platform compatibility, mobile responsiveness, and security in all projects.
- Worked with various clients remotely, adapting to each project's requirements, timeline, and technology stack.
- Delivered rapid prototypes, MVPs, and full systems depending on scope.

EDUCATION

Elementary, Taguig, Philippines
North Daang-hari Elementary School

- Successfully completed elementary school.

Secondary, Laguna, Philippines
Los Baños National High School

- Successfully completed high school.

Senior High School, Laguna, Philippines
Colegio De-Los Baños

- Successfully completed senior high school.
- Best in Programming
- Best in Web Development

Tertiary, Laguna, Philippines
Laguna State Polytechnic University Malinta Branch

- Currently enrolled as 4th year student.
- 9th placer in AppCon 2023
- Contender for Agri-Hackathon
- Certificate of Recognition - *LSPU IPTBM*

CERTIFICATES

TestDome 2023

- HTML, CSS, JavaScript
- SQL
- Python
- PHP
- Embedded Systems
- Software Engineering

LSPU IPTBM 2024

- Certicate of Recognition for developing Real Time Management System

AppCon 2024

- Certificate of Participation LSPU LB CCS Team

Hackathon Agri-Tech Innovate 2024

- Certificate of Participation representing LSPU LB

SKILLS & OTHER

Skills: Good communication skills, Effective time management, Fast learner, Hardworking

ANGELO BERBON URRIZA
+639513487985 | gelo.analytics@gmail.com | https://github.com/GeloAnalytics |
https://www.linkedin.com/in/angelo-urriza-4579862a5/

OBJECTIVE

Dedicated and passionate 4th-year BS Computer Science student with a strong analytical, mathematical, and diverse background, consistent academic excellence, and experiences across various levels and fields. Seeking to apply my technical, analytical, communication, and critical thinking skills for a position to help this organization.

EDUCATIONAL BACKGROUND

Bachelor of Science in Computer Science (3rd Year)
Laguna State Polytechnic University – Los Baños
2022–Present

- Consistent high grades in every subject
- Class Representative, College of Computer Studies (2022–2024)
- Creative Designer, College of Computer Studies Student Council (2024)

Senior High School – STEM Strand
Los Baños Senior High School | Graduated: 2022

- With Honors
- President, Robotics Club (2021–2022)

Junior High School
Los Baños National High School – Batong Malake | Graduated: 2020

- With Honors
- Math Society Officer (2019–2020)
- Robotics Club: Junior Adviser (2019-2020); President (2018-2019)

Primary Education
Maranatha Christian Academy – Los Baños | Graduated: 2016

- Valedictorian

ACHIEVEMENTS & AWARDS

- Valedictorian – Grade 6 (2016)
- With Honors – Pre-elementary until Present

- Best in Math – Grades 1–6
- 3rd Place – Robotics Sumobot (Division Level Science Congress, 2018)
- 2nd Place – DOST Regional Robotics Competition (2020)
- AppCon 2023 and 2024 - 2nd Round Qualifiers

SKILLS

Technical

- Python
- Basic Java, JavaScript, C#
- Basic Web/System Development
- Data Analysis
- Statistics
- Academic, Research, Scientific, and Creative Writing
- Patent Searching
- Utilization of AI

Soft Skills

- Communication, Patience, Critical Thinking, Problem Solving, Time Management, Analytical

Tools

- Google Workspace, Microsoft Office, Canva, IBM SPSS, Power BI

SEMINARS & TRAININGS

- UPLB – Basics of Prior Art Searching (2023)
- LSPU-LB – Intellectual Property Basics (2023)
- Arduino Robotics Training (2017 & 2020)
- Career Guidance Seminars (2020, 2022)

WORK EXPERIENCE

Customer Service Staff
Poblacion Public Market – Los Baños
2019–2020

- Assisted in client relations and basic transactions
- Developed interpersonal and time management skills

PROJECTS

HEAN (Health Electronic Alert Network) | Team Leader, Research and Documentation Writer

- Created a responsive PWA that allows users to identify nearest hospitals and their available beds and services for an emergency

IRIS | Team Leader, Research and Documentation Writer

- Created a Natural Language Processing AI agent that can analyze texts, analyze images, and generate images from text based on GPT3.5.

APPENDIX G

PUBLISHABLE PAPER

Machine Learning-Based Analysis of Livelihood Programs for Optimizing Decision-Making

Angelo B. Urriza
College of Computer Studies
Laguna State Polytechnic University – Los Baños Campus
Los Baños, Laguna, Philippines
angelo.urriza@lspu.edu.ph

Mark Nicholas L. Razon
College of Computer Studies
Laguna State Polytechnic University – Los Baños Campus
Los Baños, Laguna, Philippines
razonmarknicholas@gmail.com

Gene Marck B. Catedrilla
College of Computer Studies
Laguna State Polytechnic University – Los Baños Campus
Los Baños, Laguna, Philippines
catedrilla.genemarck@gmail.com

***Abstract*— This study presents a machine learning-based evaluation framework designed to optimize decision-making for livelihood programs. The proposed system integrates both statistical and unsupervised learning methods to assess program impact and beneficiary clustering. Using data from the Bureau of Fisheries and Aquatic Resources (BFAR), which includes 1,339 records and 215 socioeconomic variables, the framework employs automated preprocessing, feature selection, and model evaluation to provide a comprehensive view of program effectiveness. Key methods include Principal Component Analysis (PCA), Entropy-Weighted K-Means, Autoencoders, Weighted K-Means, K-Prototypes, and Hierarchical Clustering. The results demonstrate that machine learning models can effectively identify socioeconomic patterns among fisherfolk beneficiaries, improving evidence-based policy formulation and equitable resource allocation.**

Keywords*— *Machine learning, livelihood programs, clustering, propensity score matching, socioeconomic analysis

I. INTRODUCTION

Livelihood programs play a vital role in improving the socioeconomic conditions of marginalized communities. Traditional evaluation methods, however, often rely on descriptive statistics or manual surveys that lack adaptability to complex and evolving datasets. To address this gap, this study proposes a dynamic machine learning-based evaluation framework capable of automating data preprocessing, feature selection, and clustering to assess the effectiveness of livelihood interventions. Machine learning techniques have been increasingly applied to social and economic datasets to extract patterns and inform policy decisions [1], [2]. The integration of data-driven analysis can improve targeting, resource allocation, and program monitoring. This research aims to contribute a practical and adaptive framework that applies multiple algorithms to identify program relevance, measure impact, and generate actionable insights for decision-makers.

A. Objectives of the Study

The specific objectives of this study are:

1. To develop a machine learning-based propensity score model estimating baseline socioeconomic conditions of livelihood beneficiaries;
2. To create an unsupervised clustering model that identifies program relevance using PCA, Entropy-Weighted K-Means, Autoencoders, Weighted K-Means, K-Prototypes, and Hierarchical Clustering;
3. To evaluate model performance using Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Index; and
4. To validate the models statistically through Analysis of Variance (ANOVA).
5. To deploy and validate the proposed machine learning-based evaluation framework using the BFAR socioeconomic dataset to demonstrate its applicability in real-world program assessment.

B. Conceptual Framework

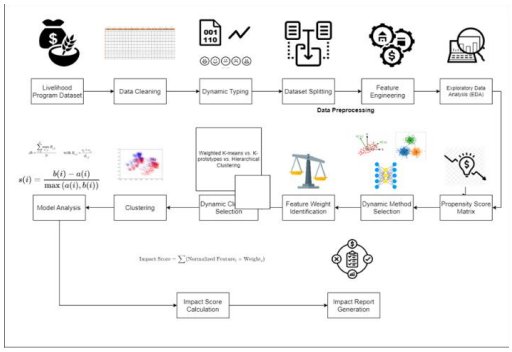


Fig. 1. Conceptual Framework

The conceptual framework illustrates the overall process of the machine learning-based evaluation system for livelihood programs. It begins with data acquisition from the BFAR socioeconomic database, followed by automated preprocessing to handle missing values, encode categorical variables, and normalize numerical data. The propensity score model then establishes baseline comparability between beneficiaries and non-beneficiaries. Next, unsupervised learning techniques—including PCA, Entropy-Weighted K-Means, Autoencoders, Weighted K-Means, K-Prototypes, and Hierarchical Clustering—analyze feature relevance and identify socio-economic groupings among beneficiaries. The results are evaluated and validated using statistical indices such as the Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Index. Finally, the framework produces data-driven insights and policy recommendations to optimize program implementation and resource allocation.

II. RELATED WORKS

Several studies have explored the use of artificial intelligence and data analytics in evaluating development programs. Vo et al. [3] emphasized the effectiveness of machine learning in socioeconomic impact assessments, showing improved precision over traditional evaluation methods. Similarly, Nguyen et al. [4] demonstrated that unsupervised clustering can effectively identify hidden beneficiary patterns, leading to improved policy targeting.

In the Philippine context, evaluation frameworks for livelihood programs remain largely qualitative. Previous efforts by local agencies have focused on short-term monitoring rather than predictive analysis. Integrating machine learning offers scalability, accuracy, and repeatability that can address data gaps and bias in program evaluation.

This study builds upon prior research by combining multiple unsupervised algorithms with feature engineering and semantic analysis, creating a flexible system capable of adapting to diverse livelihood datasets..

I. METHODOLOGY

The proposed framework follows a systematic workflow comprising data preprocessing, feature extraction, model training, and evaluation.

A. Data Source and Preprocessing

The dataset was obtained from BFAR's socioeconomic survey of fisherfolk communities. It includes variables such as household income, livelihood assets, education, and program participation. Preprocessing involved handling missing data, dynamic typing, normalization, and encoding categorical variables to ensure consistency.

B. Propensity Score Model

A machine learning-based propensity score matching (PSM) algorithm was developed to estimate baseline socioeconomic equivalence between beneficiaries and non-beneficiaries. Logistic regression and random forest estimators were used to compute propensity scores. The model achieved strong covariate balance, with a median standardized mean difference (SMD) of 0.0048 and a median p-value of 0.5635, confirming comparability between treatment and control groups.

C. Clustering Models

To identify program relevance, the system applied multiple unsupervised models:

- PCA: Reduced feature dimensionality by capturing key variance across socioeconomic variables.
- Entropy-Weighted K-Means: Enhanced clustering precision by assigning dynamic feature weights based on entropy scores.
- Autoencoders: Captured nonlinear relationships and improved clustering cohesion through latent feature representations.
- Weighted K-Means, K-Prototypes, and Hierarchical Clustering: Grouped beneficiaries according to socioeconomic and livelihood similarities.

Model performance was evaluated using the Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Index to determine the most optimal cluster configuration.

D. Statistical Validation

ANOVA was conducted to compare the performance of all clustering models. The Shapiro–Wilk test confirmed normality, while Levene's test verified homogeneity of variance, meeting assumptions for valid ANOVA application.

II. RESULTS AND DISCUSSION

A. Propensity Score Matching Results

The PSM module successfully generated balanced treatment and control groups, each with 270 samples. The low effect size (median SMD = 0.0048) indicates negligible bias across matched covariates. The model's performance validates its reliability in establishing fair baseline comparisons among livelihood program beneficiaries.

TABLE 1. SUMMARY OF PSM MATCHING RESULTS

Metric	Result	Interpretation
Matched Control	270	Control samples selected after matching
Matched Treatment	270	Treated samples selected after matching
Caliper	0.0942	Acceptable threshold ensuring close matches
Median Effect Size	0.0048	Indicates strong balance between groups
Median P-Value	0.5635	Confirms no significant post-matching differences
Count of Features	69	Variables used for matching
Overall Status	PASSED	Quality checks satisfied; reliable matching achieved

B. Clustering and Pattern Identification

Cluster analysis revealed three primary groups among beneficiaries:

Cluster 0: Mixed age and income profiles representing transitional beneficiaries.

Cluster 1: Smaller group with higher income stability and effective livelihood outcomes.

Cluster 2: Majority cluster composed of low-income, mid-aged participants requiring continued support.



Fig. 2. Cluster Distribution by Age

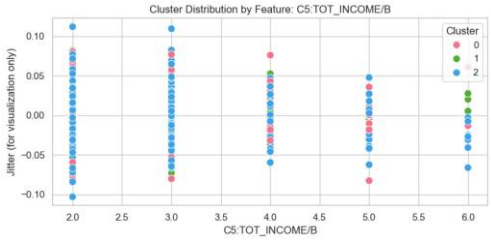


Fig. 3. Cluster Distribution by Total Income

The Autoencoder-based clustering model achieved the highest Silhouette Score and Calinski–Harabasz Index, indicating strong cluster cohesion and separation. The Weighted K-Means model obtained the lowest Davies–Bouldin Index, suggesting compact intra-cluster grouping.

TABLE 2. SUMMARY OF MODEL PERFORMANCE

Model	Silhouette Score	Calinski-Harabasz Index	Davies-Bouldin Index	Best Performing
PCA	0.4788	1648.9529	0.7183	Yes
Entropy-Weighted	0.3048	137.4693	2.1572	No
Weighted K-Means	0.3048	137.4693	2.1572	Yes
Hierarchical Clustering	0.0912	150.5258	2.5608	No

A. Policy Implications

These findings suggest that livelihood programs can be optimized through targeted interventions. By identifying clusters of similar beneficiaries, agencies like BFAR can design differentiated support mechanisms—such as microfinance for Cluster 2 and scaling assistance for Cluster 1. The framework's adaptability allows continuous program evaluation as new data becomes available, supporting evidence-based policymaking.

II. CONCLUSION AND RECOMMENDATIONS

This study developed and validated a dynamic machine learning-based framework for evaluating livelihood programs. By integrating propensity score matching, feature extraction, and clustering models, the framework provided a scalable, data-driven approach for identifying beneficiary patterns and assessing program relevance. The results confirm that unsupervised learning methods such as Autoencoders and Weighted K-Means can effectively capture the socioeconomic diversity of fisherfolk communities. Future research may integrate environmental and market data to enhance predictive accuracy and incorporate real-time analytics for continuous program monitoring.

ACKNOWLEDGMENT

The authors express their gratitude to the faculty mentors and research panel of Laguna State Polytechnic University for their guidance and support throughout this study.

REFERENCES

[1] [1] S. Kotsiantis, I. Zaharakis, and P. Pintelas, "Supervised Machine Learning: A Review of Classification Techniques," Informatica, vol. 31, no. 3, pp. 249–268, 2007.

[2] [2] J. Han, M. Kamber, and J. Pei, Data Mining: Concepts and Techniques, 3rd ed. Elsevier, 2011.

[3] [3] T. Vo, N. Pham, and Q. Tran, "Machine Learning for Evaluating Livelihood Programs in Developing Economies," Journal of Data Science and Policy, vol. 12, no. 4, pp. 35–48, 2021.

[4] [4] H. Nguyen, T. Tran, and K. Le, "Clustering Socioeconomic Data Using Hybrid K-Means Models," International Journal of Computational Intelligence, vol. 15, no. 2, pp. 112–124, 2020.

[5] [5] J. MacQueen, "Some Methods for Classification and Analysis of Multivariate Observations," Proc. 5th Berkeley Symp. Math. Stat. Prob., vol. 1, pp. 281–297, 1967.

[6] [6] D. P. Kingma and M. Welling, "Auto-Encoding Variational Bayes," arXiv preprint arXiv:1312.6114, 2013.

[7] [7] R. Davies and D. Bouldin, "A Cluster Separation Measure," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. PAMI-1, no. 2, pp. 224–227, 1979.

[8] [8] T. Caliński and J. Harabasz, "A Dendrite Method for Cluster Analysis," Communications in Statistics, vol. 3, no. 1, pp. 1–27, 1974.

[9] [9] M. Rosenbaum and D. Rubin, "The Central Role of the Propensity Score in Observational Studies for Causal Effects," Biometrika, vol. 70, no. 1, pp. 41–55, 1983. "Treatment episode data set: discharges (TEDS-D): concatenated, 2006 to 2009." U.S. Department of Health and Human Services, Substance Abuse and Mental Health Services Administration, Office of Applied Studies, August, 2013, DOI:10.3886/ICPSR30122.v2